BT102 - FINALTERM MERGED PPT (LECTURE 53 TO 104) Microbiology

Prepared By:

All About Biotech

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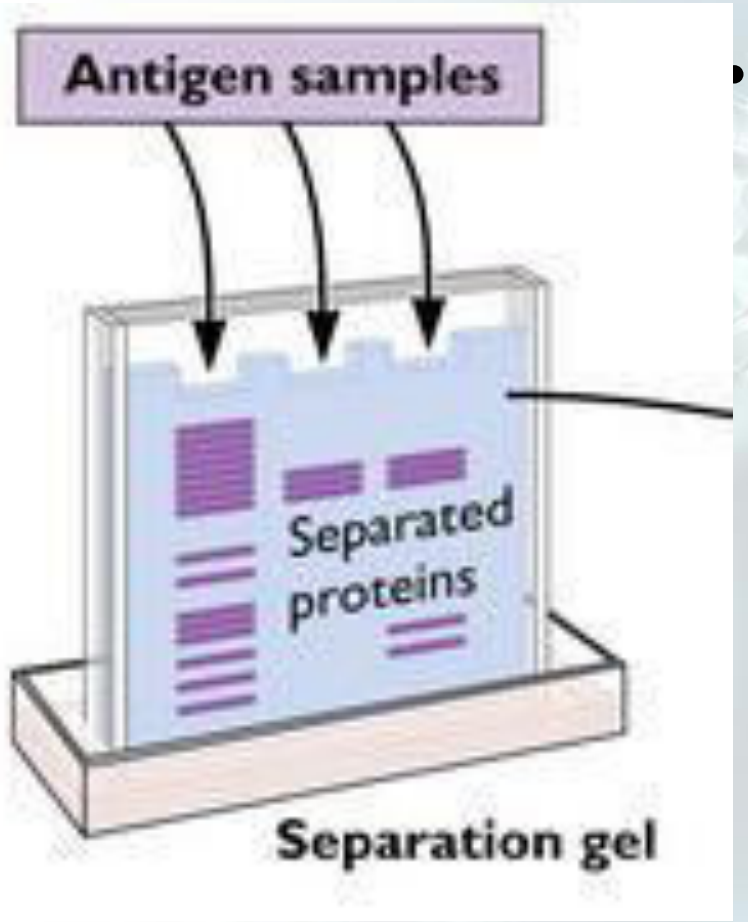
Bacterial Identification V

Learning Objectives

- Western blotting

Bacterial Identification V

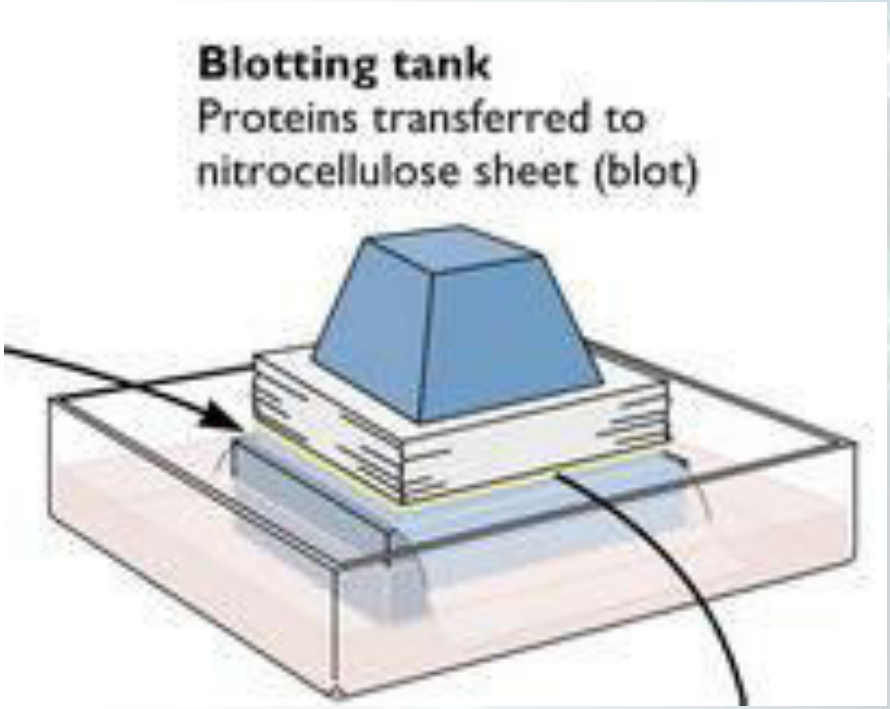
Western Blotting



- Proteins from microbes are separated by electrophoresis

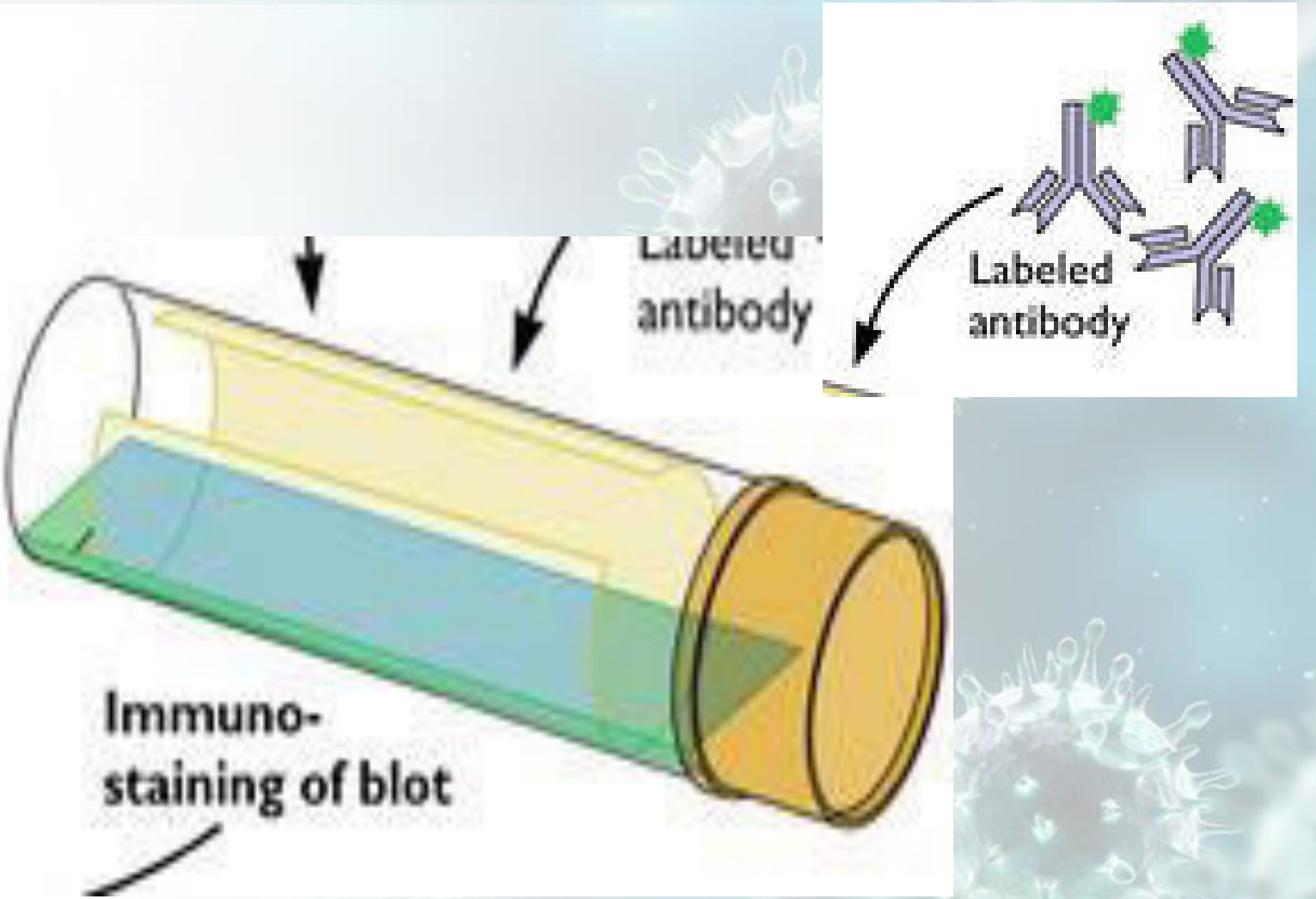
Bacterial Identification V

Serological Testing

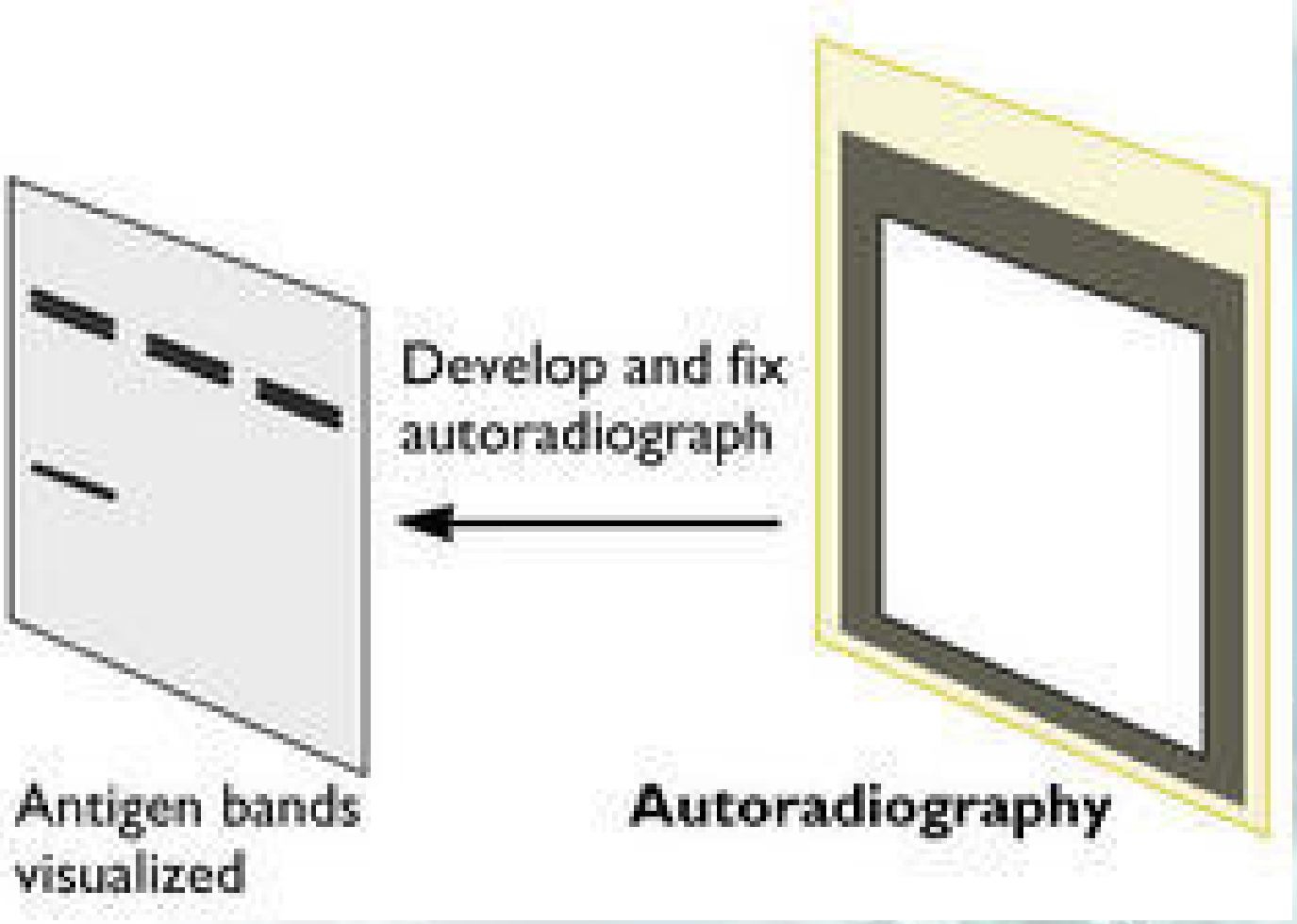


Protein bands are blotted onto a nitrocellulose membrane.

Bacterial Identification V



Bacterial Identification V



Bacterial Identification V

Summary

- Proteins could be detected by electrophoresis and blotting.
- Blots can be detected by specific antibodies labeled with radioisotopes etc.

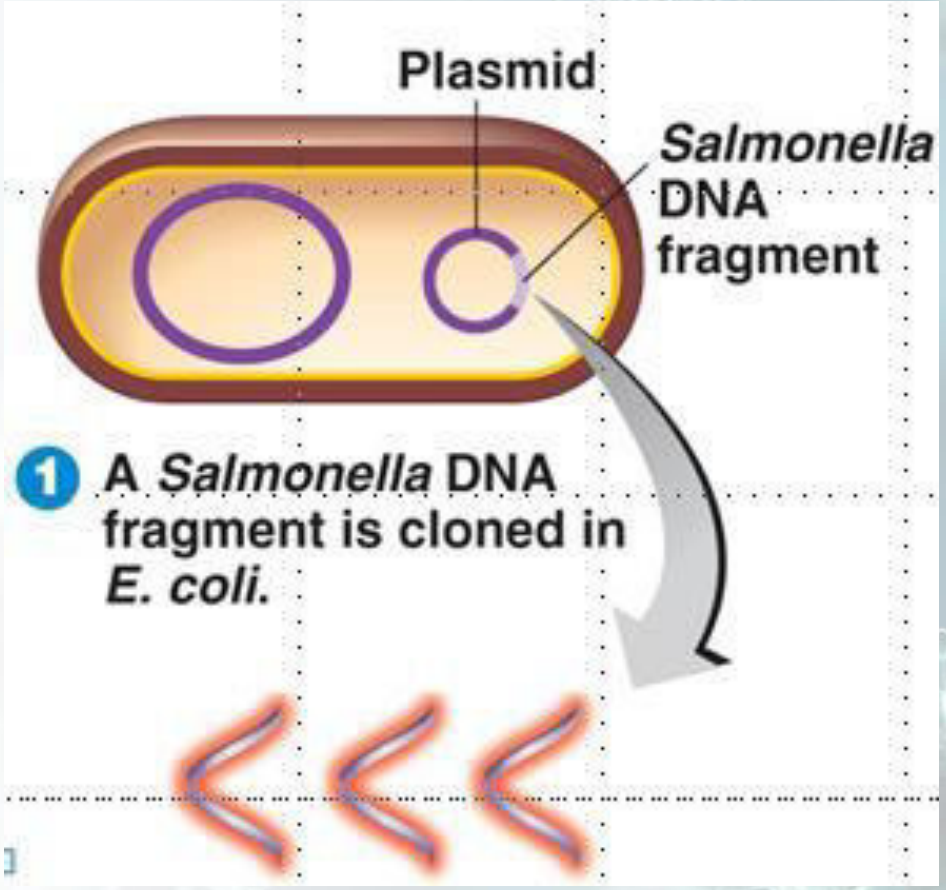
Bacterial Identification VI

Learning Objectives

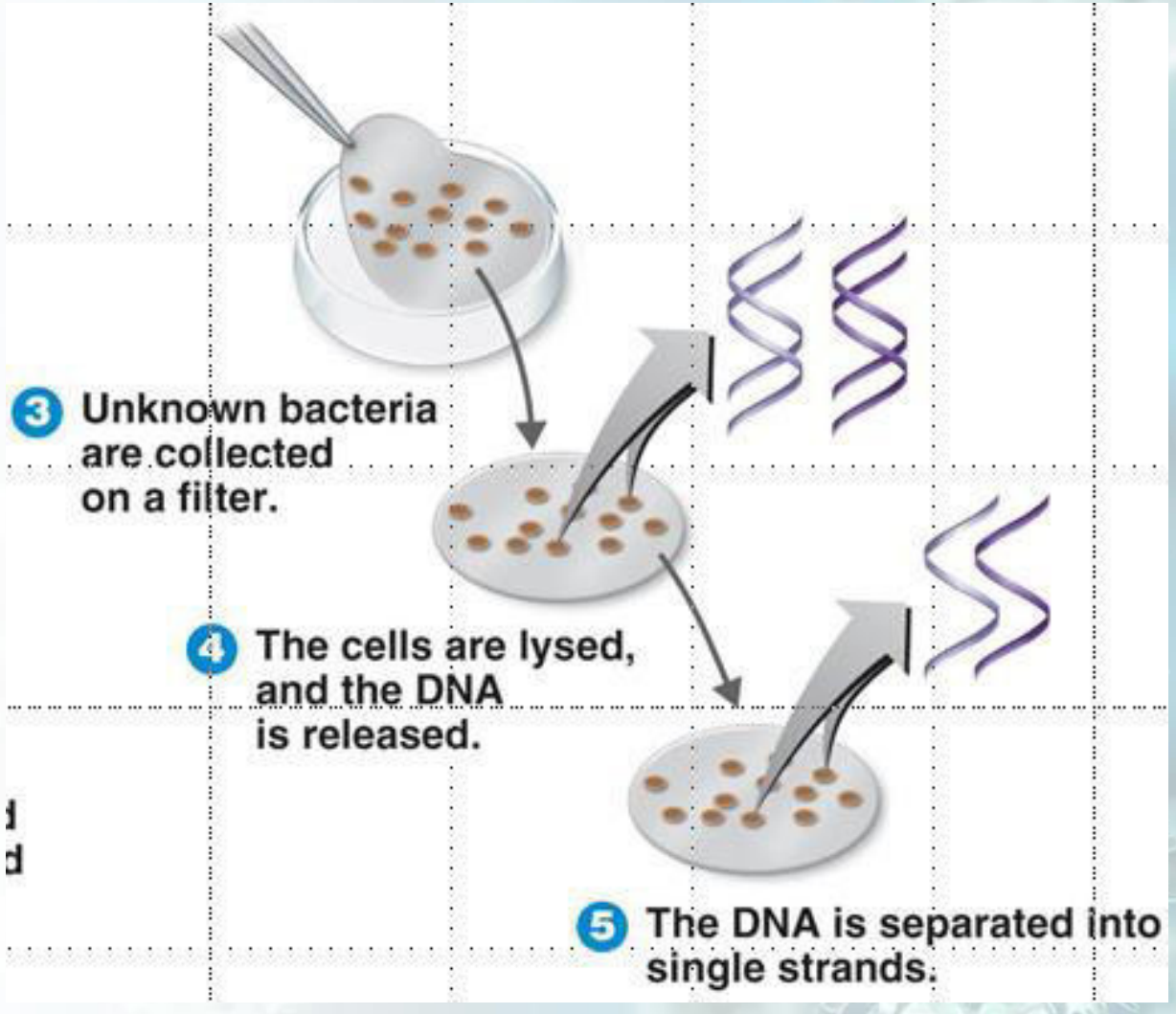
- Nucleic acid base microbial detection methods
  - DNA probes
  - DNA chip
  - PCR

Bacterial Identification VI

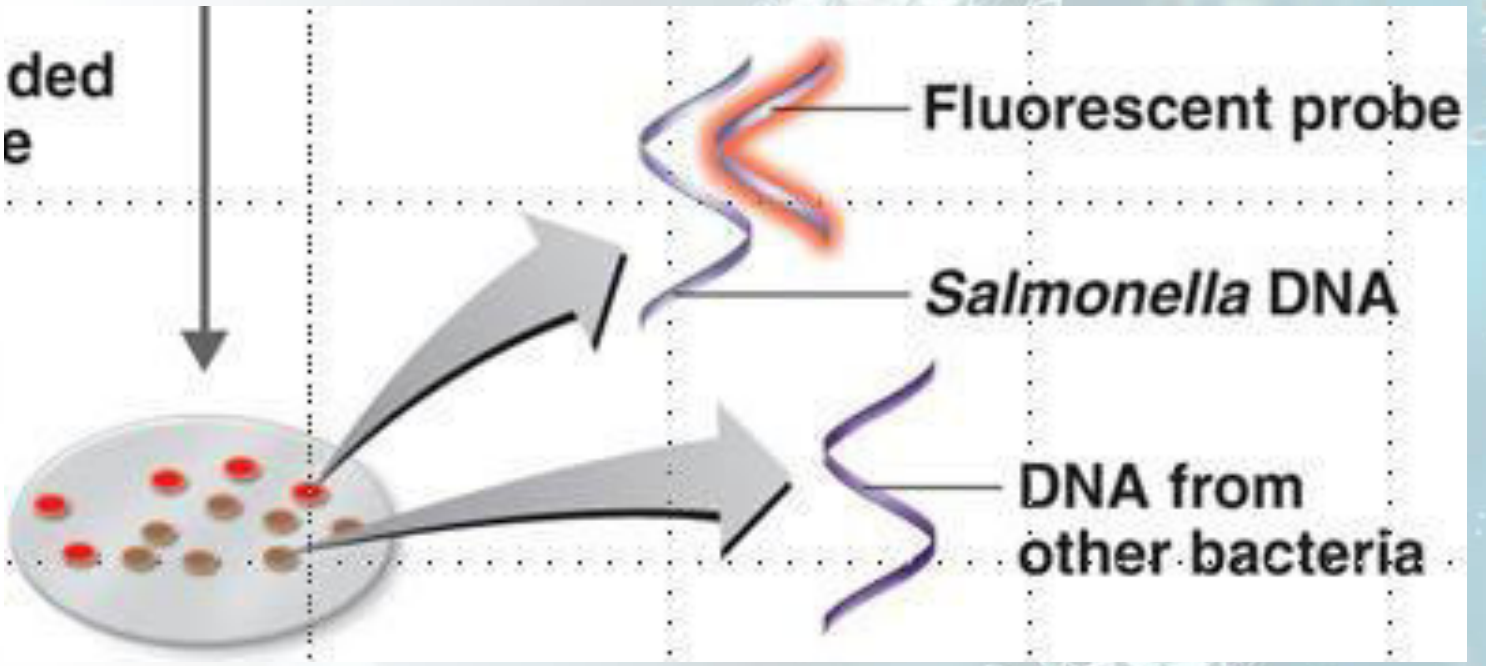
DNA probes



Bacterial Identification VI



Bacterial Identification VI



Bacterial Identification VI

DNA Chip



(a) A DNA chip can be manufactured to contain hundreds of thousands of synthetic single-stranded DNA sequences. Assume that each DNA sequence was unique to a different gene.

Bacterial Identification VI

Summary

- DNA probes can be used for diagnosing various diseases.

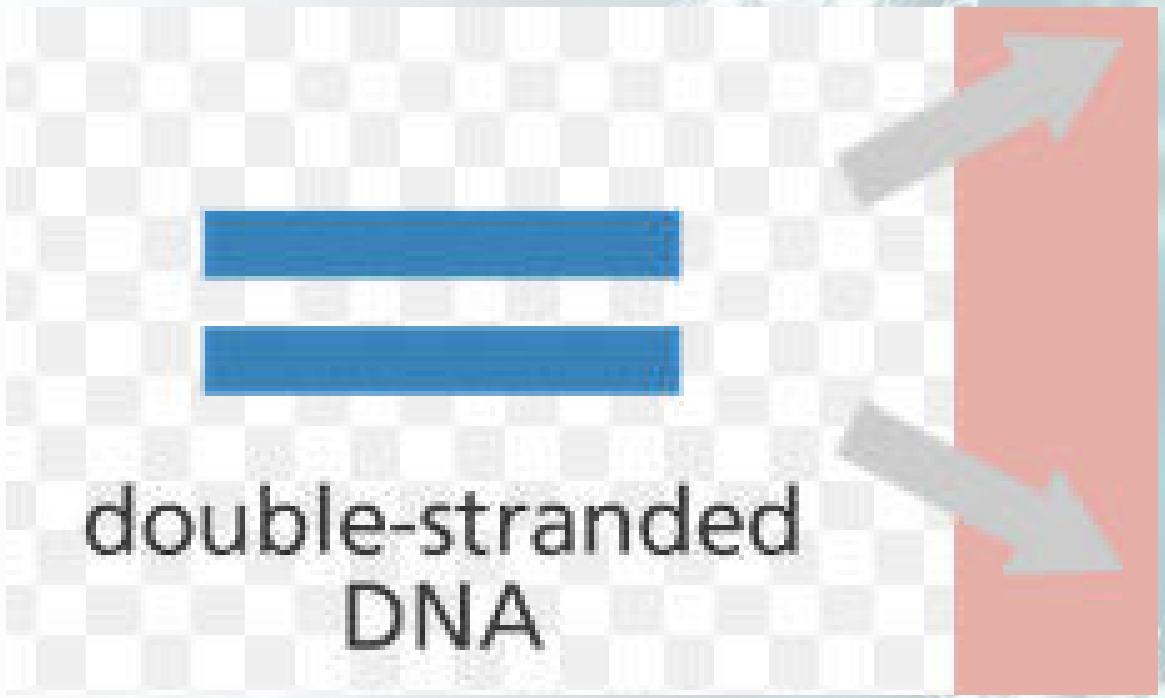
Bacterial Identification VII

Learning Objectives

- Nucleic acid base microbial detection methods
  - PCR

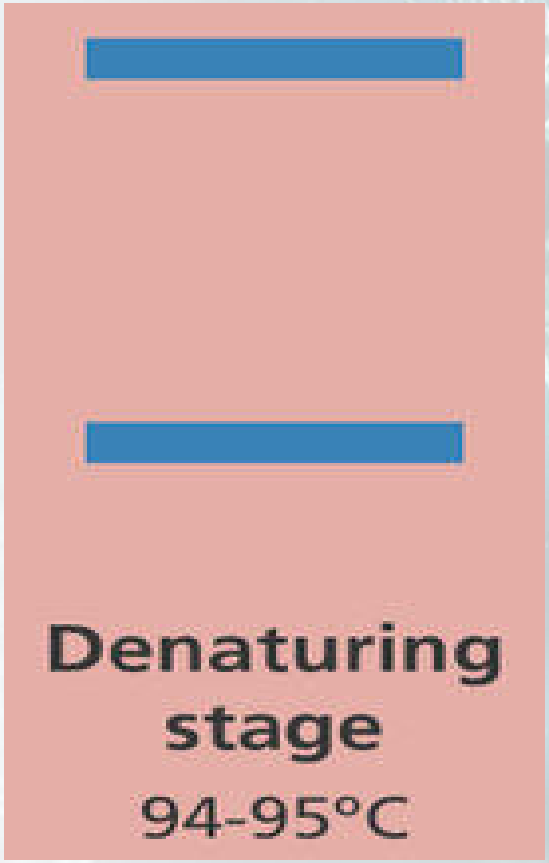
Bacterial Identification VII

PCR Basics



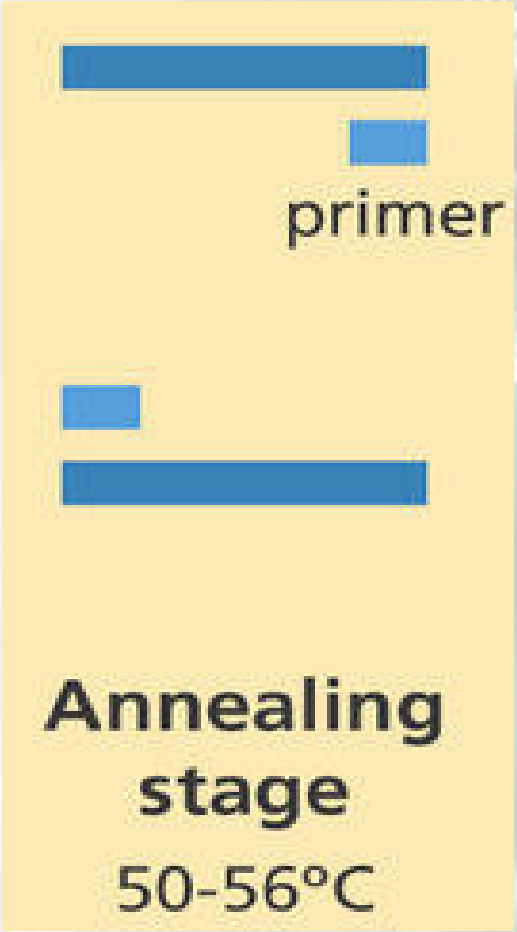
Bacterial Identification VII

PCR Basics



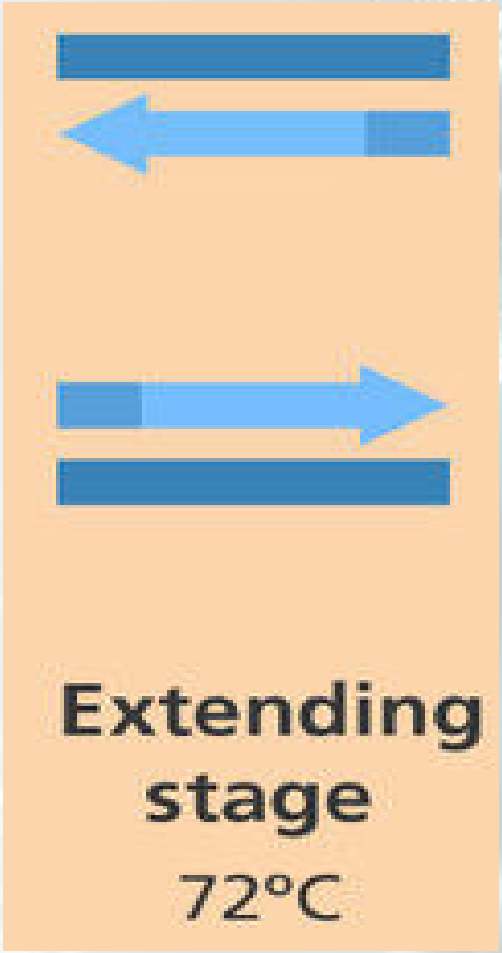
Bacterial Identification VII

PCR Basics

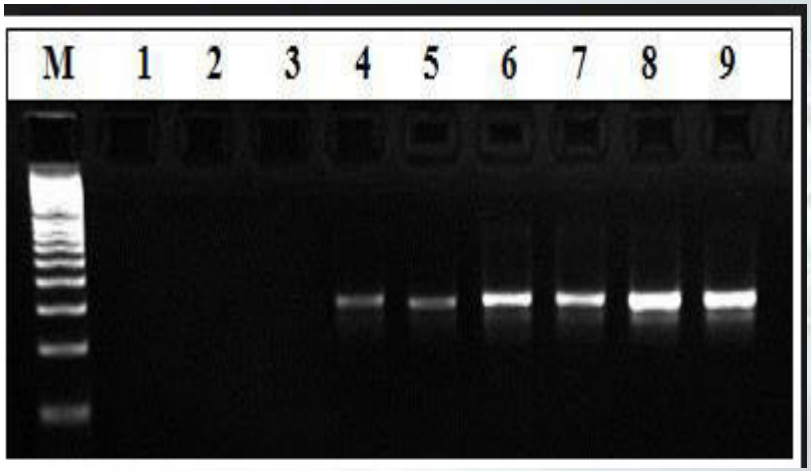


Bacterial Identification VII

PCR Basics



Bacterial Identification VII



PCR Basics

- After 30 cycles, enough target DNA accumulates which could be detected by electrophoresis.

Bacterial Identification VII

pcr_components.html

Bacterial Identification VII

Summary

- PCR provides a quicker way of identification of microbes.

Fungi I

Learning Objectives

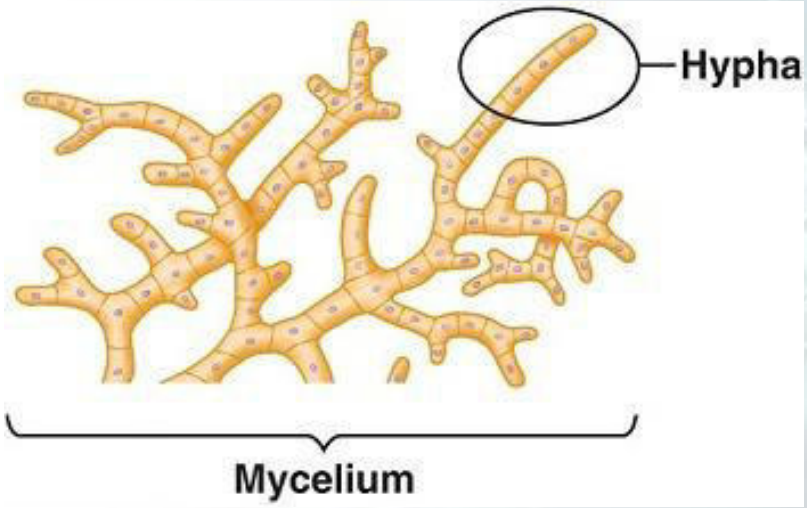
- What are fungi
- Nomenclature used to define fungi

Fungi I

General Features

- Kingdom: Fungi
- Chemoheterotrophs
- Acquire nutrients by absorption
- Asexual and Sexual reproduction

Fungi I



Definitions

- **Hyphae:** Thread like filaments of a fungus
- **Mycelium:** Collection of hyphae
- **Thallus:** Collection of mycelia

Fungi I

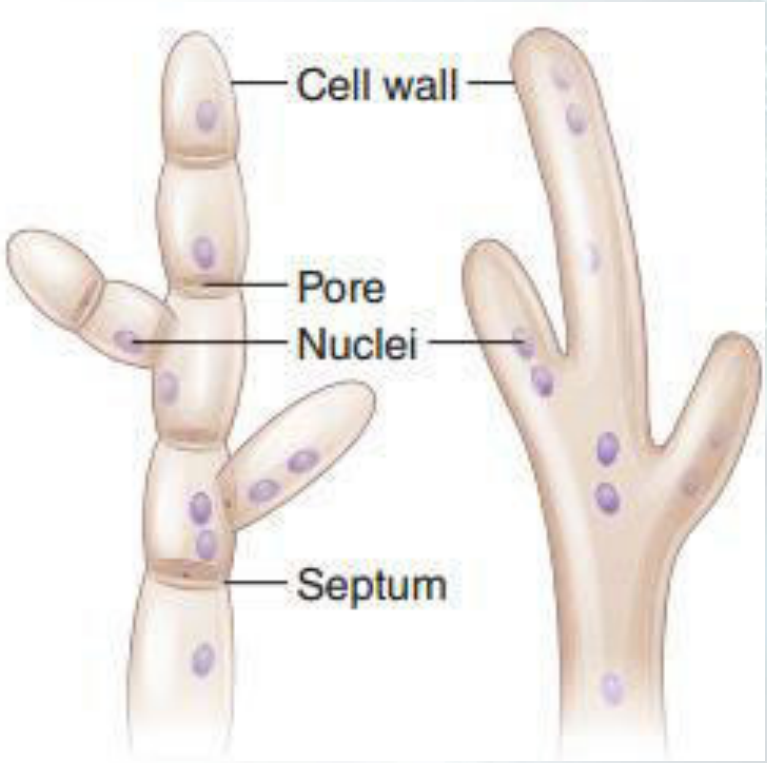




Definitions

- **Mold:** Filamentous fungi
- **Yeast:** unicellular fungus
- **Mycology:** Study of fungi

Fungi I

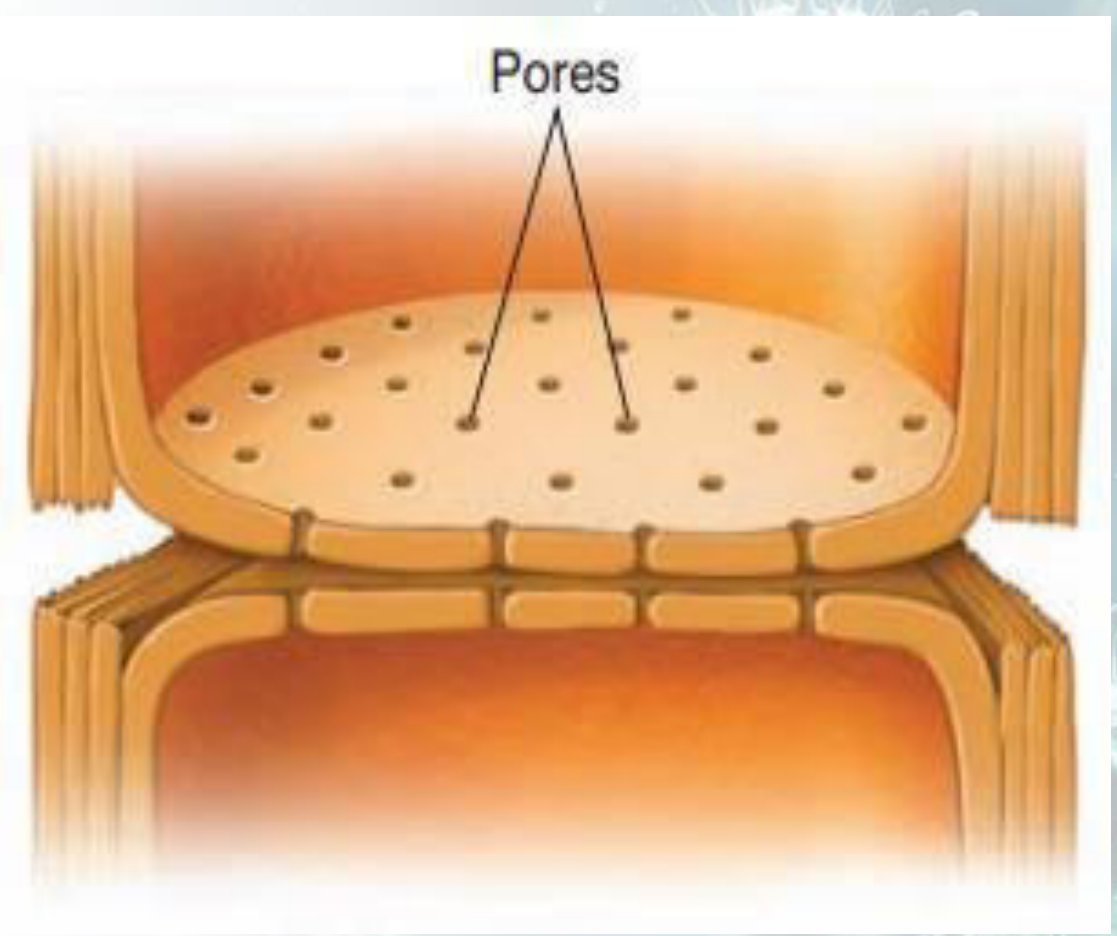


Types of Hyphae

- **Septate:** Cross-walls, divide hyphae into uninucleate cell-like units
- **Non-septate** or coenocytic:
  - No septum

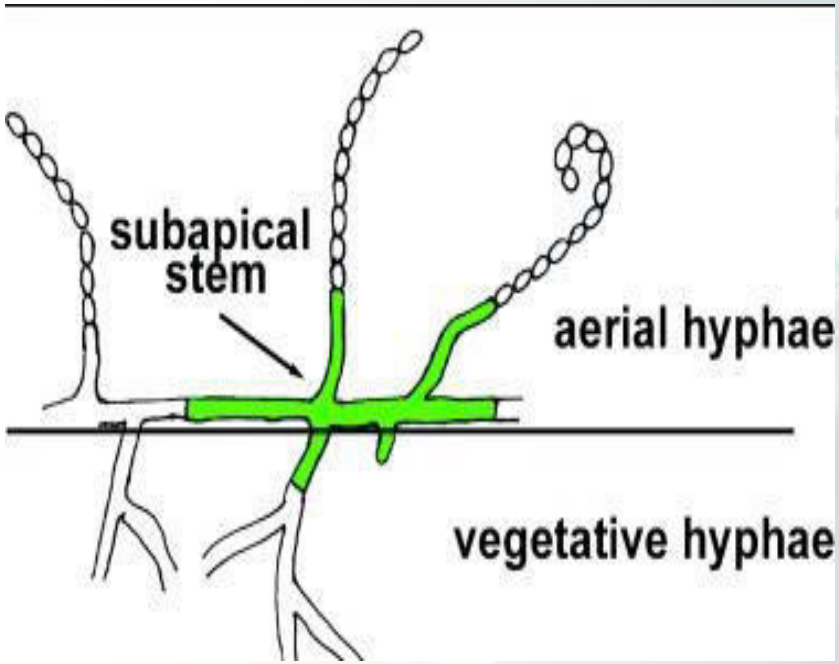
Fungi I

Septal Pores



Fungi I

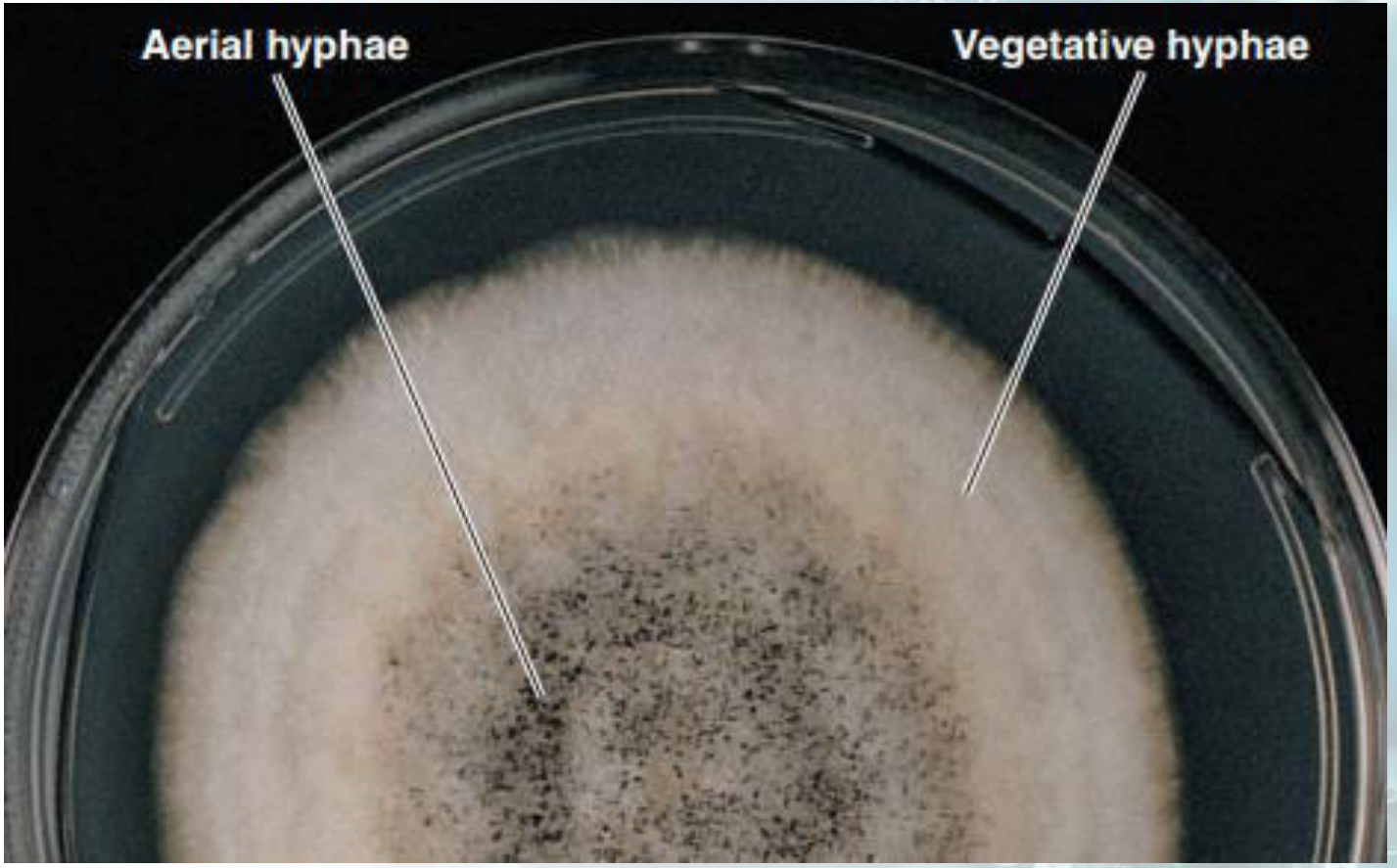
Types of Hyphae



- **Vegetative Hypha:** Absorbs nutrients
- **Aerial or reproductive hypha:** Bears reproductive spores

Fungi I

Types of Hyphae



Fungi I

Summary

- Fungi are eukaryotes
- Multicellular except yeasts
- Molds and Yeasts
- Septate and non-septate hyphae

Fungi II

Learning Objectives

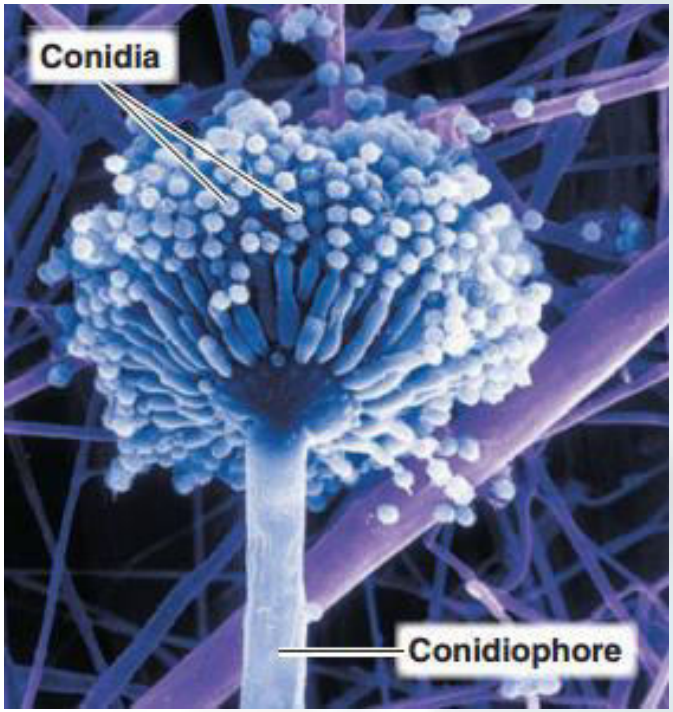
- Reproductive spores of filamentous fungi

Fungi II

Types of Spores

- **Asexual** spores formed by mitosis of hyphal cells
- **Conidiospores:** Unicellular or multicellular spore not enclosed by a sac
- **Sporangiospores:** Enclosed by a sac

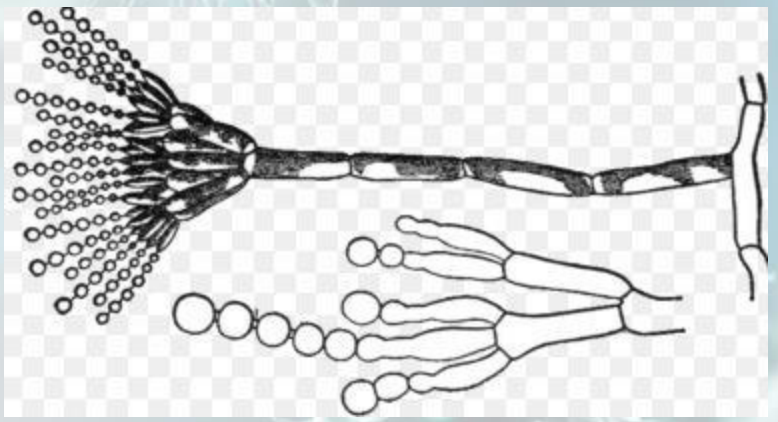
Fungi II



Conidiospores

- **Conidia:** Chain of spores at the end of a conidiphore

Aspergillus



Fungi II

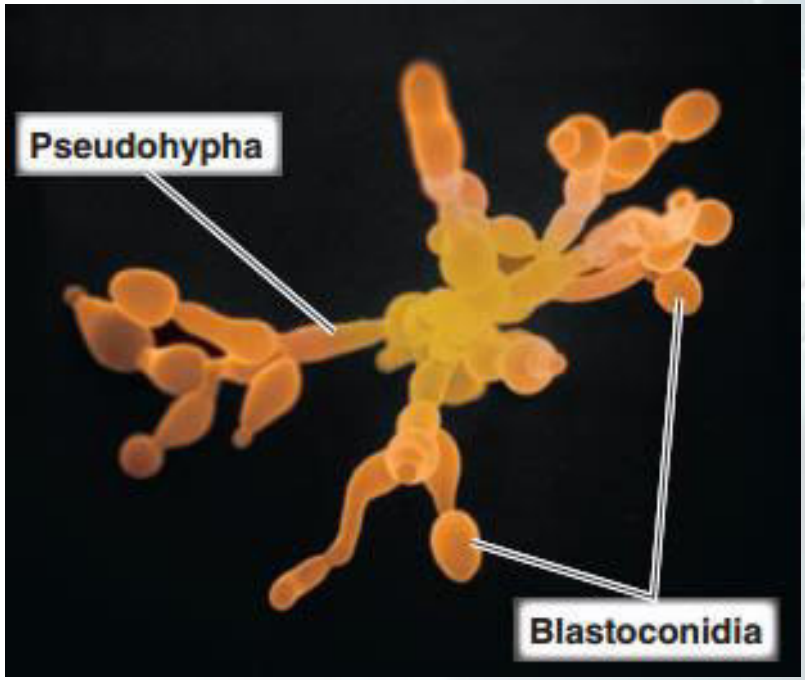


Conidiospores

- **Arthroconidia:** spores formed by fragmentation of septate hypha
  - Coccidioides immitus

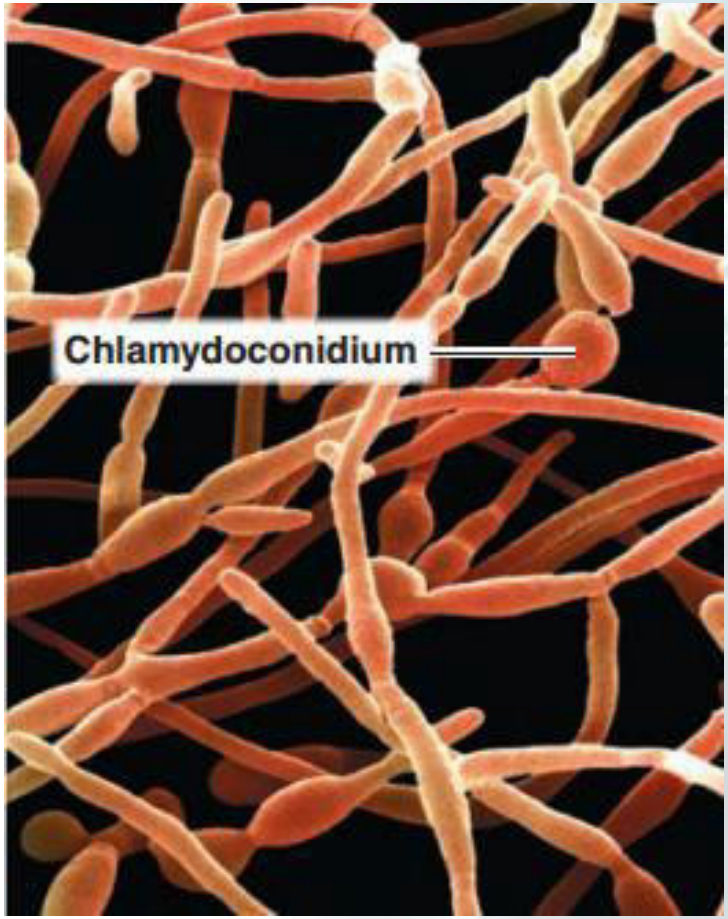
Fungi II

Conidiospores



- **Blastoconidia:** spores as budding of the parent cells
  - Candida albicans

Fungi II



Conidiospores

- **Chlamydioconidia:** spores formed by rounding and enlargement within a hyphal segment

  – Candida albicans

Fungi II

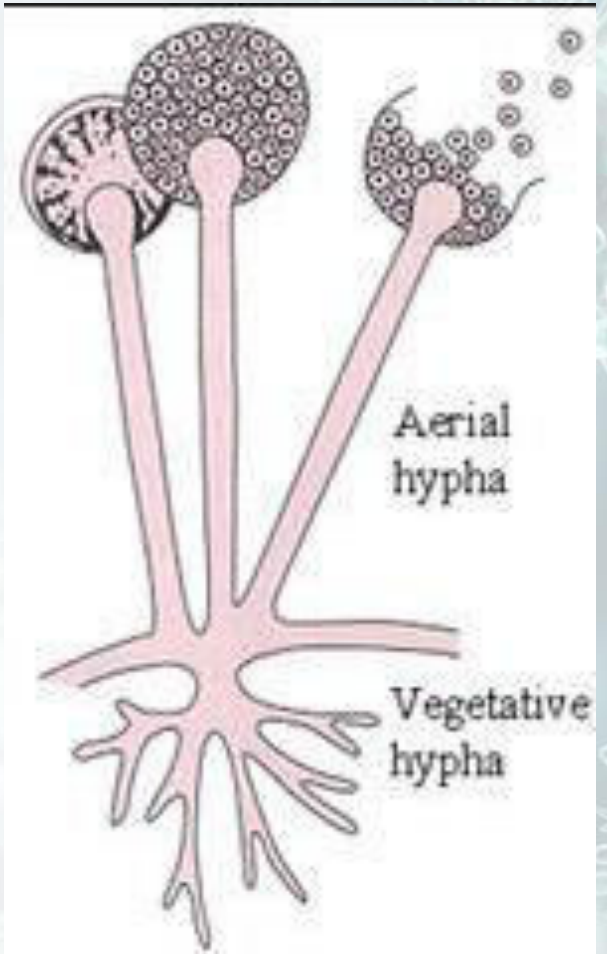


Sporangiospores

- Formed within a sporangium or sac at the end of an aerial hypha called sporangiophore
  - Rhizopus

Fungi II

Sporangiospores



Fungi II

Summary

- Fungi reproduce by sexual and asexual spores
- Conidiospores
- Sporangiospores

Fungi III

Learning Objectives

- Reproductive spores of filamentous fungi
  - Sexual spores

Fungi III

Sexual Spores

- **Plasmogamy:** A haploid nucleus of a donor cell (+) penetrates the cytoplasm of a recipient cell (-).
- **Karyogamy:** The (+) and (-) nuclei fuse to form a diploid zygote nucleus.

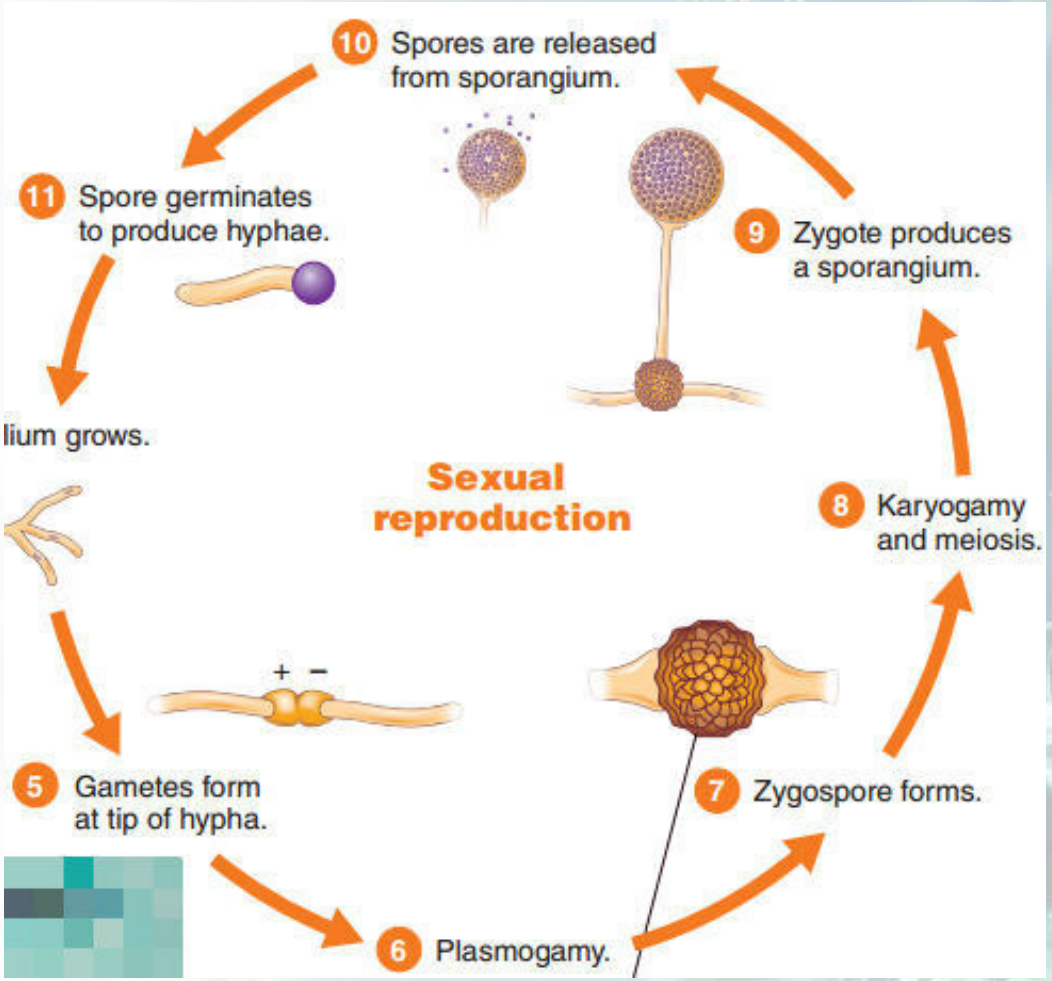
Fungi III

Sexual Spores

- **Meiosis:** The diploid nucleus gives rise to haploid nuclei (sexual spores)

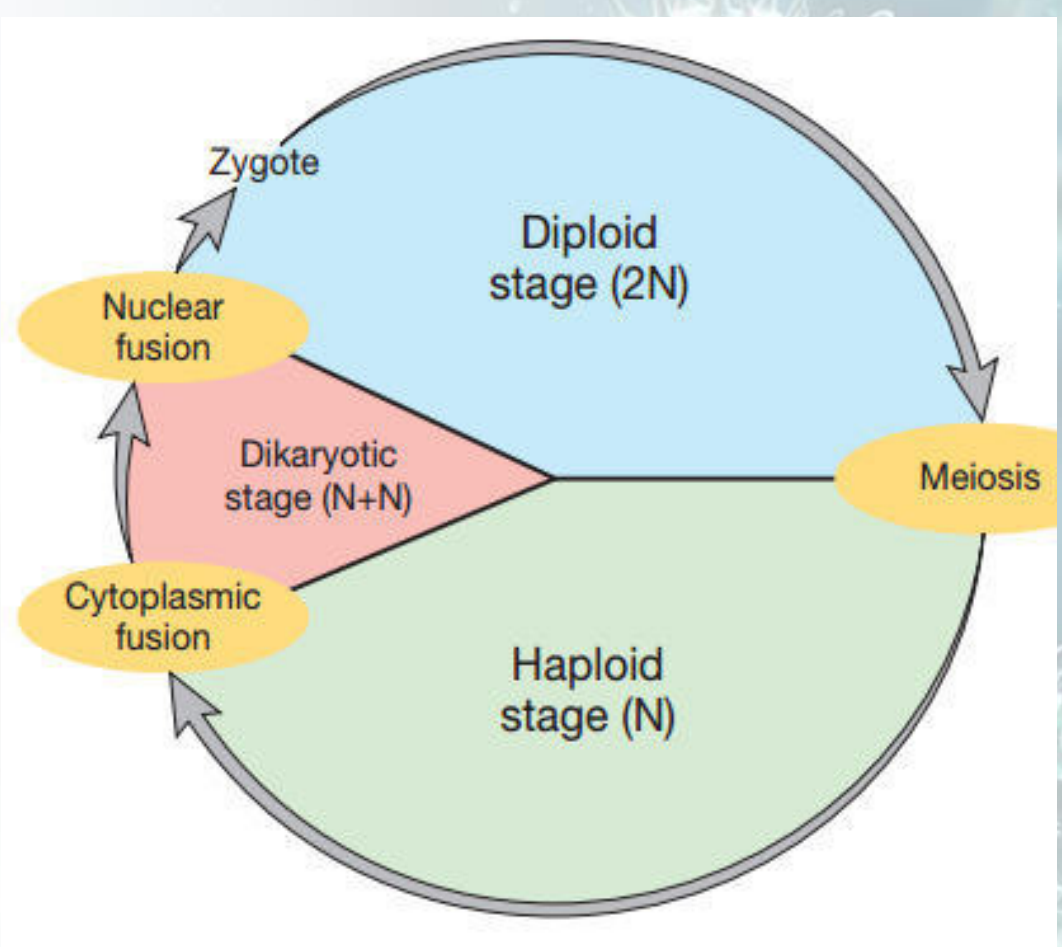
Fungi III

Sexual Spores



Fungi III

Sexual Spores



Fungi III

Summary

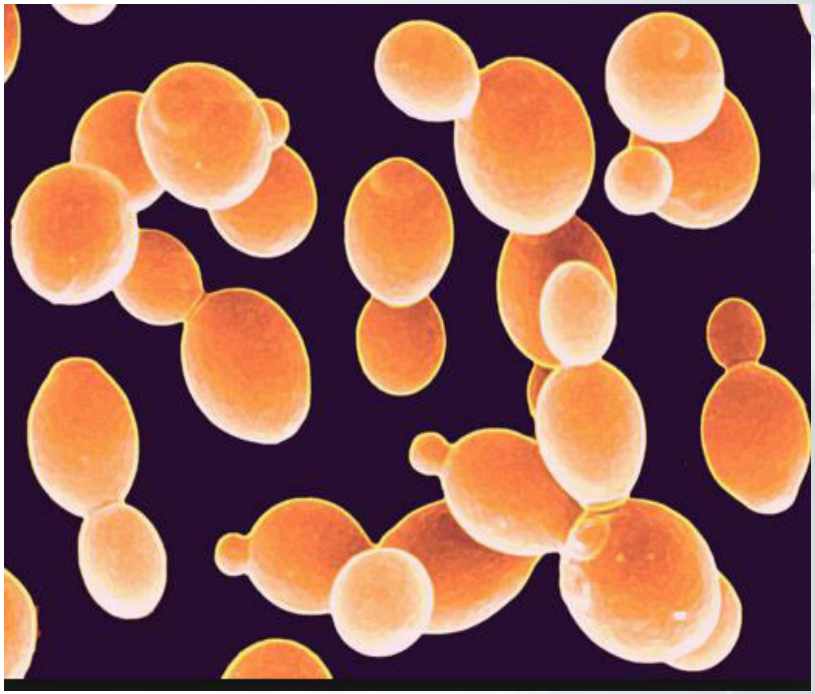
- Fungi reproduce by sexual and asexual spores
- Plasmogamy
- Karyogamy
- Meiosis

Fungi IV

Learning Objectives

- What is a yeast
- Dimorphic fungi

Fungi IV



Sexual Spores

- Non-filamentous fungi
- Unicellular
- Reproduce by budding (asexually) or sexually
- Some are dimorphic
  - Yeast: at 37°C
  - Molds: Room temp

Fungi IV

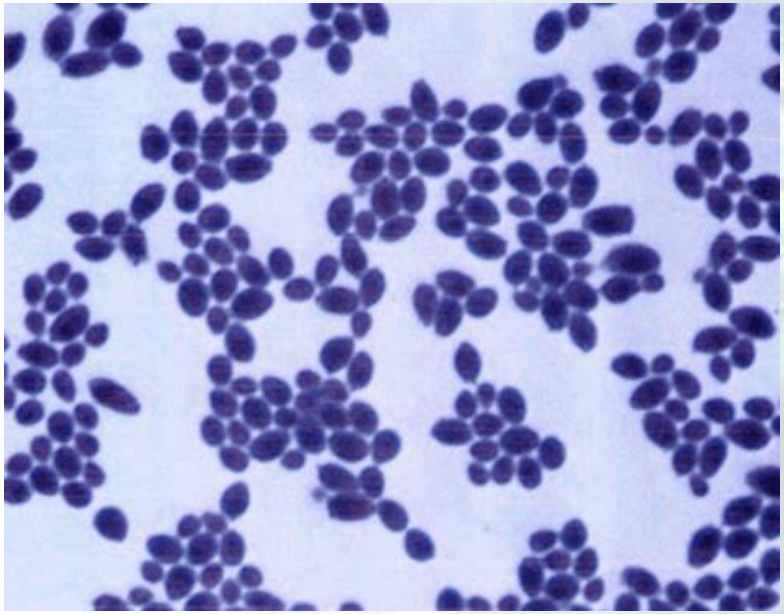
Yeasts



- Larger than bacteria
- No flagella or cilia
- Facultative anaerobes

Fungi IV

Yeasts



- Larger than bacteria
- No flagella or cilia
- Facultative anaerobes
- Gram positive

Fungi IV

Summary

- Yeasts are unicellular
- Some are dimorphic
- Both sexual and asexual reproduction
- Bigger than bacteria

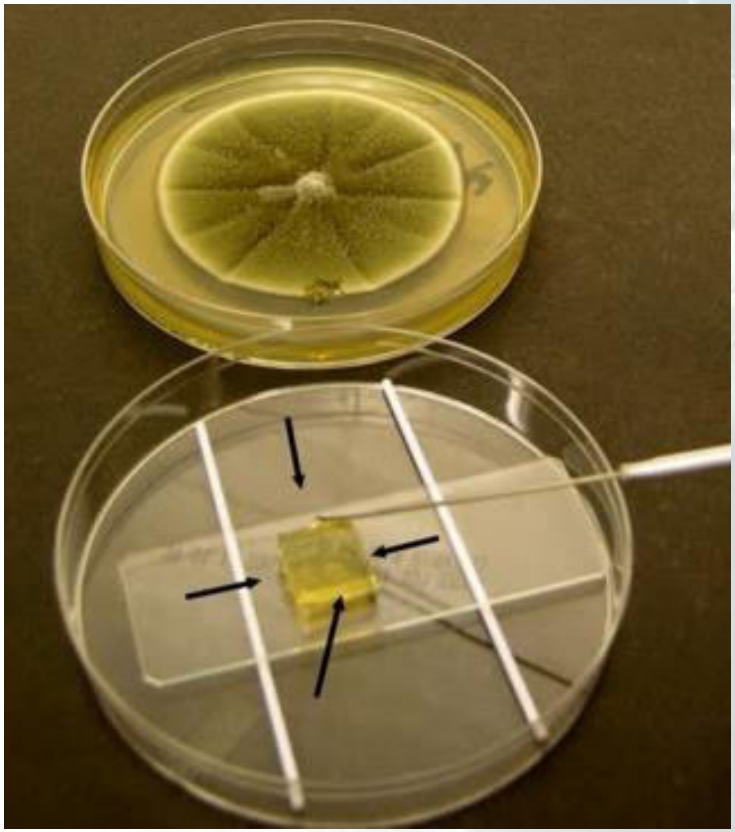
Fungi V

Learning Objectives

- How to study fungi

Fungi V

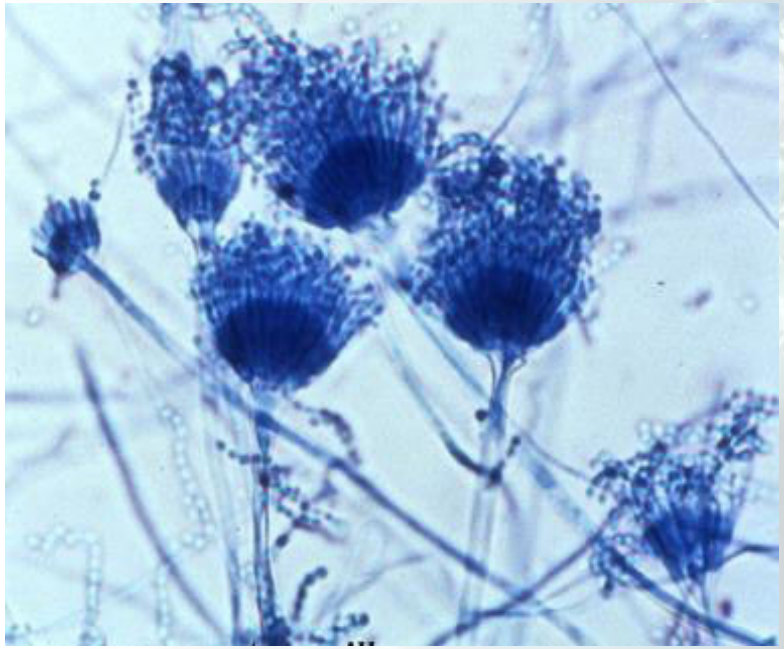
Filamentous fungi



- Identified by morphology, not by biochemical testing
- Slide culture is made using Sabouraud's agar
- Spores are inoculated
- Slides stained with lactophenol cotton blue

Fungi V

Aspergillus





Fungi V

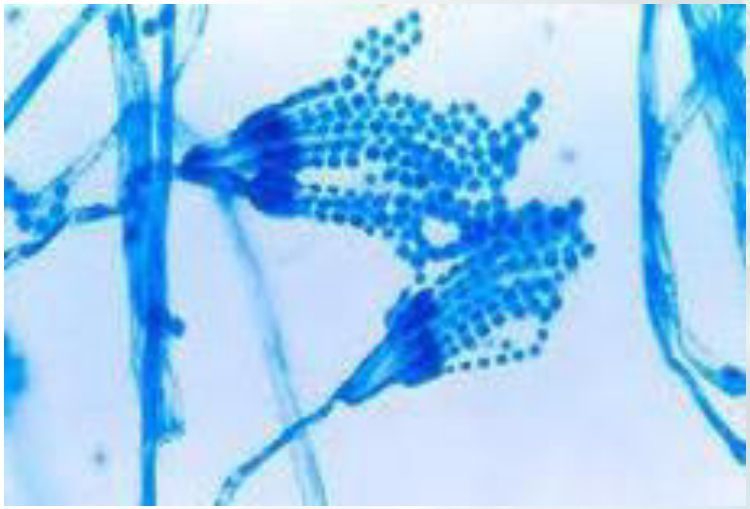
Trichophyton





Fungi V

Aspergillus





Fungi V

Yeast

- Yeast can ferment and can produce various products
- Biochemical testing is used for yeast identification

Fungi V

How to Culture Fungi?

Fungi V

Summary

- Morphology is used to identify filamentous fungi
- Biochemical testing is used for yeast identification

Fungi VI

Learning Objectives

- Economic benefits of fungi
- Diseases by fungi

Fungi VI

Uses of Yeast

- Saccharomyces for alcoholic beverages
  - Anaerobic conditions for alcohol
  - Aerobic conditions
    - Bread making

Fungi VI

Yeast uses

- Biotechnology: Hepatitis B vaccine
- Research

Fungi VI

Fungal uses

- **Citric acid:** Aspergillus niger
- **Cellulase:** Trichoderma (plant cell digestion to clear fruit juices)
- **Taxol:** Taxomyces (anticancer)

Fungi VI

Fungal uses

- **Biological pest control:** Entomophaga (kills gypsy moth)
- **Penicillin:** Penicillium

Fungi VI

Diseases by Fungi

- **Mycosis** is a general term used for fungal diseases

- **Systemic mycosis:** Deep within the body

  – histoplasmosis and coccidioidomycosis

Fungi VI

Diseases by Fungi

- **Subcutaneous Mycosis:** Beneath the skin
  - Sporotrichosis in gardeners and farmers

Fungi VI

Diseases by Fungi

- **Cutaneous Mycosis:**
  - Trichophytons use keratinase to degrade keratin in nails, hair and skin
  - Ringworm infections

Fungi VI

Summary

- Fungi are useful mostly
- A few cause infections

Lichen I

Learning Objectives

- Lichens
- Uses of lichens

Lichen I



Lichens

- Combination of green alga or a cyanobacterium (phycobiont) and a fungus (Mycobiont)
- Symbiotic relationship
- Partners do not grow alone
- Grow where no other microbes can grow

Lichen I

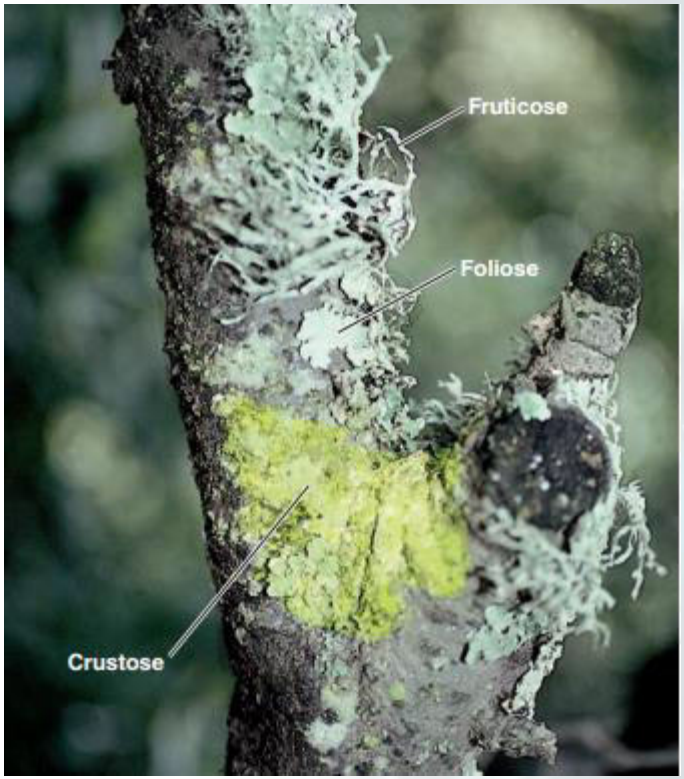
Lichens

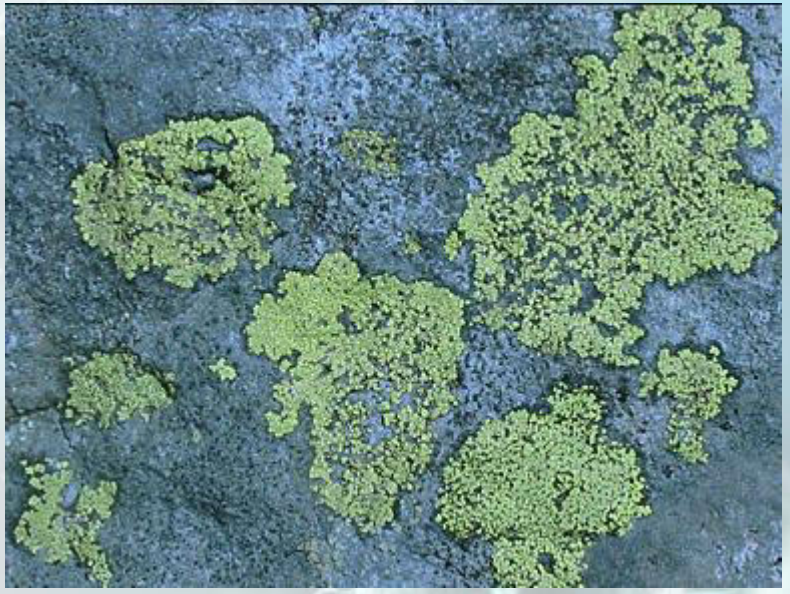
- Grow on bare rocks, tree trunks and house roofs and bare soils
- Typically grow extremely slowly
  - 1 mm to 3 cm per year
- Classified on the basis of the fungus

Lichen I

Types of Lichens

- Crustose: forms a thin crust





Lichen I

Types of Lichens

- **Foliose: Leaf-like**





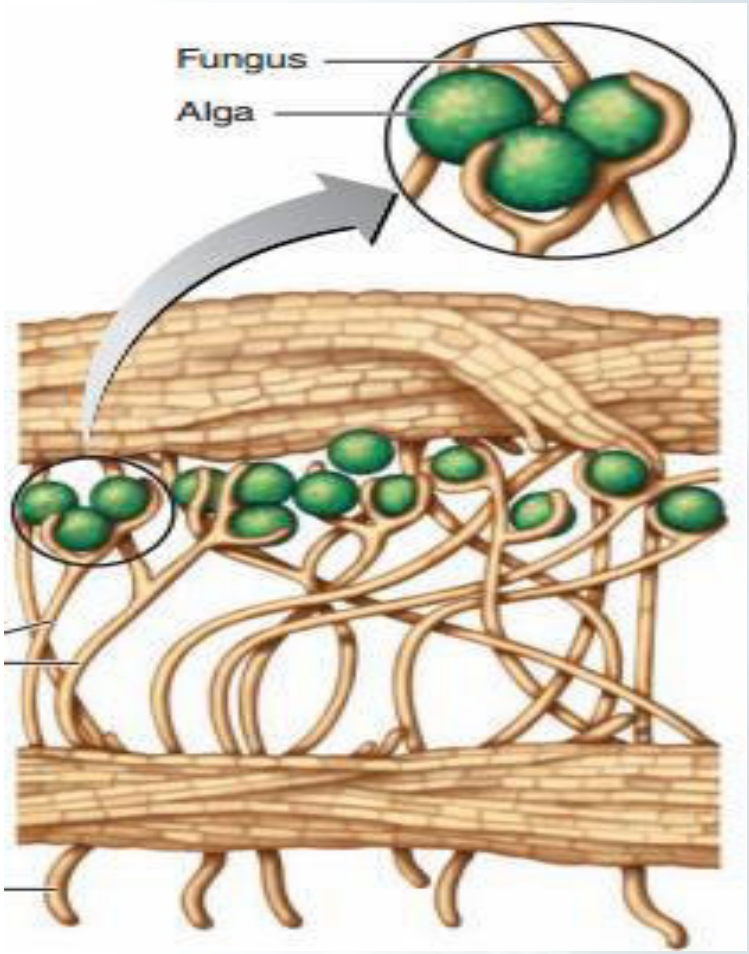
Lichen I



Types of Lichens

- **Fruticose: Finger-like projections**

Lichen I



Lichen Thallus

- **Medulla:** Algal layer and fungal hyphae
- **Cortex:** Could be two, upper layer and lower layer
- **Rhizine:** Roots or Holdfasts

Lichen I



Lichen uses

- Dyes for clothes
- Usnic acid as antiseptic in China
- Erythrolitmin: for Litmus paper
- Can cause allergic contact dermatitis

Lichen I

Summary

- Fungus and alga or cyanobacterium
- Cannot grow alone
- Have various uses
- Some can cause contact dermatitis

Algae I

Learning Objectives

- Algae
- Reproduction
- Uses of algae

Algae I

Algae

- Simple eukaryotic cells
- Some are unicellular, others are multicellular (thallus)
- Lack tissues (roots, stem and leaves) seen in plants

Algae I

Algae

- Absorb nutrients from water through their surfaces
- Mostly photoautotrophs, a few are chemoheterotrophs

Algae I

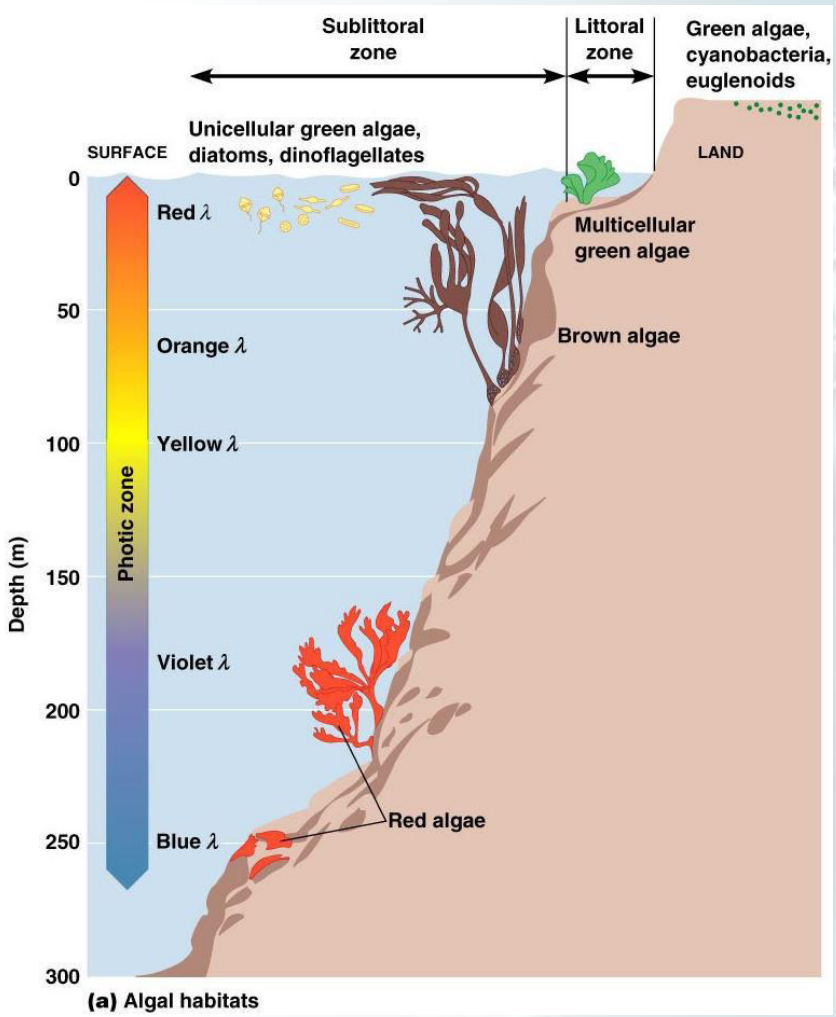


Seaweed

Algae

- Mostly aquatic (water is necessary for physical support, reproduction and diffusion of nutrients)
- Cell covering thallus can carry photosynthesis
- Some algae float by gas-filled bladders

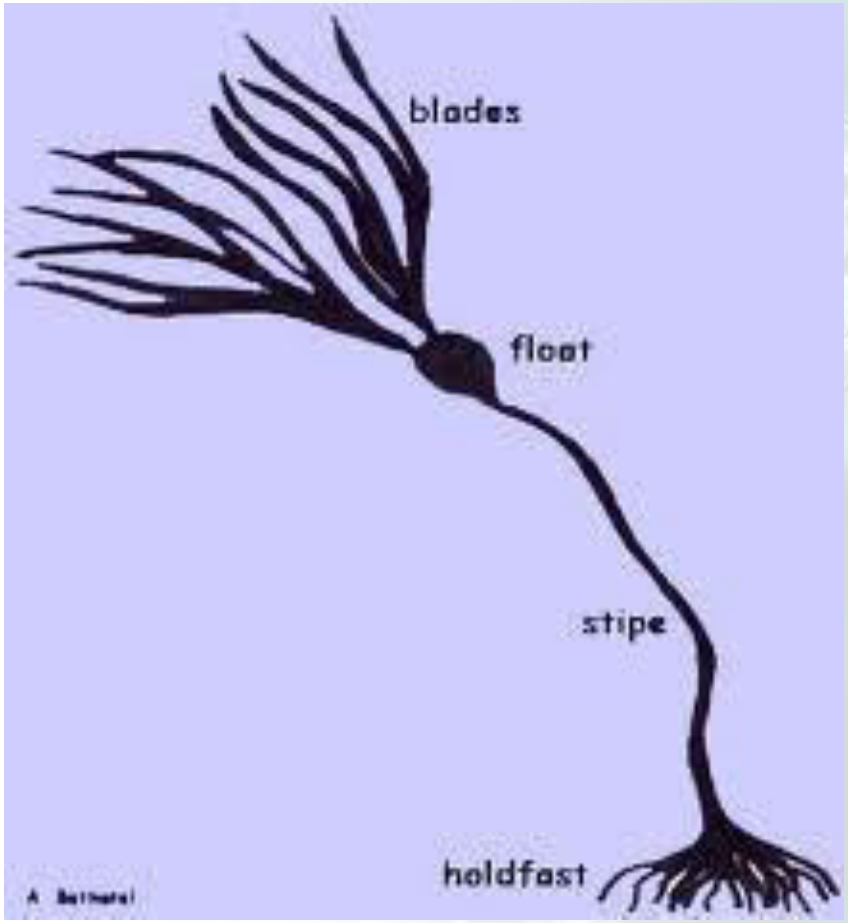
Algae I



Algae

- Their location in oceans depends on the availability of nutrients, wavelength of light and the surface on which to grow.

Algae I

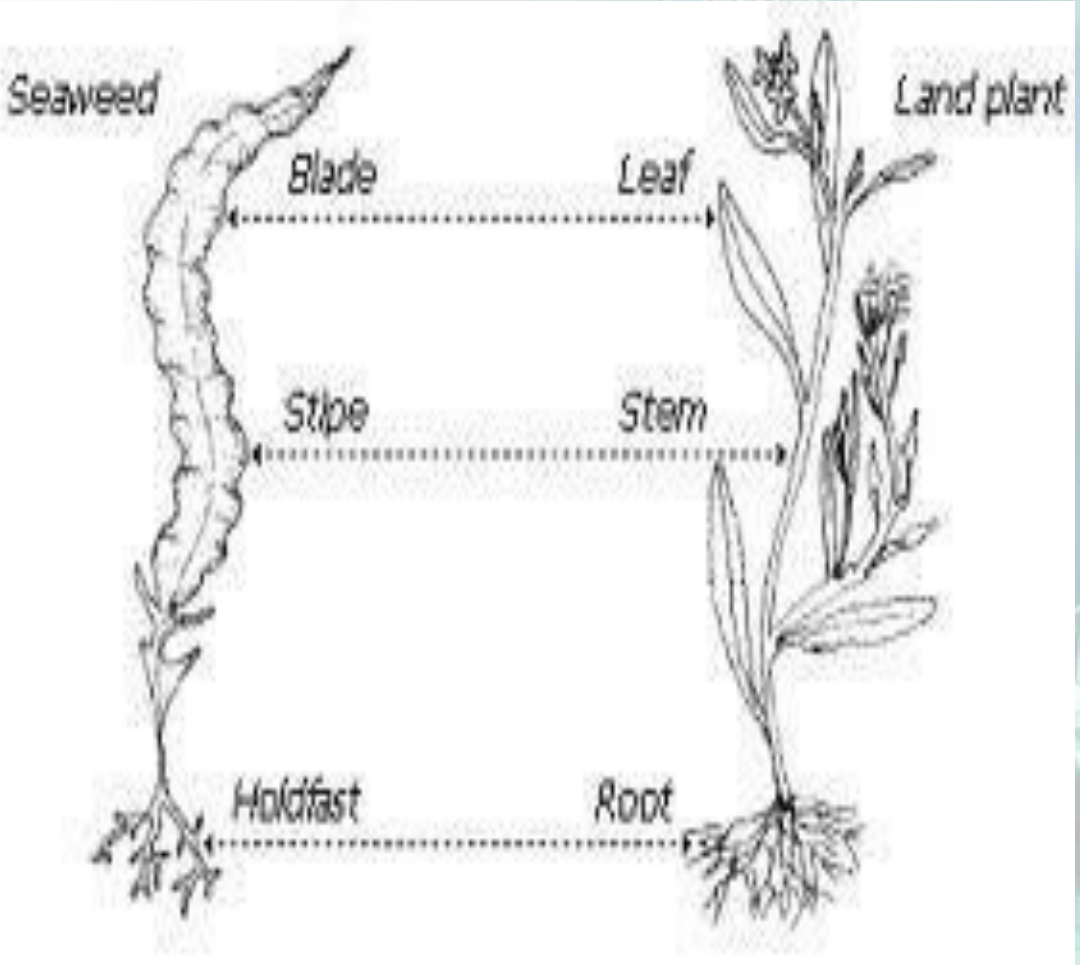


Algae

- Body of multicellular alga is called a thallus which consists of branched holdfasts (anchor alga to rock) stemlike hollow stipes and leaflike blades.

Algae I

Algae



Algae I

Reproduction

- All algae can reproduce asexually
  - Multicellular algae: a fragment can form a new thallus
  - Unicellular algae: mitosis followed by cytokinesis

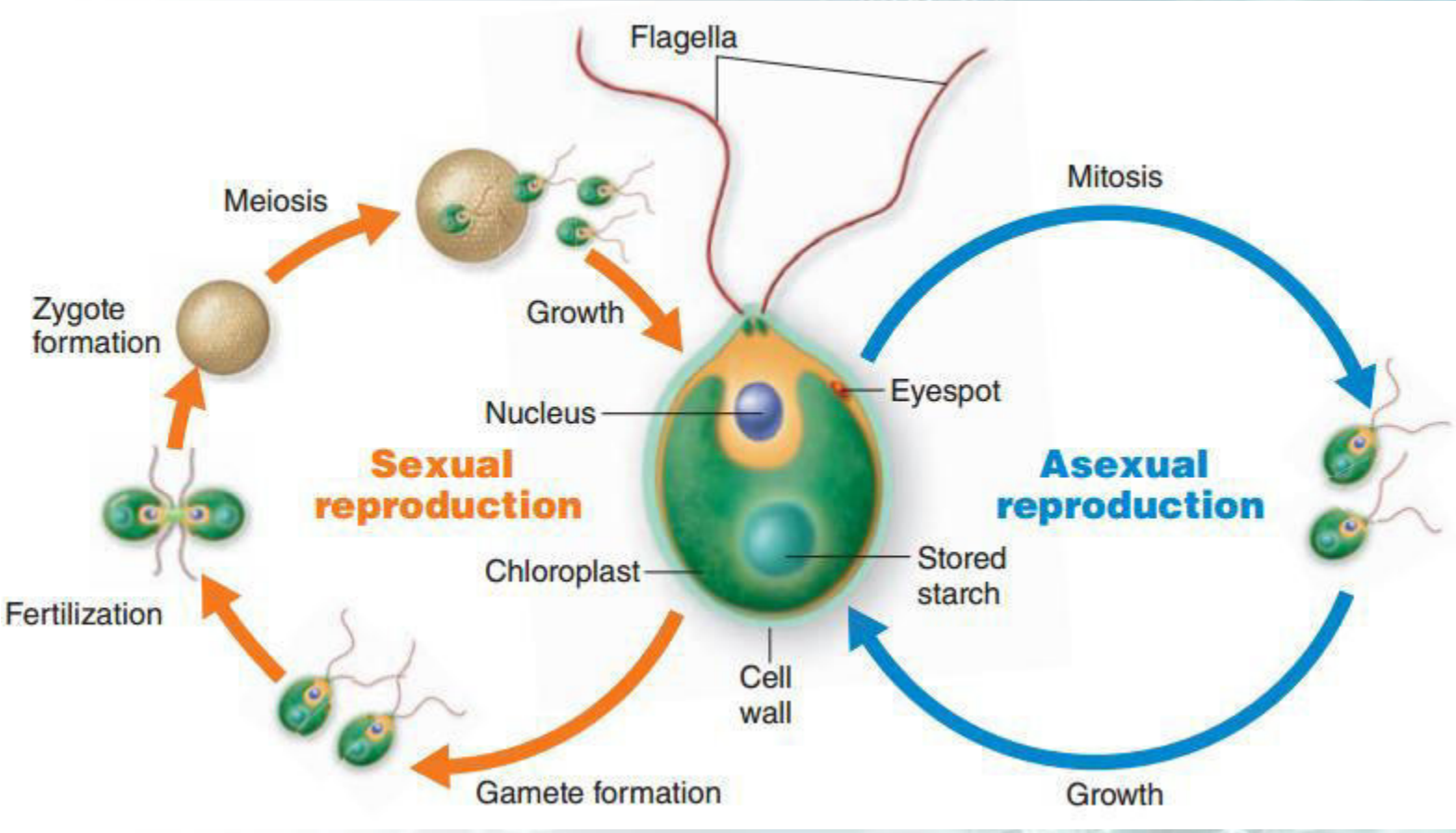
Algae I

Reproduction

- Sexual reproduction: Alternation of generations of haploid and diploid individuals

Algae I

Reproduction



Algae I

Summary

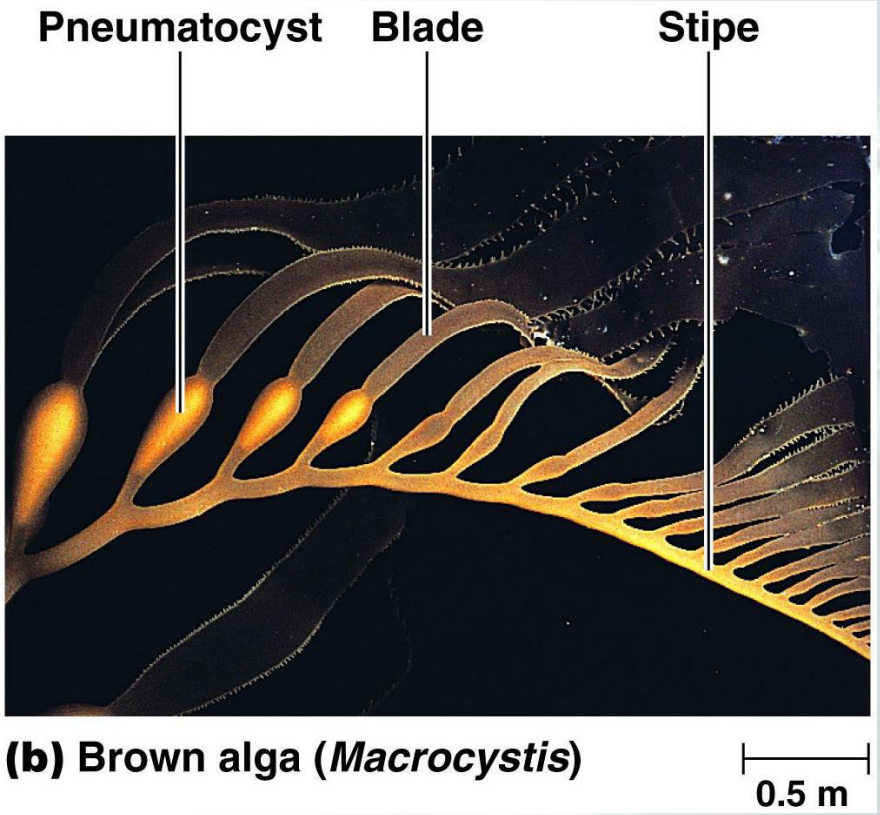
- Eukaryotes
- Unicellular and multicellular
- Lack tissues
- Mostly aquatic
- Reproduce sexually and asexually

Algae II

Learning Objectives

- Algae
- Types
- Uses of algae

Algae II



(b) Brown alga (*Macrocystis*)

Phaeophyta (Kelp)

- Brownish in color
- Cellulose and alginic acid cell walls
- Multicellular
- Chlorophyll *a* and *c*, xanthophylls
- Store carbohydrates
- Harvested for algin

Algae II

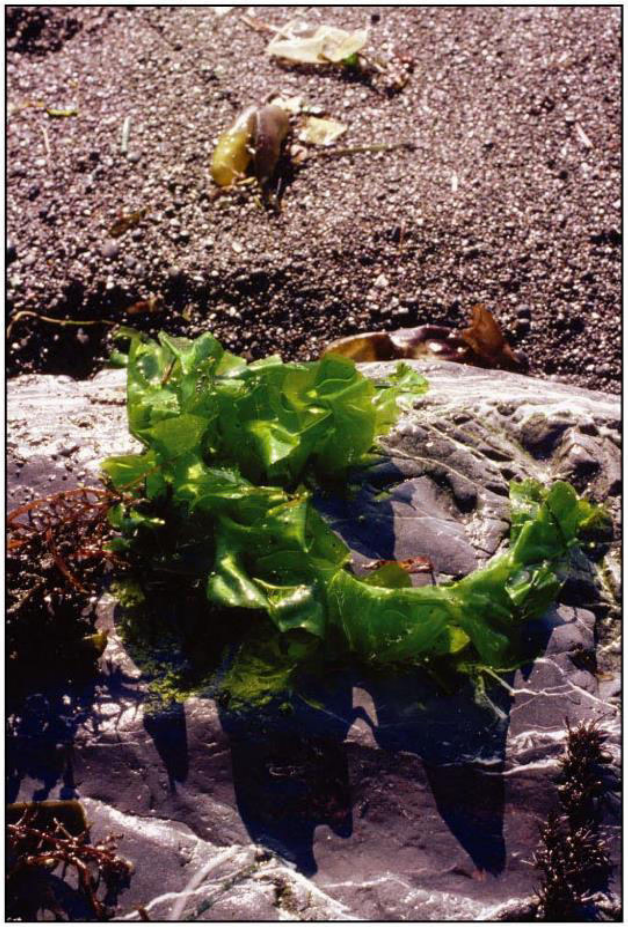


(c) Red alga (*Microcladia*)

Rhodophyta

- Reddish in color
- Cellulose cell walls
- Most are multicellular
- Chlorophyll *a* and *d*, phycobiliproteins
- Store glucose polymer
- Harvested for agar and carrageenan

Algae II



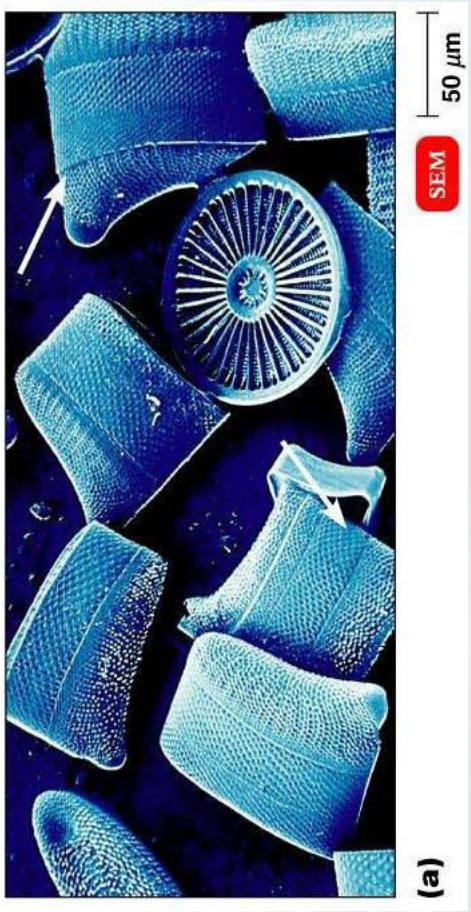
(a) Multicellular green alga (*Ulva*)

10 cm

Chlorophyta

- Green algae
- Cellulose cell walls
- Unicellular or multicellular
- Chlorophyll *a* and *b*
- Store glucose polymer
- Believed to have given rise to terrestrial plants

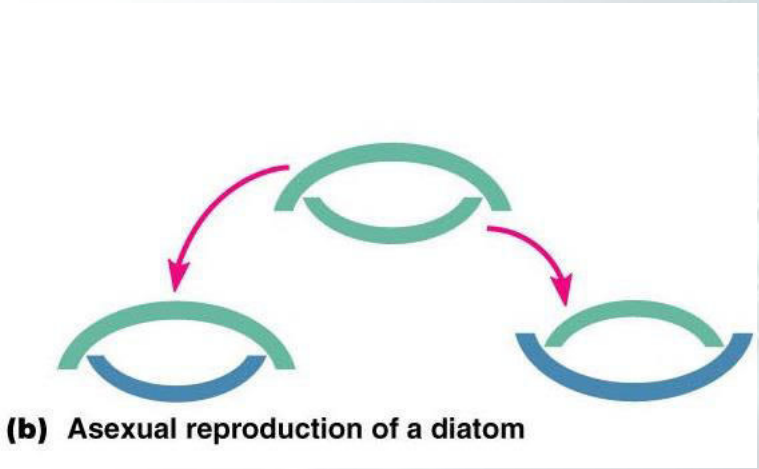
Algae II



Bacillariophyta

- Brownish in color
- Pectin and silica cell walls
- Unicellular
- Chlorophyll *a* and *c*, carotene, xanthophylls

Algae II

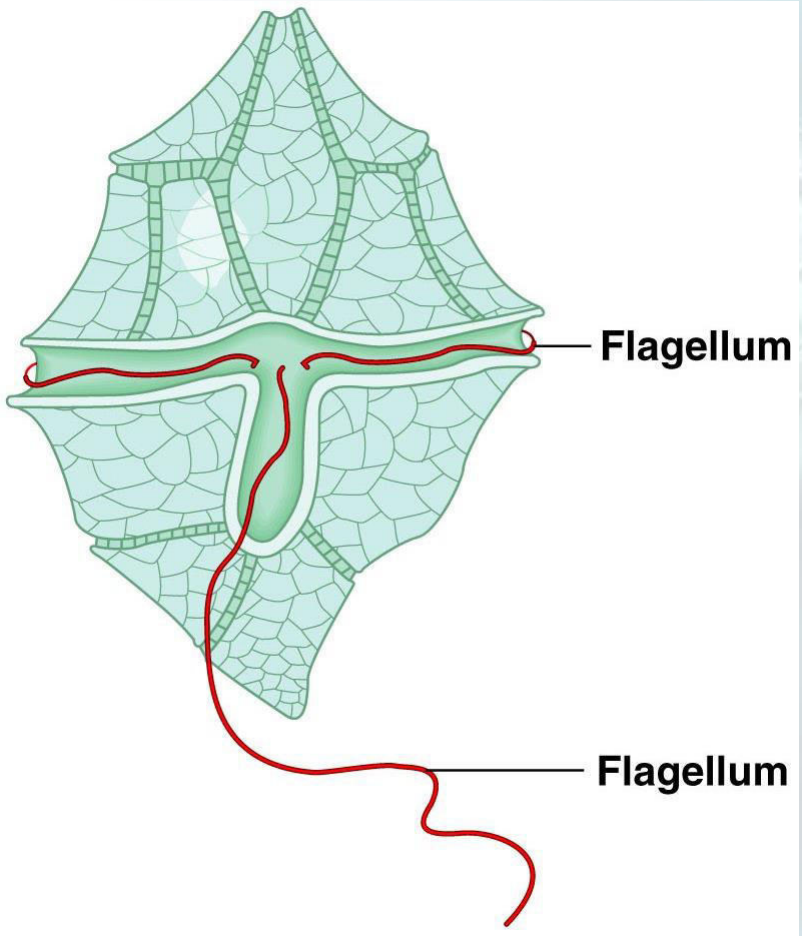


(b) Asexual reproduction of a diatom

Bacillariophyta

- Store oil
- Fossilized diatoms formed oil
- Produce domoic acid

Algae II



Dinoflagellata

- Brownish in color
- Cellulose in plasma membrane
- Unicellular
- Chlorophyll *a* and *c*, carotene, xanthins
- Store starch

Algae II

Dinoflagellata

- Some are symbionts in marine animals
- Neurotoxins cause paralytic shellfish poisoning

Algae II



Oomycota

- Colorless, white
- Cellulose cell walls
- Multicellular
- Chemoheterotrophic
- Produce zoospores

Algae II



Oomycota

- Decomposers and plant parasites
  - *Phytophthora infestans:* responsible for Irish potato blight

Algae II



Oomycota

- Decomposers and plant parasites
  - *P. cinnamomi* infects Eucalyptus.
  - *P. ramorum* causes sudden oak death.

Algae II

Role of Algae

- Important part of aquatic food chain
- Fix CO_2 into organic compound
- 80% earth O_2 produced by planktonic algae

Algae II

Role of Algae

- Petroleum and natural gas are the residues (hydrocarbons) of diatoms and other planktonic organisms.
- Unicellular algae are symbiont in animals.

Algae II

Summary

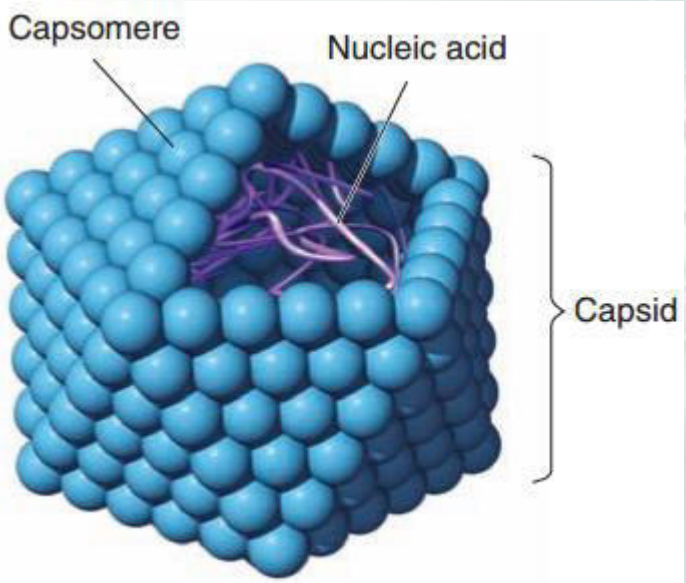
- Various types of algae
- Have variety of chlorophylls
- Have various uses
- Some cause plant and fish diseases

Virus I

Learning Objectives

- Definitions
- Capsid architecture
- Envelope

Virus I



Definitions

- **Virus: infectious viral particle, composed of nucleic acid, surrounded by a protein coat called capsid.**

Virus I

Virion

- Viruses have an extracellular and an intracellular state.
- Outside of a cell, in the extracellular state, a virus is called a virion.

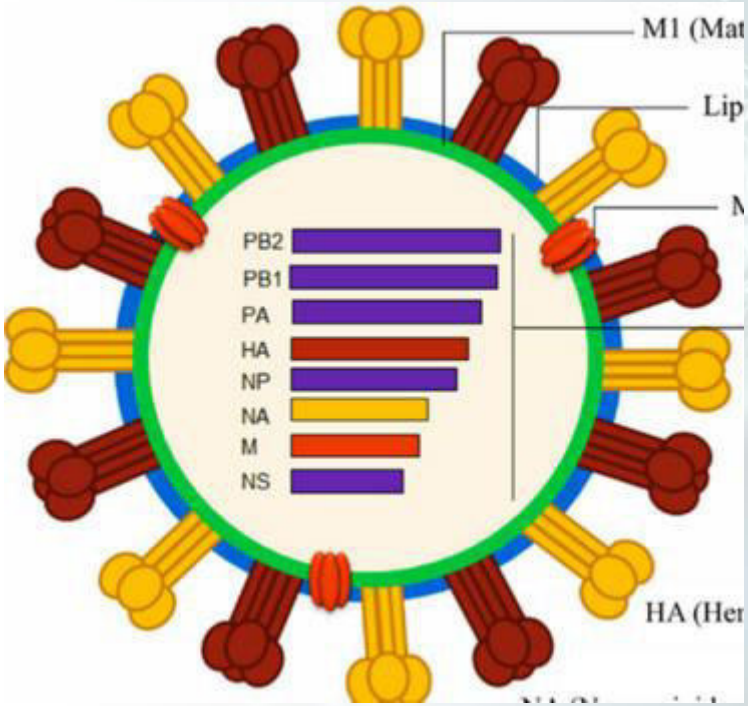
Virus I



Viroid

- Infectious circular RNA of plants
- Formed of only RNA
- Protein coat is absent
- Smaller in size than a typical virus
- Infects plants only

Virus I



Virus Structure

- Nucleic Acid: Either DNA or RNA (never both)
  - Single or double stranded
  - Circular or linear
  - One segment or multiple segments

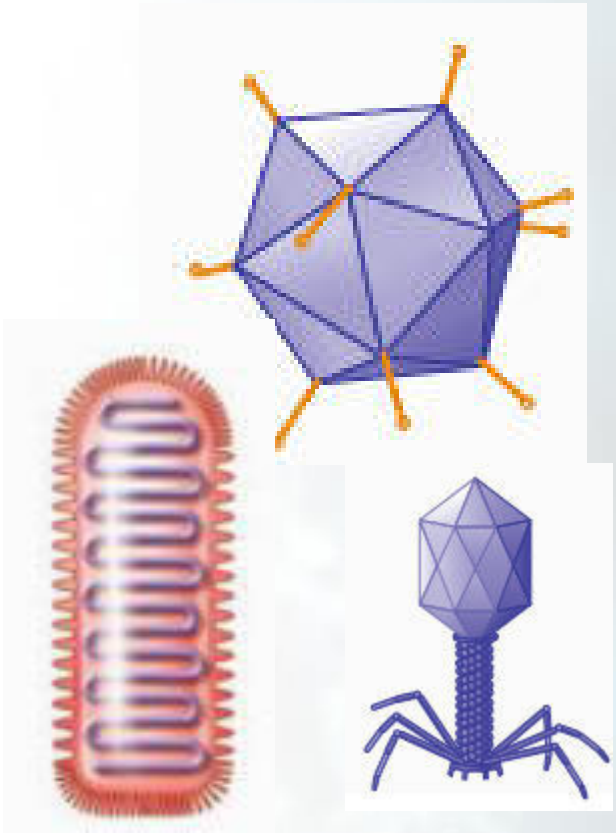
Virus I



Virus Structure

- Capsid: made up of protein subunits, capsomeres.
  - Capsomeres: subunit proteins can be single type or multiple types

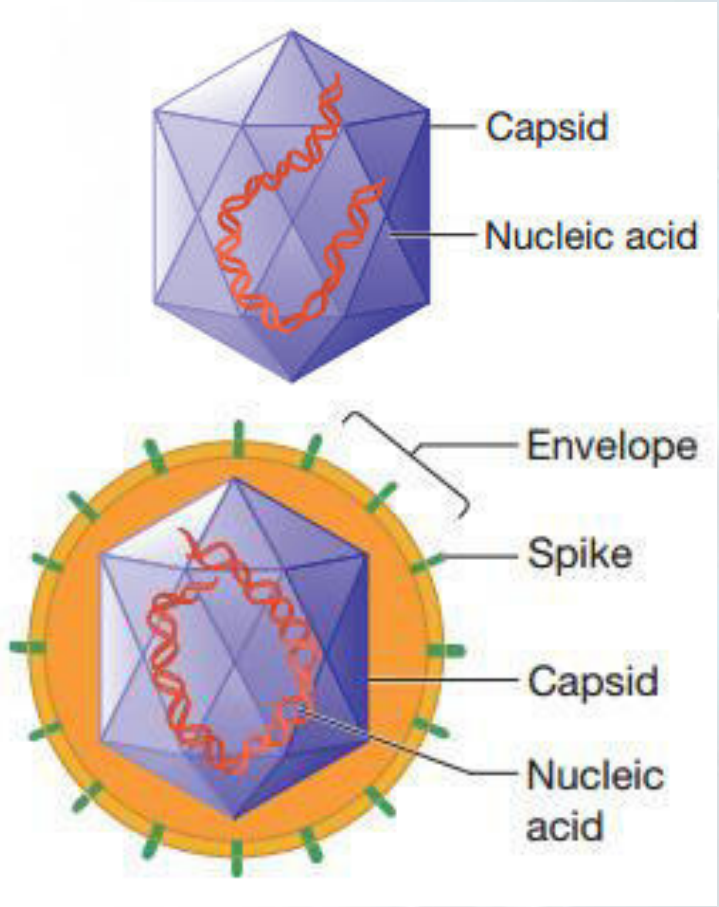
Virus I



Virus Structure

- Capsid: made up of protein subunits, capsomeres.
  - Arrangement of capsomeres is characteristic of a particular type of virus

Virus I



Virus Envelope

- In some viruses, the capsid is covered by an envelope (lipid, protein and carbs)

Virus I

Virus Envelope

- May or may not have spikes (carbohydrate-protein complexes) projecting from the surface of the envelope.

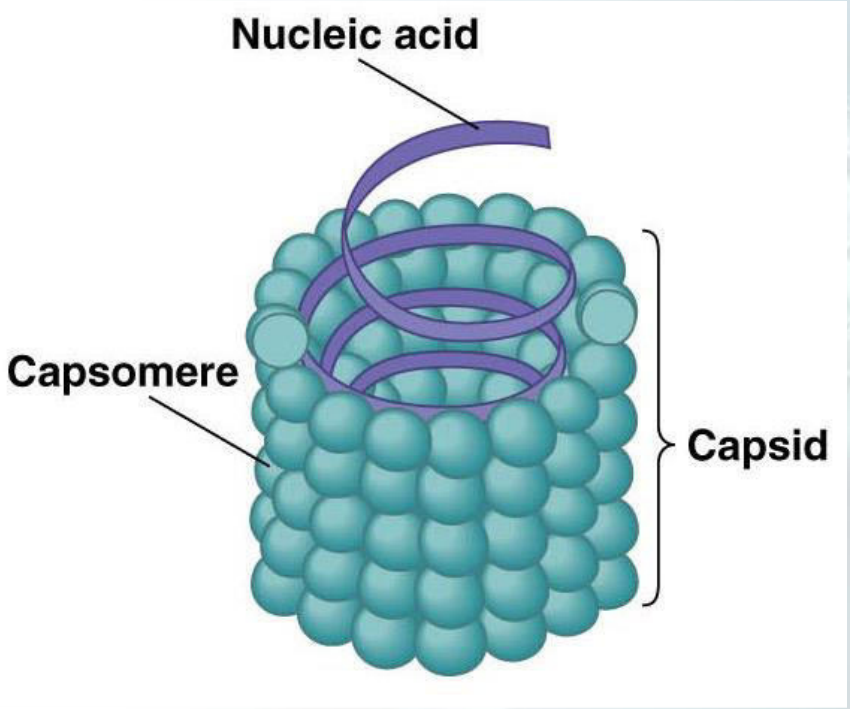
Virus I



Virus host range

- Most viruses infect only specific types of cells
  in one host
- Host range is determined by specific host attachment sites and cellular factors

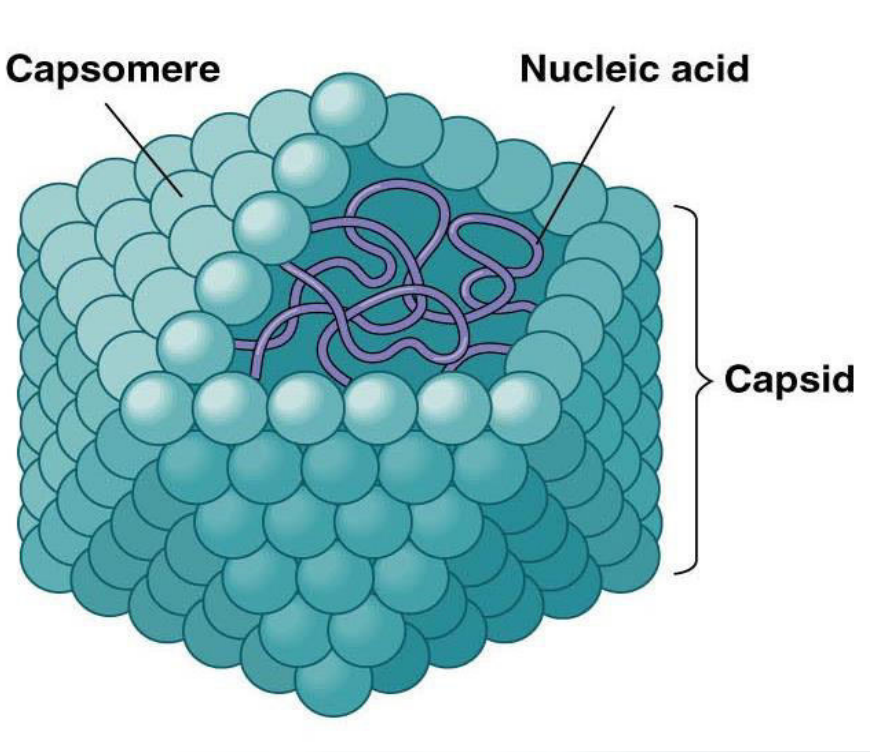
Virus I



Capsid Architecture

- Helical Viruses
- Resemble long rods, rigid or flexible
  - Rabies virus

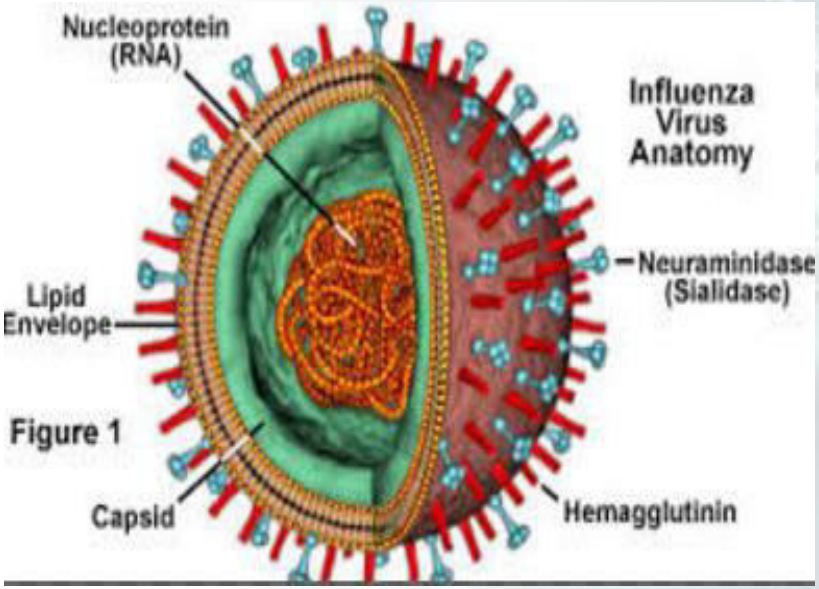
Virus I



Capsid Architecture

- Polyhedral Viruses
- Icosahedron: Regular polyhedron with 20 triangular faces and 12 corners
- Example: Poliovirus

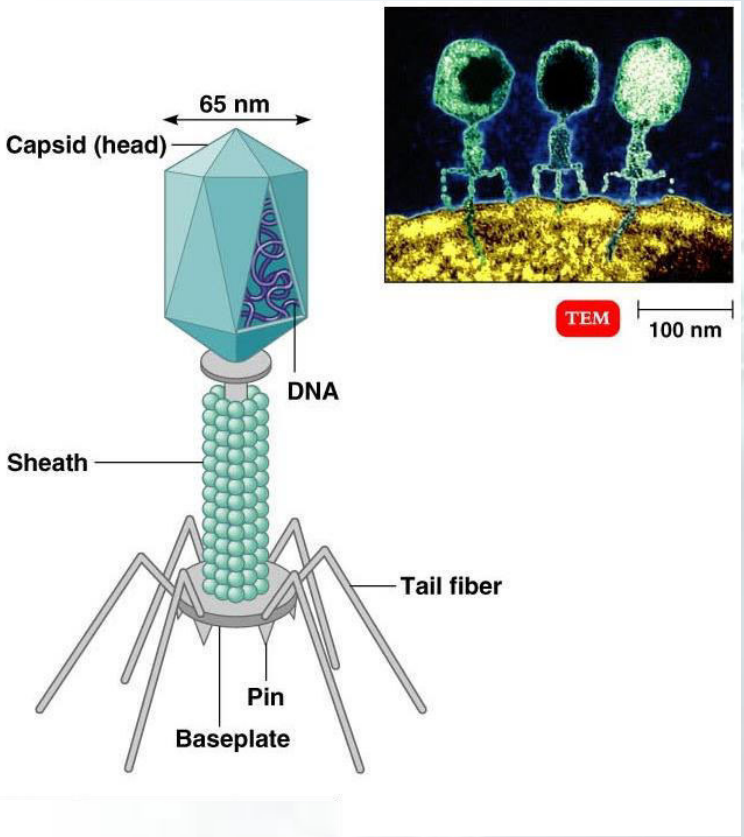
Virus I



Enveloped Viruses

- Enveloped Viruses
- Roughly spherical
  - Influenza virus

Virus I



Complex Viruses

- Head is polyhedral.
  - N.Acid
- Tail sheath is helical.
  - Bacteriophage

Virus I

Viral Taxonomy

- Family names end in -*viridae.*
- Genus names end in -*virus.*
  - Herpesviridae
  - *Herpesvirus*

Virus I

Viral Taxonomy

- **Viral species: A group of viruses sharing the same genetic information and ecological niche (host).**
- **Common names are used for species.**

Virus I

Viral Taxonomy

- Subspecies are designated by a number.
  - Human herpes virus HHV-1, HHV-2, HHV-3

Virus I

Summary

- Virus, virion and viroid
- DNA or RNA
- Symmetry of viruses
- Capsomeres

Virus II

Learning Objectives

- Viral Replication

Virus II

Viral Multiplication

- **Attachment:** Viruses attach to cell membrane.
- Host cell receptors (determines the range of infectivity)

Virus II

Viral Multiplication

- **Entry:** Penetration by **pinocytosis** (plasma membrane folds inwards to from a vesicle)

Virus II

Viral Multiplication

- **Fusion:** Viral envelope fuse with plasma membrane and releases the capids into the cell's cytoplasm

Virus II

Viral Multiplication

- **Uncoating:**
  Separation of viral NA from its protein coat by viral or host enzymes.
  - (undergoes eclipse period)

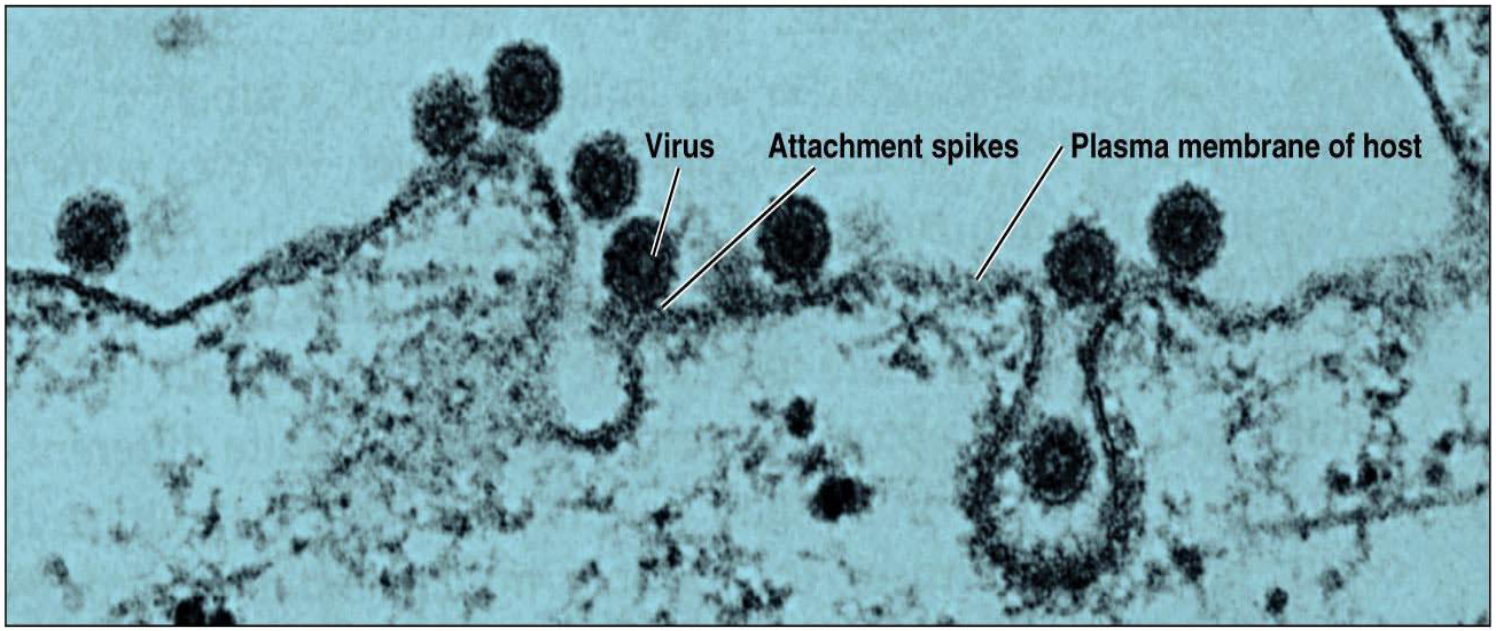
Virus II

Viral Multiplication

- **Biosynthesis:** Production of nucleic acid and proteins.
- **Maturation:** Nucleic acid and capsid proteins assemble.
- **Release** by budding (enveloped viruses) or rupture.

Virus II

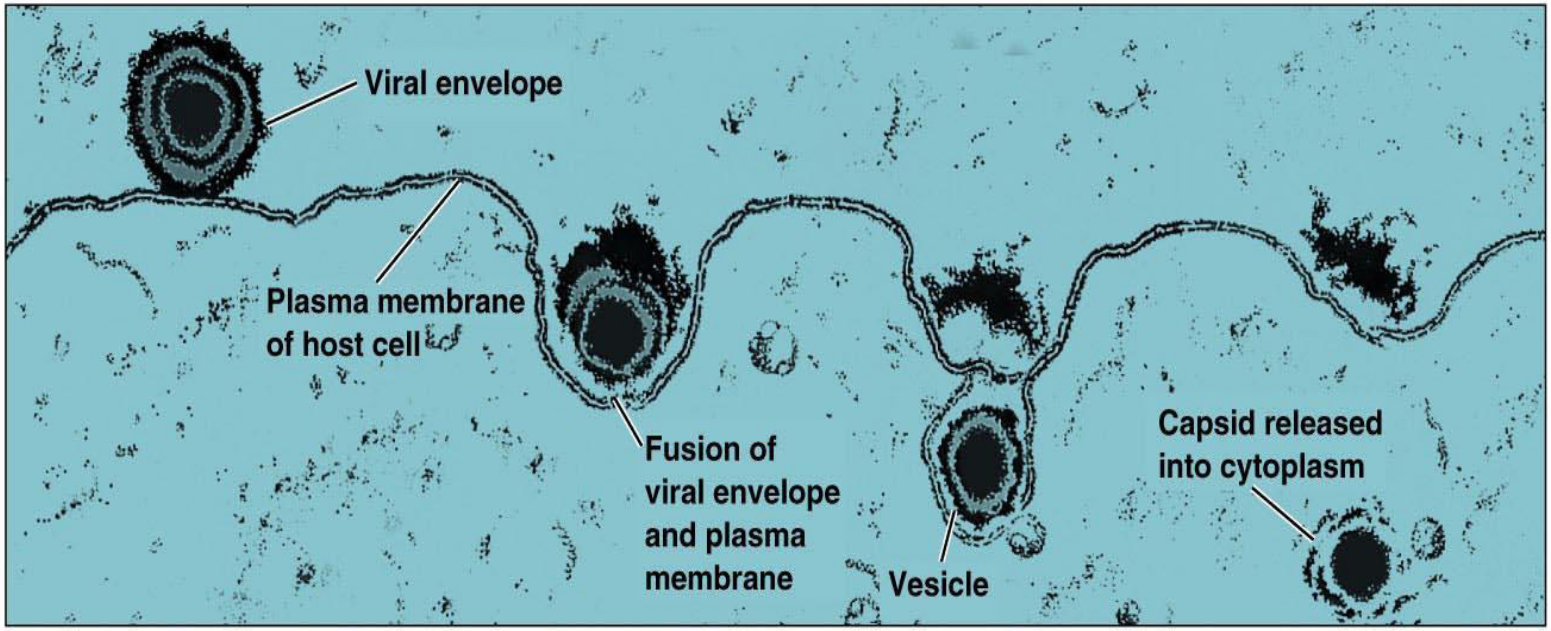
Pinocytosis



(a) Entry of togavirus

Virus II

Fusion



(b) Entry of herpesvirus

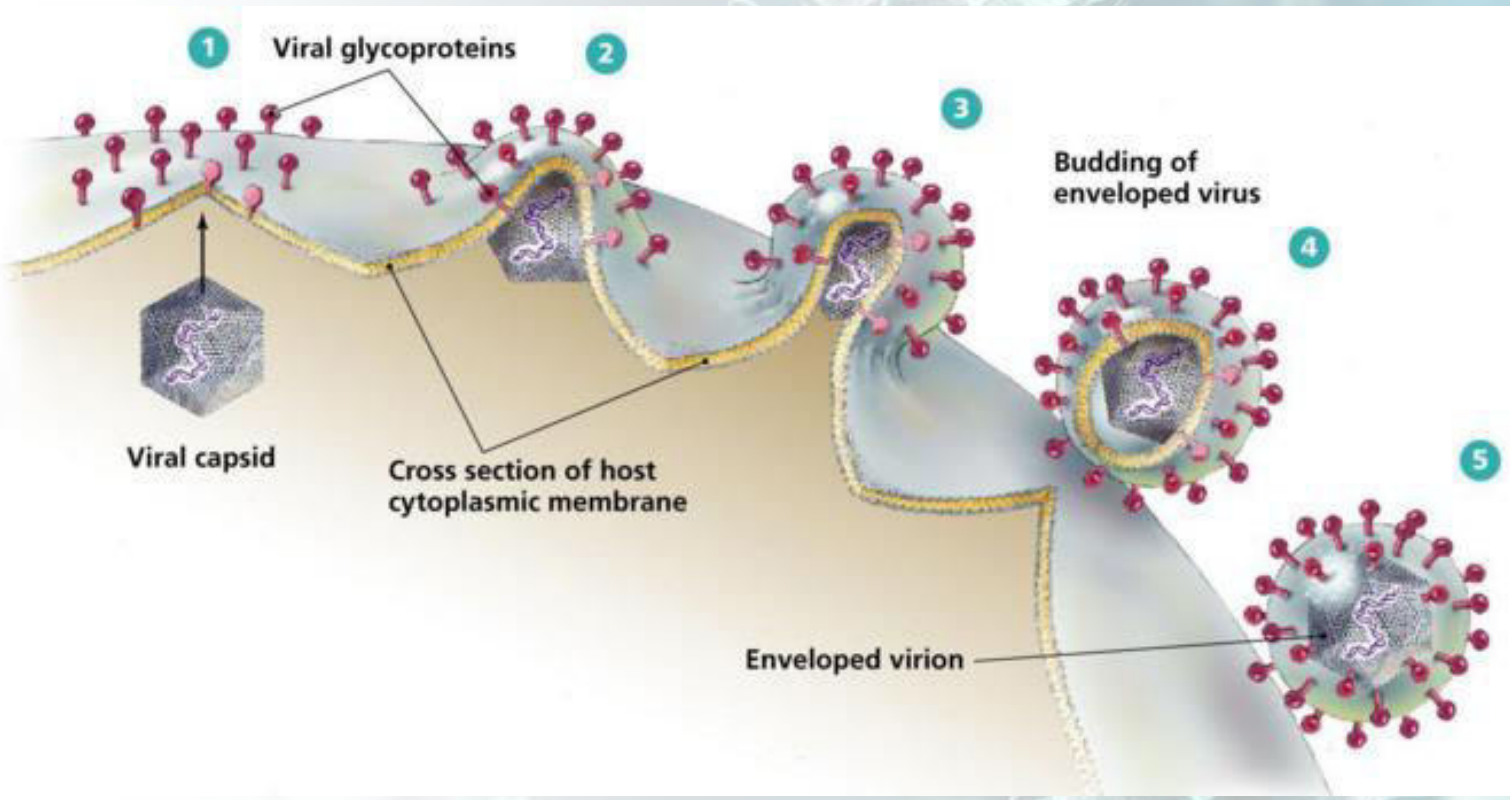
Virus II

Biosynthesis

- Early proteins
- Late proteins
- Genome replication
- Assembly
- Release

Virus II

Release of virus by budding



Virus II

Summary

- Replication of virus has many stages

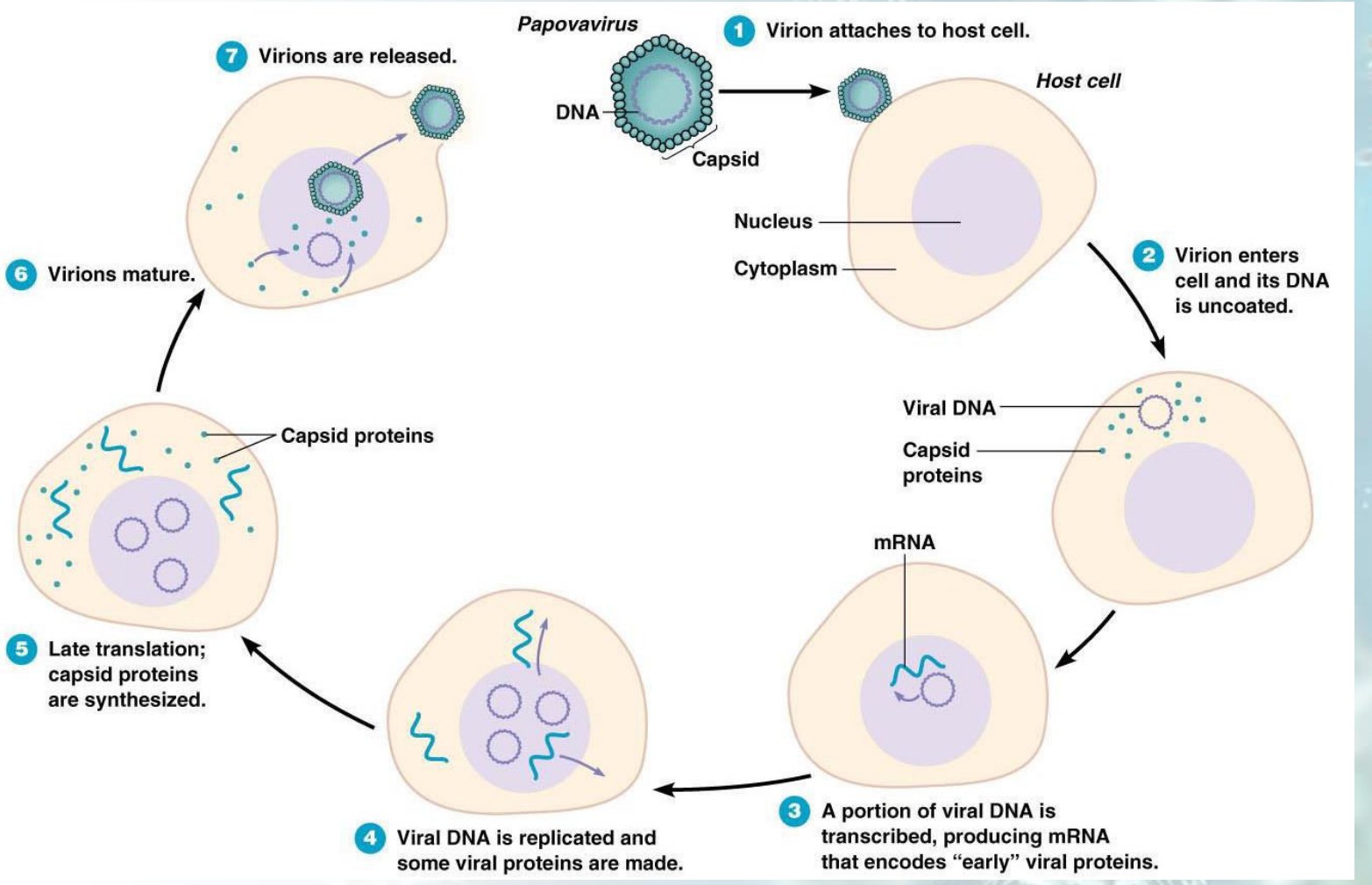
Virus III

Learning Objectives

- How do various viruses multiply?

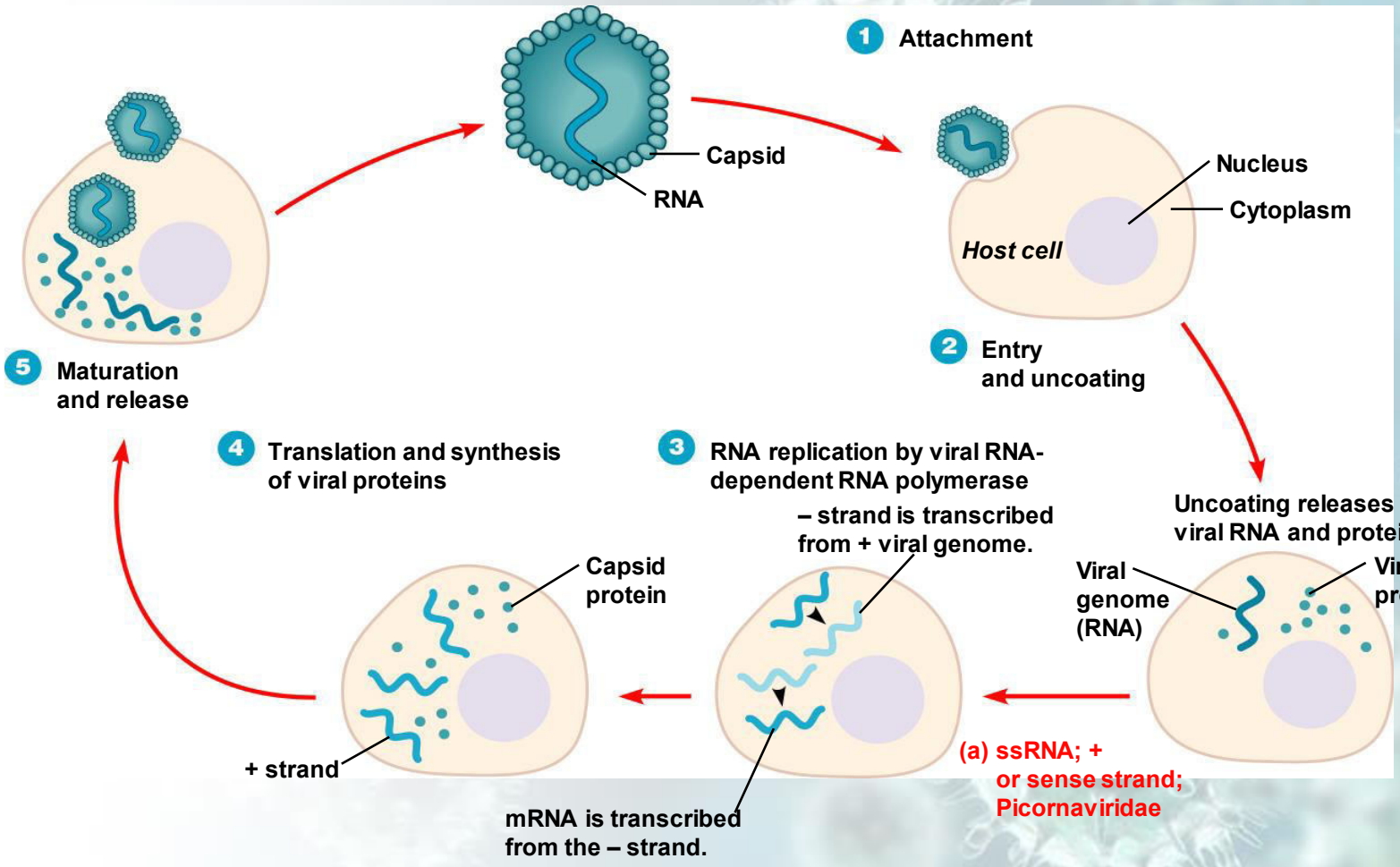
Virus III

Multiplication of DNA Viruses



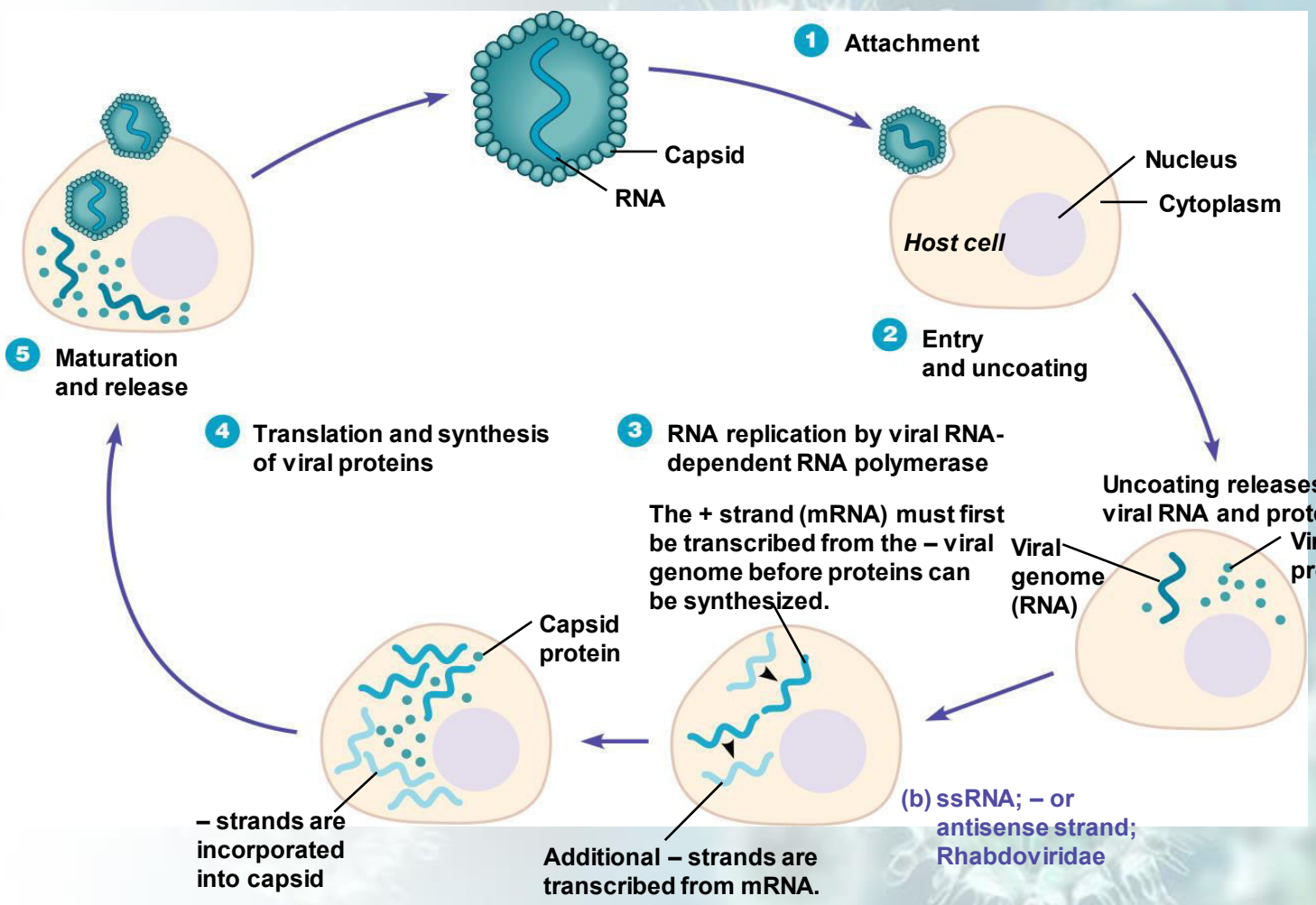
Virus III

+ or Sense Strand RNA Virus



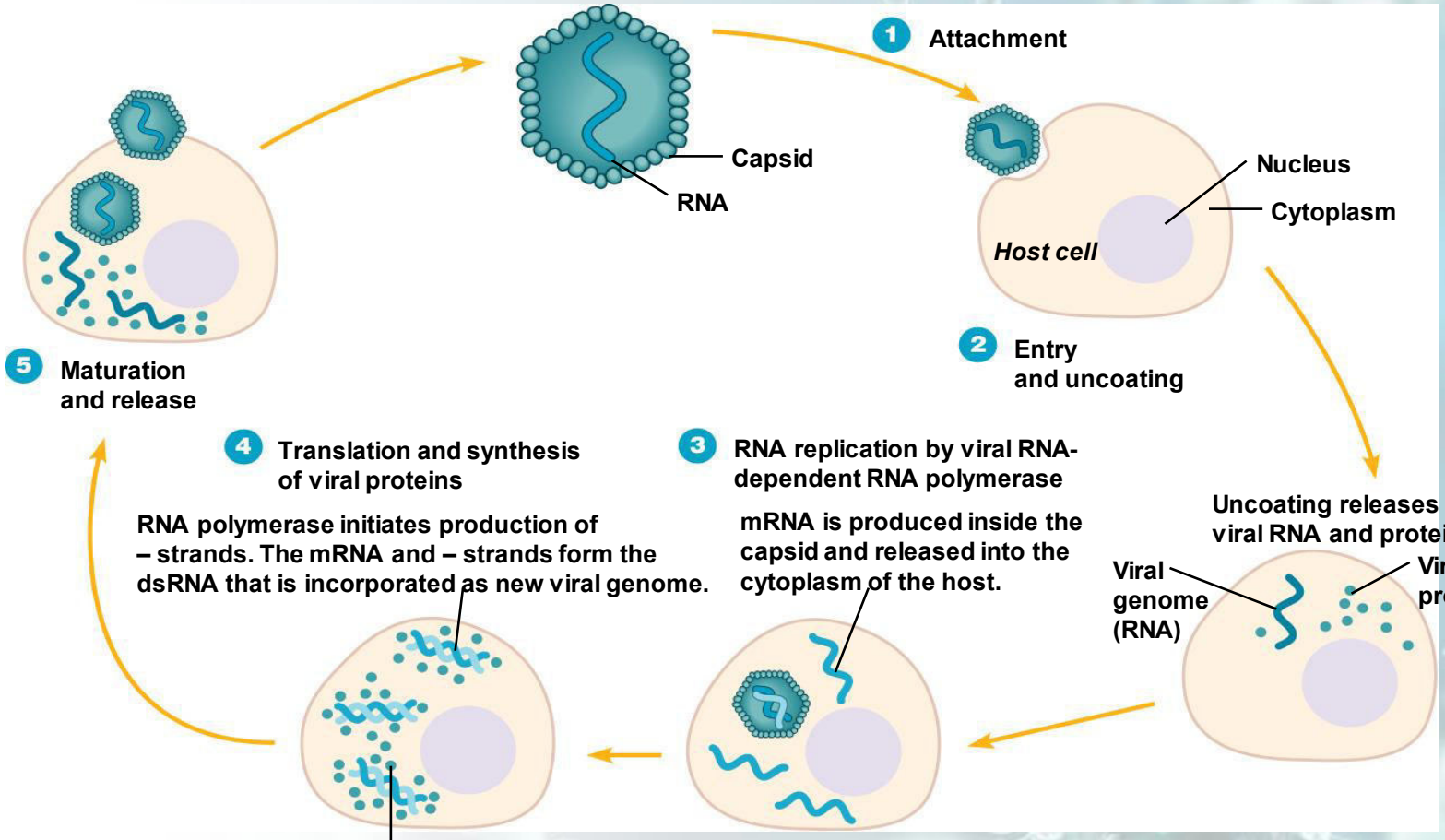
Virus III

- or Antisense Strand RNA Virus



Virus III

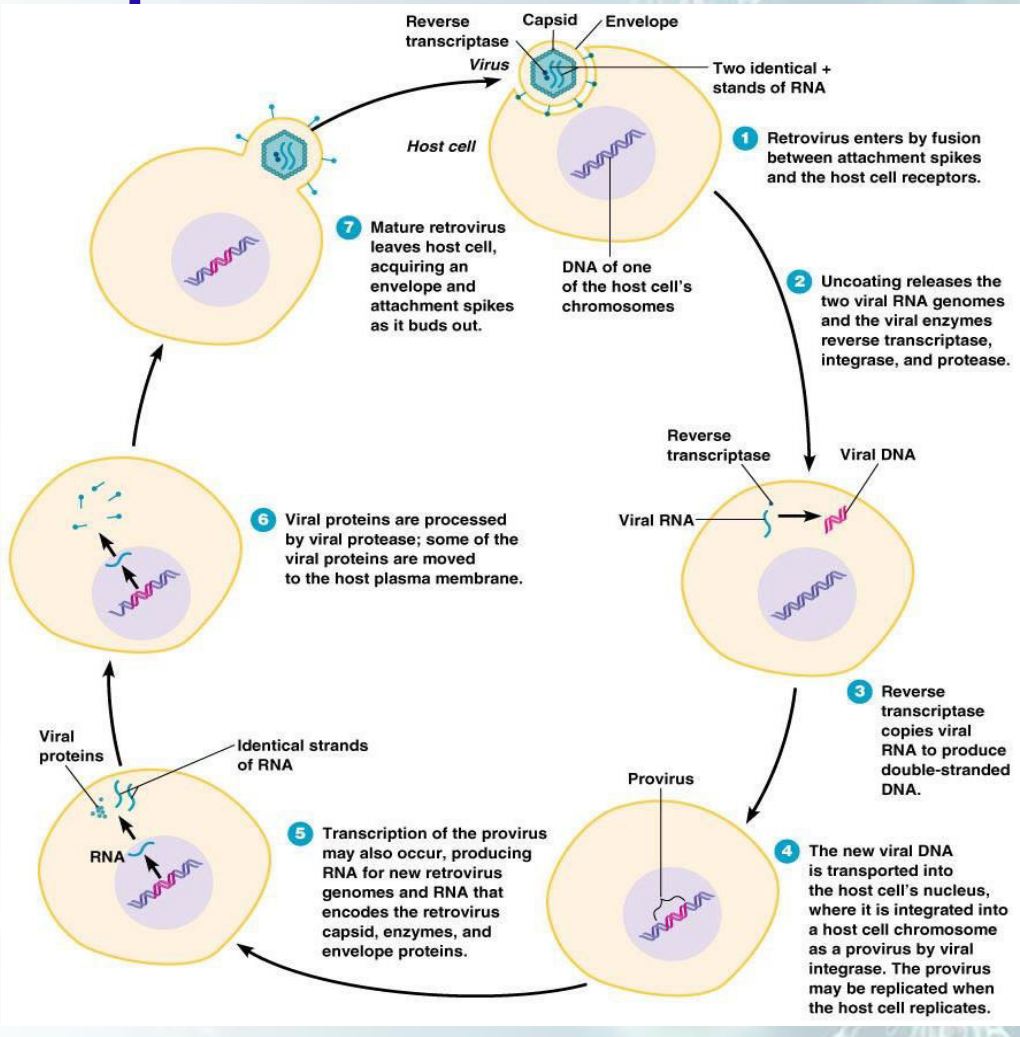
Double Stranded RNA Virus



(c) dsRNA; + or sense strand with – or antisense strand; Reoviridae

Virus III

Multiplication of a Retrovirus



Virus III

Summary

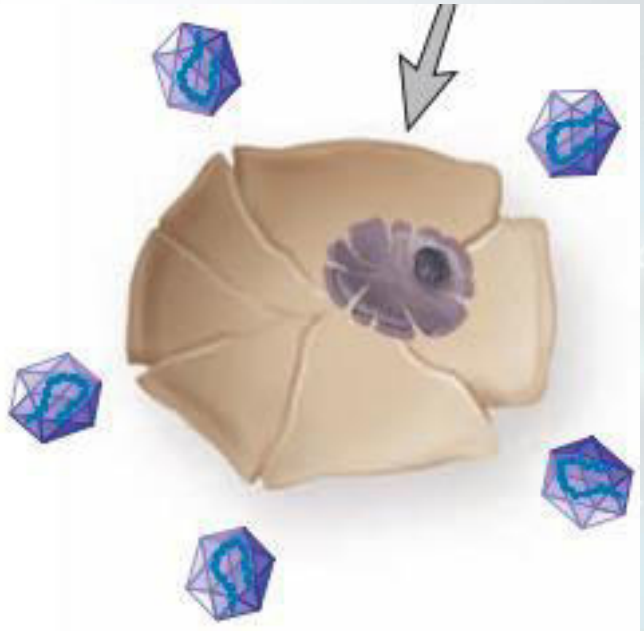
- Virus, virion and viroid
- DNA or RNA
- Symmetry of viruses
- Capsomeres

Virus IV

Learning Objectives

- Consequences of viral infection

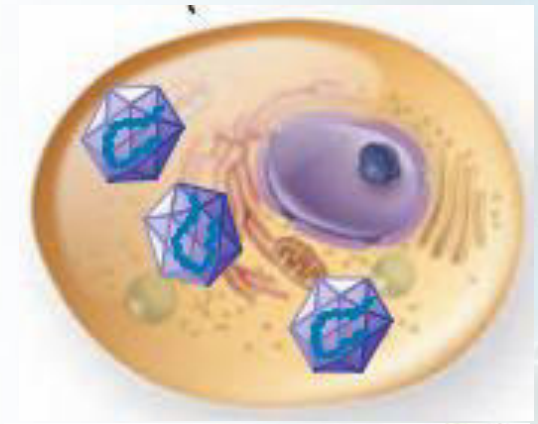
Virus IV



Consequences of Viral Infection

- Lysis of cells
- Microscopic and macroscopic Degenerative changes (Cytopathic Effects)

Virus IV



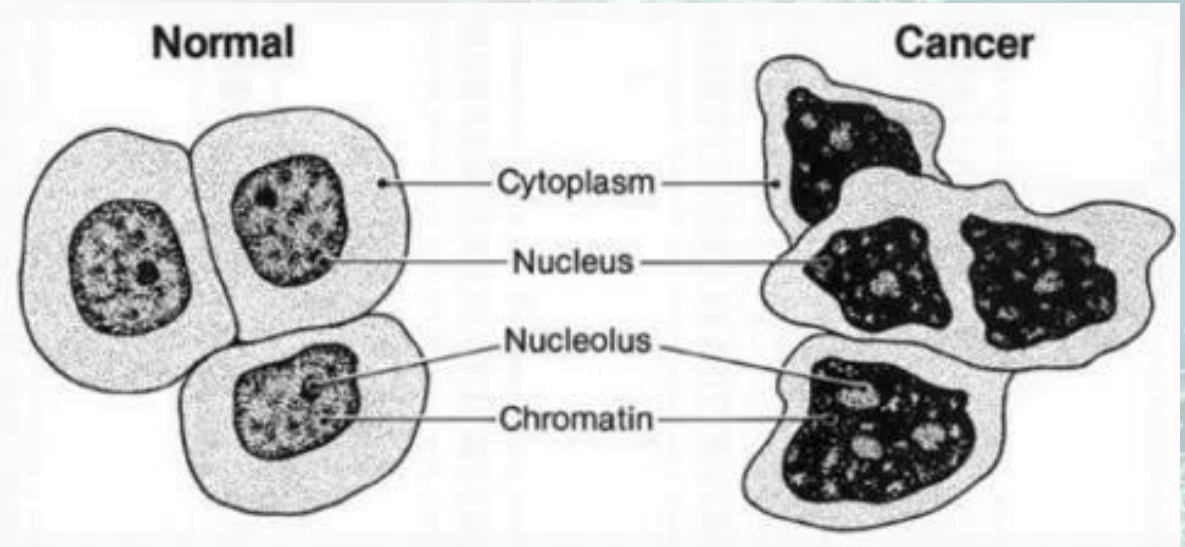
Consequences of Viral Infection

- **Virulent infection:** lysis of cells
- **Latent infection:** Virus does not replicate
- **Persistent infection:** Virus replicates but slowly

Virus IV

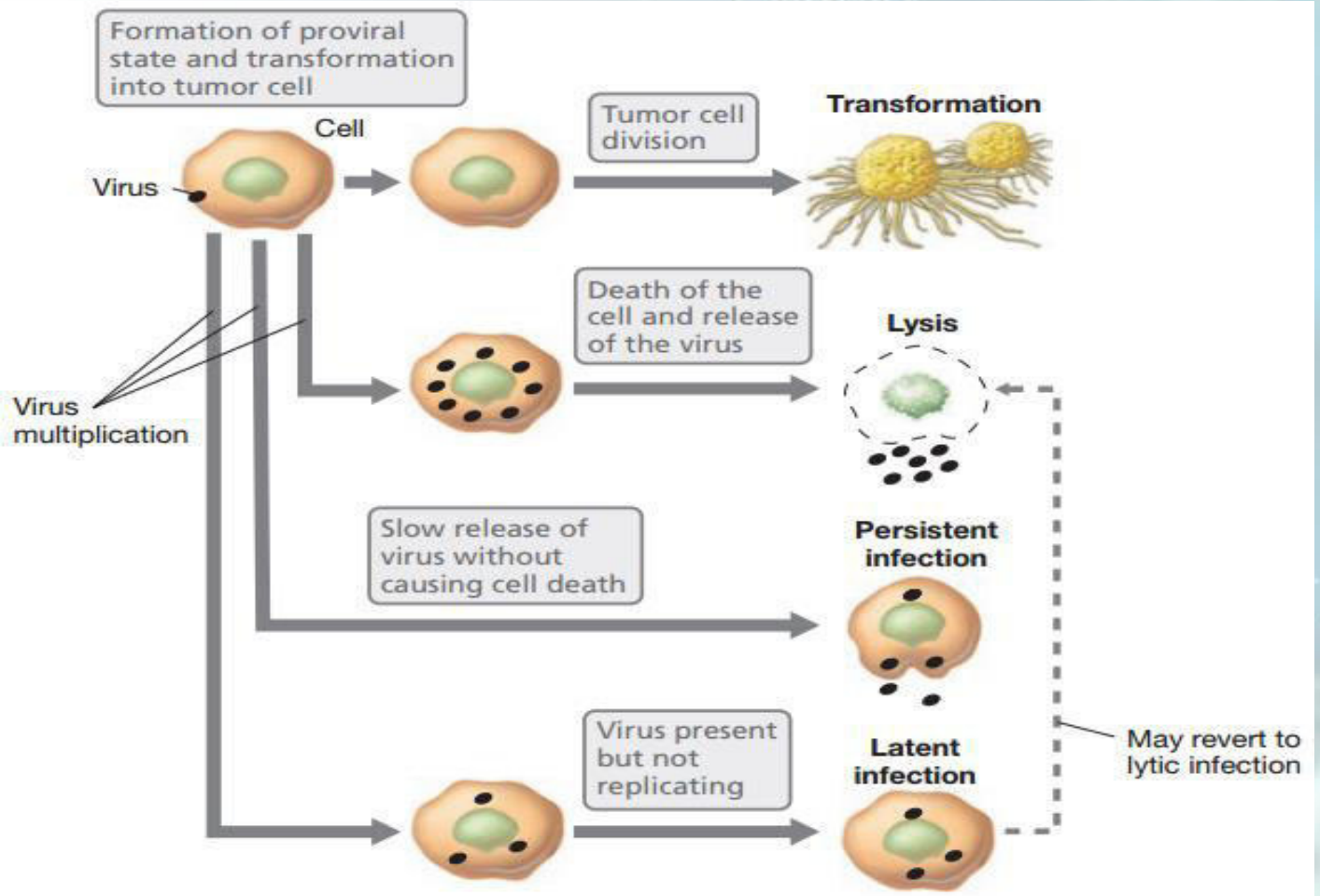
Consequences of Viral Infection

- **Transformation:**
  Tumor cells



Virus IV

Consequences of Viral Infection



Virus IV

Summary

- Lysis
- CPE

Virus V

Learning Objectives

- Virus Culture
- Virus enumeration

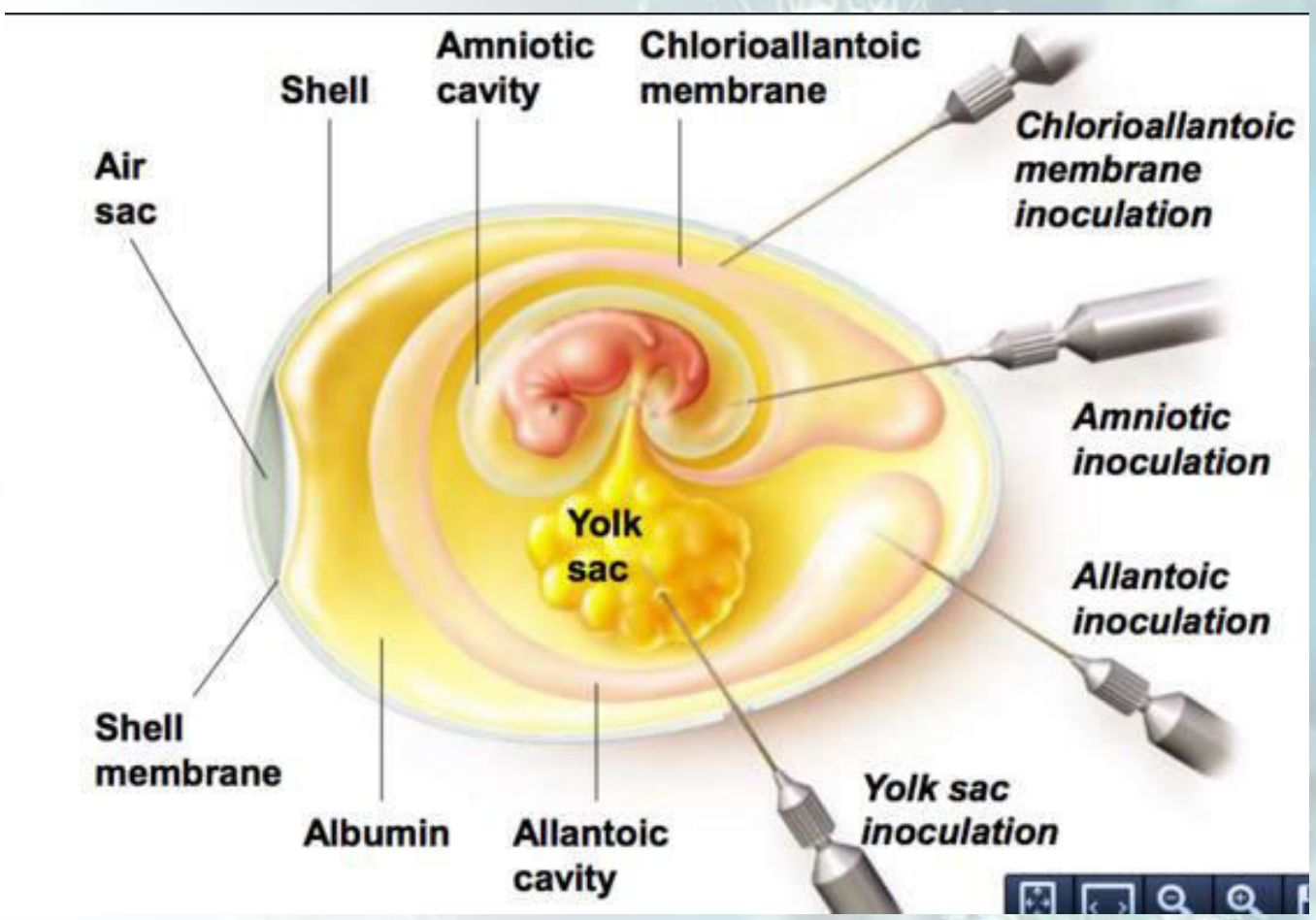
Virus V

Cultivation of Viruses

- Need live cells to grow
- Animals
- Embryonated eggs
- Cell Cultures
  - Plants and animal

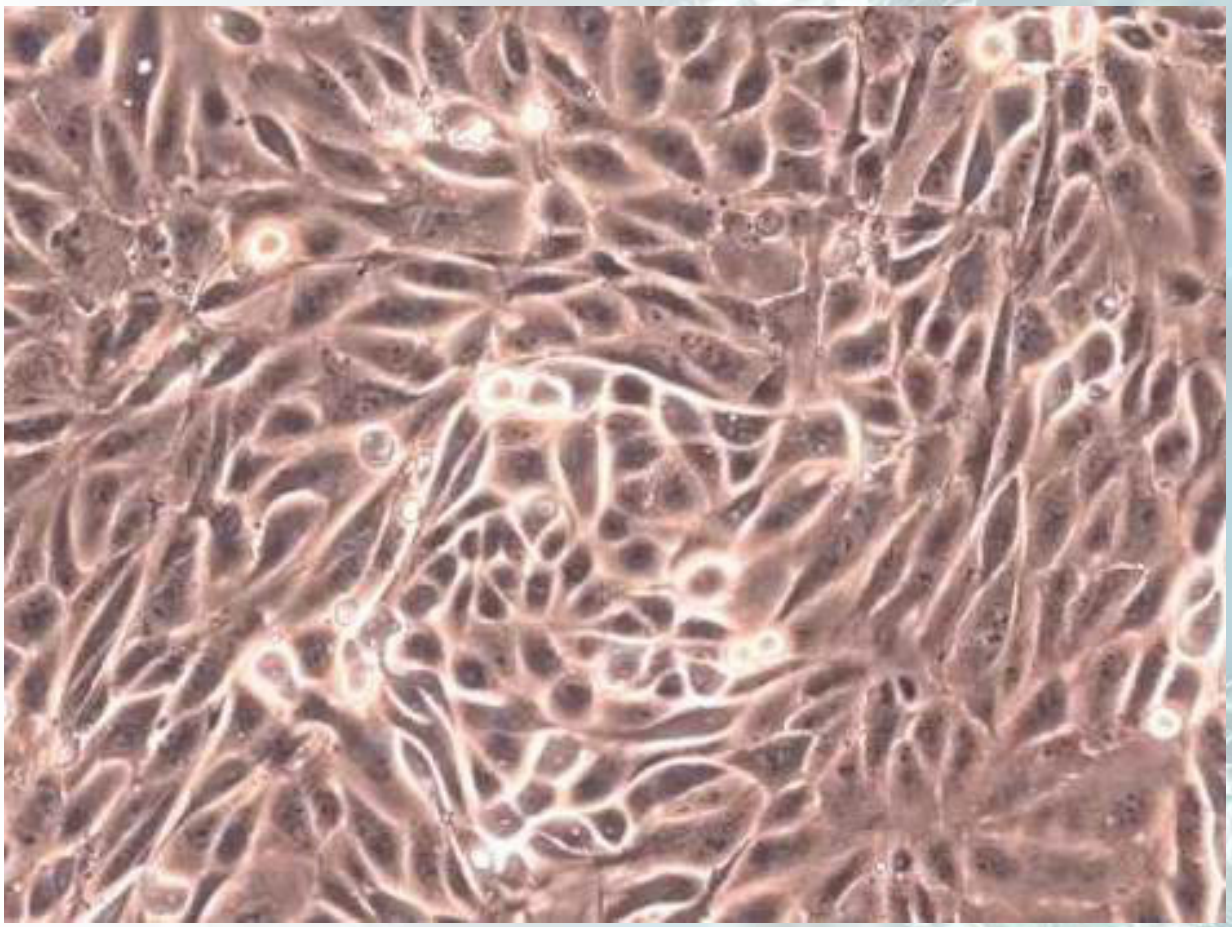
Virus V

Embryonated Hen Eggs

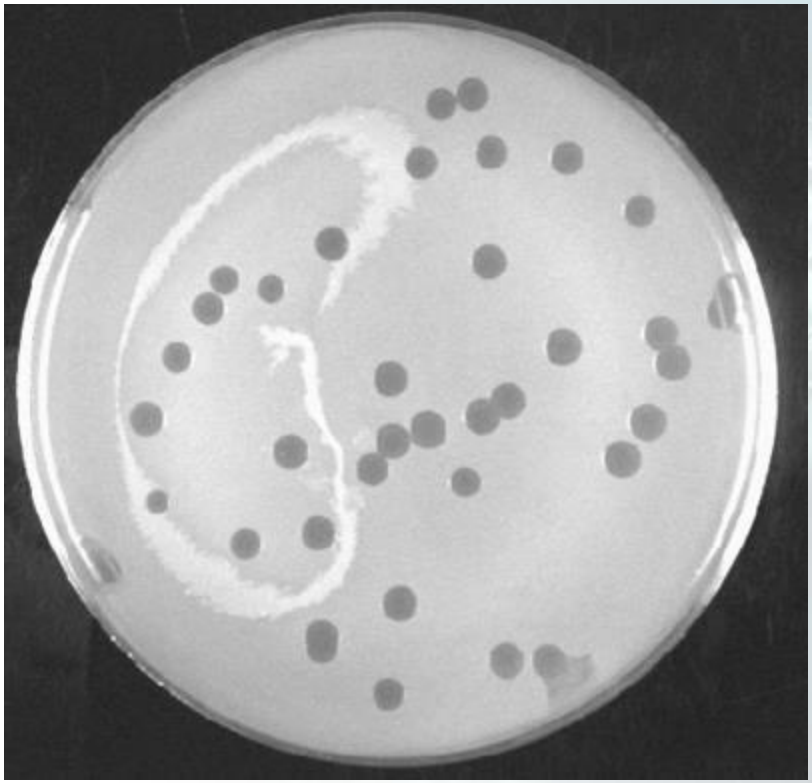


Virus V

Cell Culture



Virus V



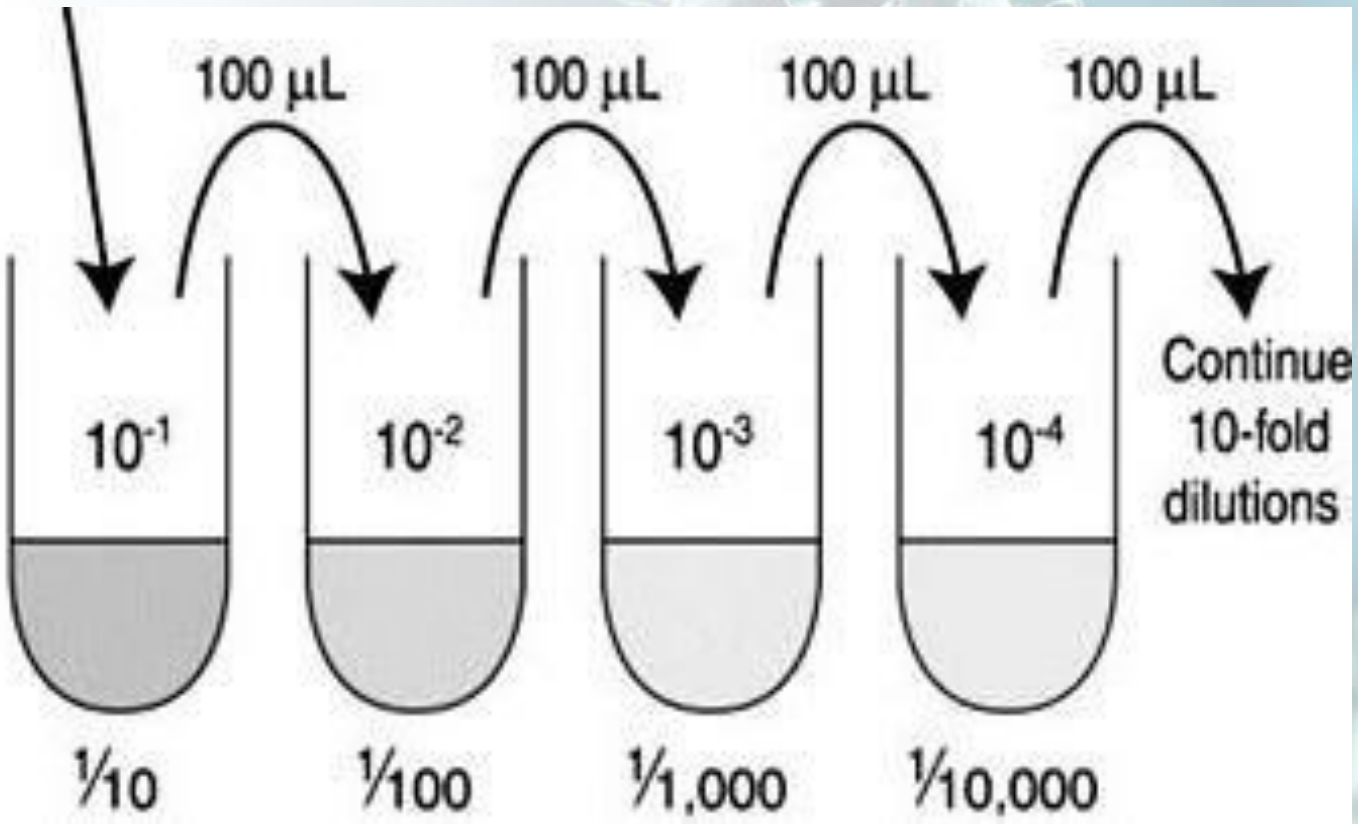
Virus Enumeration

- **Plaques formation:**
  - It is equivalent to bacterial colony
  - Plaque is an area where the virus infection has lysed the cells.

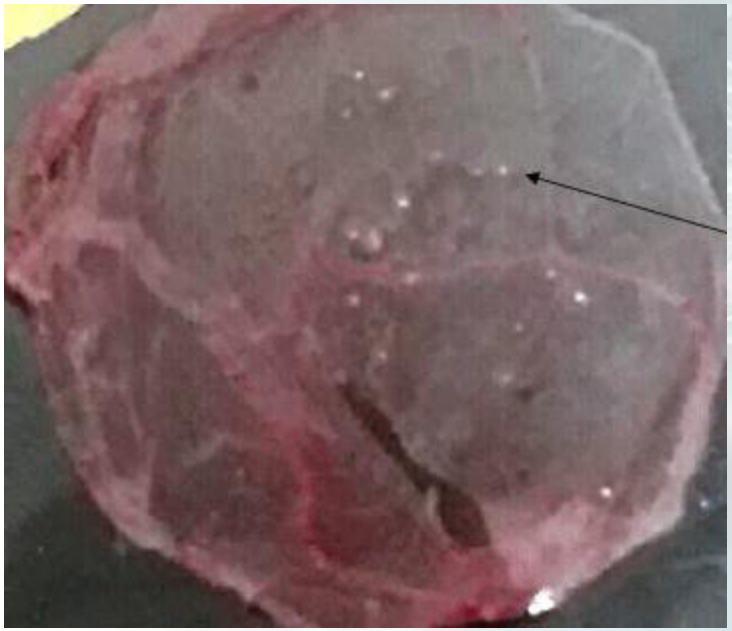
Virus V

Virus Enumeration

- **Serial Dilutions:**



Virus V



Enumeration

- Plaque assay
  - Plaque forming units are counted
- Infectious dose 50% (ID50)
  - $TCID_{50}$
- Lethal dose 50% (LD50)

Virus V

Summary

- Viruses can be cultured
  - Counted
  - CPE

Virus VI

Module 70

Virus VI

Learning Objectives

- Virusoids
- Prions

Virus VI

Virusoids

- Also called Satellite RNA
- circular ssRNA
- Encode one or two genes products
- Need helper virus to infect host cells

Virus VI

Virusoids

- Human hepatitis D virusoid
- Uses hepatitis B virus as helper

Virus VI

Prions

- Proteinaceous infectious particles
- Cause neurodegenerative diseases
  - Scrapie in sheep
  - Bovine spongiform encephlopthy
- Kuru in humans

Virus VI

Prions

- Abnormal form of cellular proteins
- How this happens, not clear
- Genetic component involved

Virus VI

Summary

- Virusoid
- Prions

Epidemiology I

Learning Objectives

- Epidemiology
- Methods of epidemiology
- Definitions

Epidemiology I

Epidemiology

- The study of where and when diseases occur and how they are transmitted in a population

Epidemiology I

Epidemiology Methods

- **Descriptive:**
  (recording the data about disease)
  - Location and time of cases of disease
  - Gender, health, age of patients etc.

Epidemiology I

Epidemiology Methods

Analytical: (To determine the cause of disease)

- Analysis of data
- Mode of transmission and means to prevent disease

Epidemiology I

Epidemiology Methods

- **Experimental Epidemiology:**
  - Hypothesis of a disease is tested
  - Application of Koch's postulates

Epidemiology I

Epidemiology

- **Case reporting: Health care workers report specified disease to local, state, and national offices**
  - **Incidence and prevalence of disease**

Epidemiology I

Epidemiology

- Incidence and prevalence of disease
  - Origin and spread of a disease
  - Data helpful for control measures

Epidemiology I

Epidemiology

- Nationally notifiable diseases: Physicians are required to report occurrence
  - Highly communicable

Epidemiology I

Epidemiology

- Highly communicable Associated with greater economic losses
- Belong to diseases that have been eradicated

Epidemiology I

Epidemiology

- **Morbidity**: Incidence of a disease
- **Morbidity rate:** Number of people affected in relation to the total population in a given time period

Epidemiology I

Epidemiology

- **Mortality:** Deaths from a disease
- **Mortality rate:** Number of deaths from a disease in relation to the population in a given time

Epidemiology I

Epidemiology

- **Pathogenicity:** The ability to cause disease
- **Virulence:** The extent of pathogenicity

Epidemiology I

Summary

- Epidemiology
- Descriptive epi
- Analytical epi
- Definitions

Epidemiology II

Learning Objectives

- Language of disease
- Microbiota
- Factors that determine the microbiota
- Symbiosis
- Microbial antagonism

Epidemiology II

Definitions

- Pathology: Study of diseases
  - Cause of the disease
  - Pathogenesis: how disease establishes itself
  - Structural and functional changes

Epidemiology II

Definitions

- **Infection:** invasion by the microbes
- **Disease:** dysfunction
- **Microbiome:** Microbial communities that live in or on humans

Epidemiology II

Definitions

- **Normal microbiota or flora:** permanent residence but do not cause diseases under normal conditions
- **Transient microbiota:** present for several days, weeks or months and then disappears

Epidemiology II

Microbiota

- What determines the distribution and composition of it?
  - Nutrients
  - Physical and chemical factors
    - pH, O_2, CO_2
  - Host defenses
    - Stress, age

Epidemiology II

Symbiosis

- Living together is symbiosis
- **Mutualism:** both are benefited
- **Parasitism:** one is harmed
- **Commensalism:** One is benefited, other is unharmed

Epidemiology II

Microbial antagonism

- It is a competitive exclusion.
  - Compete for nutrients
  - By chemicals
    - Bacteriocins by E. coli suppresses Salmonellae

Epidemiology II

Microbial antagonism

- Clostridium difficile:
  - Normal microbiota keeps it low but antibiotics can result in diarrhea

Epidemiology II

Opportunistic pathogen

- Normal microbiota under stress conditions can cause infections.

Epidemiology II

Summary

- Some definitions
- How microbiota gets established
- Various forms of symbiosis

Epidemiology III

Learning Objectives

- Description of diseases
- Stages of diseases
- Sources of infection

Epidemiology III

Definitions

- **Endemic:** consistently present, but with low incidence
  - Infected ones are reservoir of infection
- **Epidemic:** within an area, unusually high incidence of a disease

Epidemiology III

Definitions

- **Pandemic:** World wide epidemic
- **Incidence:** # of new cases in a given time period in a population
- **Prevalence:** New + existing cases

Epidemiology III

Disease Progression

- **Infection:** colonize and grows in body
- **Incubation period:** time b/w infection and diseases
- **Acute period:** height of disease

Epidemiology III

Disease Progression

- **Decline period:** Disease symptoms subside
- **Convalescent period:** patient returns to normal

Epidemiology III

Definitions

- **Reservoir:** Sites where organisms of infections remain viable
  - Act as source of infection
- **Inanimate objects:** saprophytic (Cl. tetani)
  - Difficult to eradicate

Epidemiology III

Definitions

- **Animate:** live in the body
  - Spread from person to person
    - By direct or indirect contact
- Single host organisms
  - Can be eradicated or controlled

Epidemiology III

Zoonosis

- Primarily an animal infection, but can be transmitted to humans
- Can assume epidemic proportions in humans
- Difficult to control or eradicate

Epidemiology III

Carrier

- A living carrier is a pathogen infected individual who does not show any signs or symptoms of the disease.
- A potential source of infection for others
  - Typhoid Mary story

Epidemiology III

Summary

- Some definitions relating disease stages and microbial reservoirs

Epidemiology V

Learning Objectives

- How diseases get transmitted
- Various vehicles of infection

Epidemiology V

Contact Transmission

- **Direct:** person to person
  - Physical contact
  - Influenza

Epidemiology V

Contact Transmission

- **Indirect:** A nonliving object is involved
  - **Fomite:** any nonliving object (towel, bedding)

Epidemiology V

Contact Transmission

- **Droplet:** mucus droplets

Epidemiology V

Vehicle Transmission

- Transmission by a medium
- Waterborne
- Foodborne
- Airborne
- Blood, other body secretion

Epidemiology V

Vectors

- Arthropods:
  - Mechanical transmission (Passive)
  - Biological transmission (Active)

Epidemiology V

Definitions

- Nosocomial Infections

Epidemiology V

Summary

- Vehicles of infection

Mechanisms of Pathogenicity I

Learning Objectives

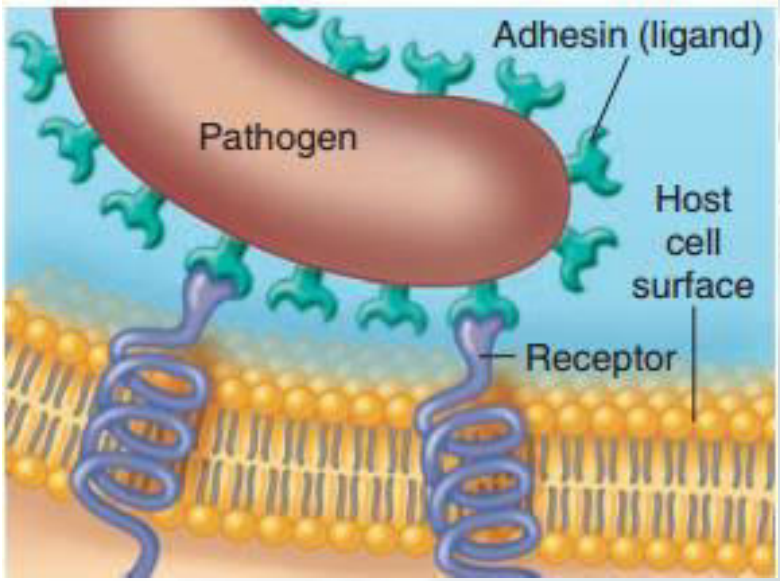
- How disease occurs
- Factors that aid in setting up of an infection

Mechanisms of Pathogenicity I

How disease occurs?

- When the microbe overpower host's body defenses, disease occurs.

Mechanisms of Pathogenicity I



How disease occurs?

- Pathogen must get access to the host.
- It must adhere to host tissues.
- It must penetrate or evade host defenses.
- It must damage the host tissues.

Mechanisms of Pathogenicity I

Portal of Entry

- **Mucus membrane:**
- Tubal organs
- Respiratory system and digestive system as largest portals

Mechanisms of Pathogenicity I

Portal of Entry

- **Skin:** It is the largest organ of the body in terms of the surface area.
- **Parenteral route:** By injury
- **Preferred route of entry:** Streptococci inhaled vs ingested

Mechanisms of Pathogenicity I

Invading Organisms

- **Number:** Few vs many
- **ID50:** Virulence can be expressed in terms of ID50
  - Depends upon the route of entry

Mechanisms of Pathogenicity I

Invading Organisms

- **Bacillus anthracis:**
  - Skin: ID50 is 10-50 spores
  - Inhalation: 10 -20 thousands spores
  - GI anthrax: ID50 is 250 to 1000 thousands spores

Mechanisms of Pathogenicity I

Invading Organisms

- **LD50:** Potency of toxin is expressed in LD50
  - Botulinum toxin in mice is 0.03 ng/kg
  - Shiga toxin: 250 ng/kg
  - Staph enterotoxin: 1350 ng/kg

Mechanisms of Pathogenicity I

Summary

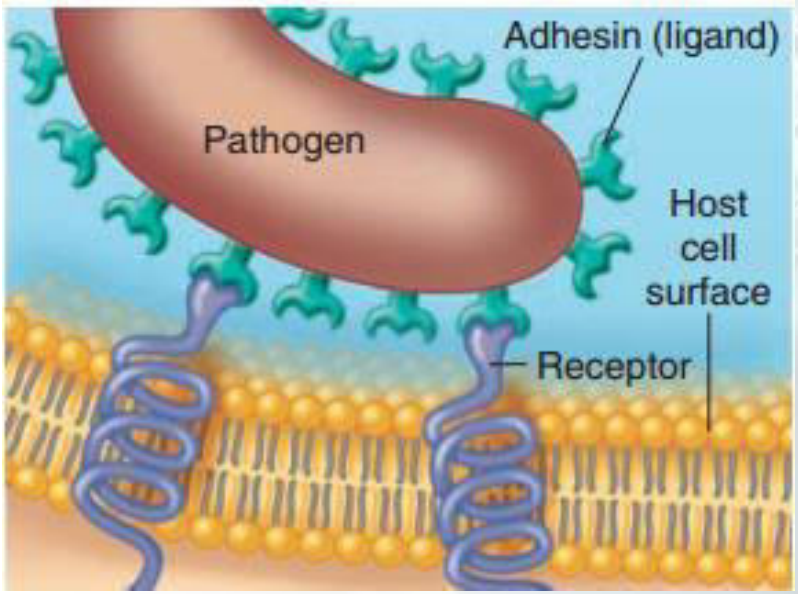
- What is needed to set up an infection
- Virulency expression
- Lethal dose and infectious dose 50

Mechanisms of Pathogenicity II

Learning Objectives

- How infectious organisms enter the body
- How organisms evade body defenses

Mechanisms of Pathogenicity II



Adherence

- Adhesion is mediated by adhesins or ligands to the receptors.
  - Adhesins present on glycocalyx, pili, fimbriae, or flagella.
  - These are glycoproteins or lipoproteins.

Mechanisms of Pathogenicity II

Biofilms

- Dental plaques is an example of biofilm.

Mechanisms of Pathogenicity II

Defense Evasion

- **Capsule:** Impairs phagocytosis
  - Antibodies can help
  - Only a few bacteria produce capsule
  - *Strep pneumoniae*
  - *Klbesiella pneumoniae*

Mechanisms of Pathogenicity II

Defense Evasion

- **Cell wall components:**
  - **M protein**
    - **Found on cell surface and fimbriae**
    - **Resists phagocytosis**
    - ***Strep pyogenes***

Mechanisms of Pathogenicity II

Defense Evasion

- **Cell wall components:**
  - **Mycolic Acid:**
    - **Resists phagocytosis**

Mechanisms of Pathogenicity II

Defense Evasion

- **Enzymes: (exoenzymes)**
  - **Coagulase:** blood clotted
  - Bacteria is trapped or hides itself in the clot
  - Staphs produce it.

Mechanisms of Pathogenicity II

Defense Evasion

- **Hyaluronidase:** Dissolves CT
  - Bacteria can spread easily.
- **Collagenase:** Breaks down collagen
- **IgA protease:** Digests IgA antibodies

Mechanisms of Pathogenicity II

Defense Evasion

- **Antigenic variation:**
  - Influenza virus
  - Neisseria gonorrheae

Mechanisms of Pathogenicity II

Summary

- Adhesins
- Capsule
- Enzymes

Mechanisms of Pathogenicity III

Learning Objectives

- How organisms cause damage to the body
- Factors that aid in setting up of an infection

Mechanisms of Pathogenicity III

Damage by Microbes

- Nutrient depletion
  - Siderophores for iron
- Direct damage by invasion
- Toxins
- Hypersensitivity reactions

Mechanisms of Pathogenicity III

Damage by Microbes

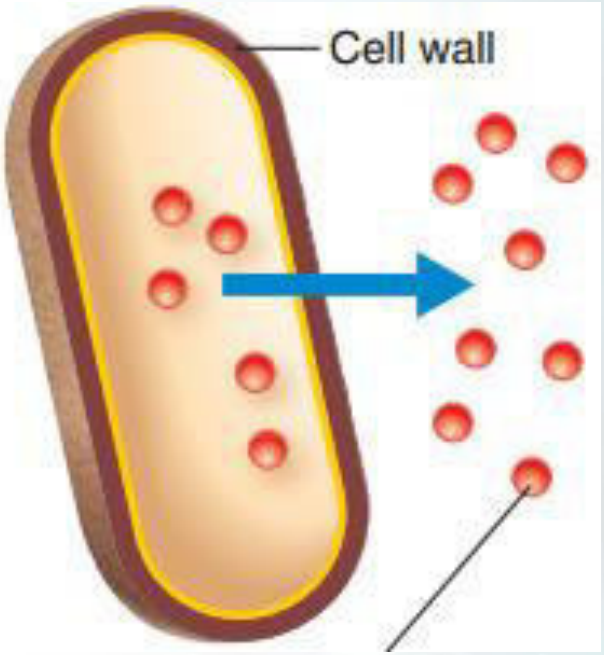
- Nutrient depletion
  - Siderophores for iron
    - Lactoferrin
    - Transferrin
    - ferritin

Mechanisms of Pathogenicity III

Damage by Microbes

- **Direct damage:**
- **Bacteria that grows in host cells rupture the cells and spread further.**

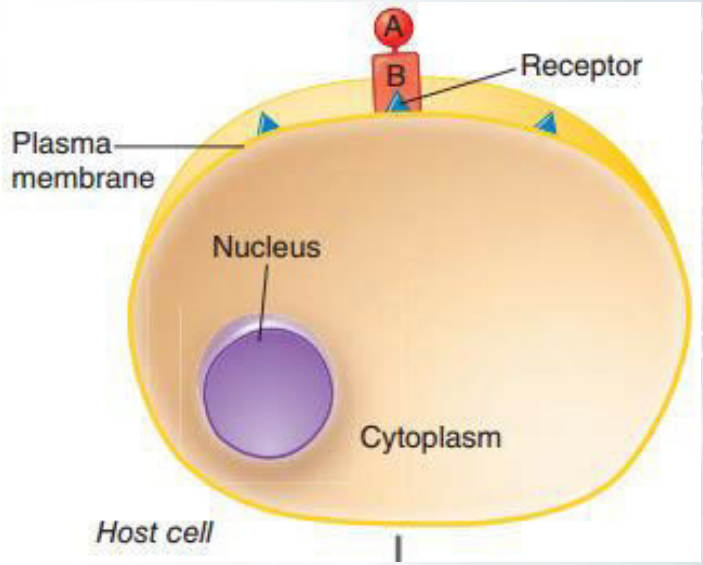
Mechanisms of Pathogenicity III



Damage by Microbes

- Toxin production:
- Exotoxins
  - Antitoxins
  - Toxoid
- A-B toxins

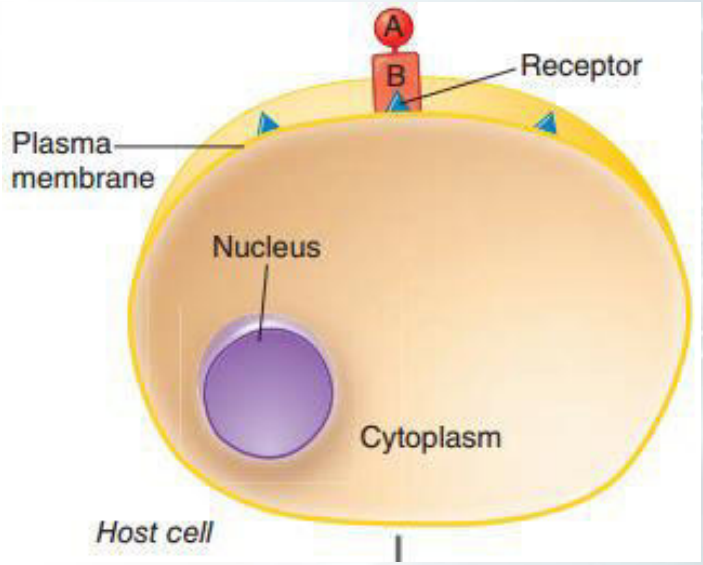
Mechanisms of Pathogenicity III



Damage by Microbes

- **Toxin Types:**
- **A-B toxins:**
  - diphtheria toxin
- **Membrane disrupting toxins:**
  - Cause lysis of cells
  - Leukocidins
  - Hemolysins

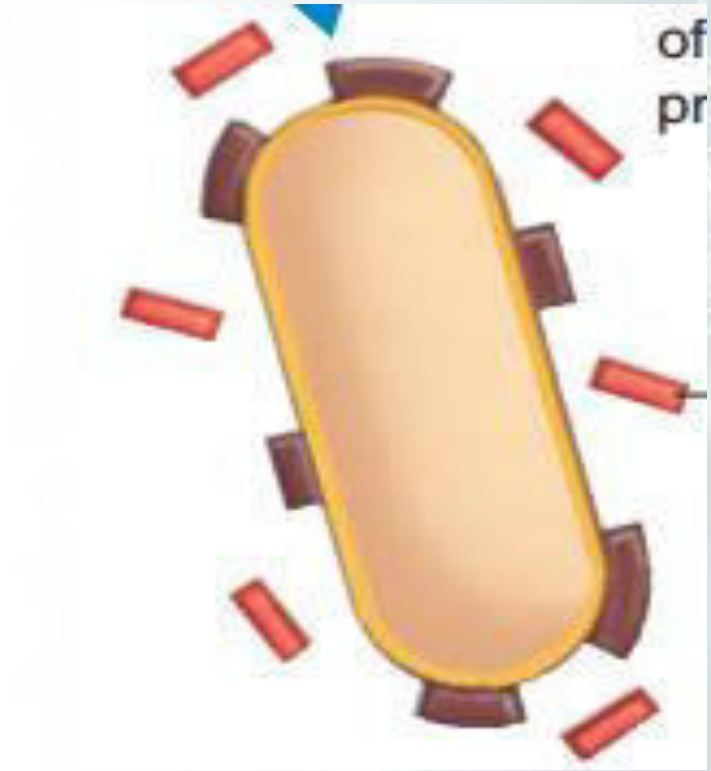
Mechanisms of Pathogenicity III



Damage by Microbes

- **Toxin Types:**
- **Superantigens:**
  - Provoke intense immune response
  - Cytokines are released by T cells.

Mechanisms of Pathogenicity III



Damage by Microbes

- **Toxin production:**
- Endotoxins
  - Fever
  - Shock
  - **Limulus amebocyte lysate assay (LAL)**
  - Hemolymph from horseshoe crab

Mechanisms of Pathogenicity III

Summary

- Damage by the organisms
- Various mechanisms

Innate Immunity I

Learning Objectives

- What is immunity
- Types of immunity

Innate Immunity I

Definitions

- **Immunity:** Ability of the body to protect itself from harmful effects of diseases.
- **Susceptibility:** lack of immunity

Innate Immunity I

Types of Immunity

- **Innate:** Present at birth
- **Adaptive:** Acquired by experience

Innate Immunity I

Innate Immunity

- Present at birth
- Rapid to respond
- Nonspecific
- Do not increase in response with repeated exposures to the same agent
- Have no memory

Innate Immunity I

Innate Immunity

- **First line of defense:** Physical barriers (skin, mucus membrane)
- **Second line of defense:** Phagocytic cells and their secretions

Innate Immunity I

Adaptive Immunity

- Specific
- Have memory
- Increase with repeated exposure
- Slower to respond
- Humoral immune response
- Cellular immune response

Innate Immunity I

Summary

- Innate immunity
- Adaptive immunity

Innate Immunity II

Learning Objectives

- Cells and secretions of innate immunity
- Lines of defense

Innate Immunity II

Innate Immunity

- **Toll like receptors (TLRs):** Protein molecules on defensive cells
- **Pathogen associated molecular patterns (PAMPs):** Formed by LPS, flagellin, peptidoglycan, DNA, RNA

Innate Immunity II

Innate Immunity

- Cytokines are released as a result of defense cell stimulation by PAMPs and TLRs interactions.
- Cytokines recruits more immune cells and also stimulate adaptive immune response.

Innate Immunity II

1st Line of Defense

- **Physical Factors:**
  - Skin
  - Mucus membrane
    - mucus
  - Lacrimal apparatus
  - Cilia

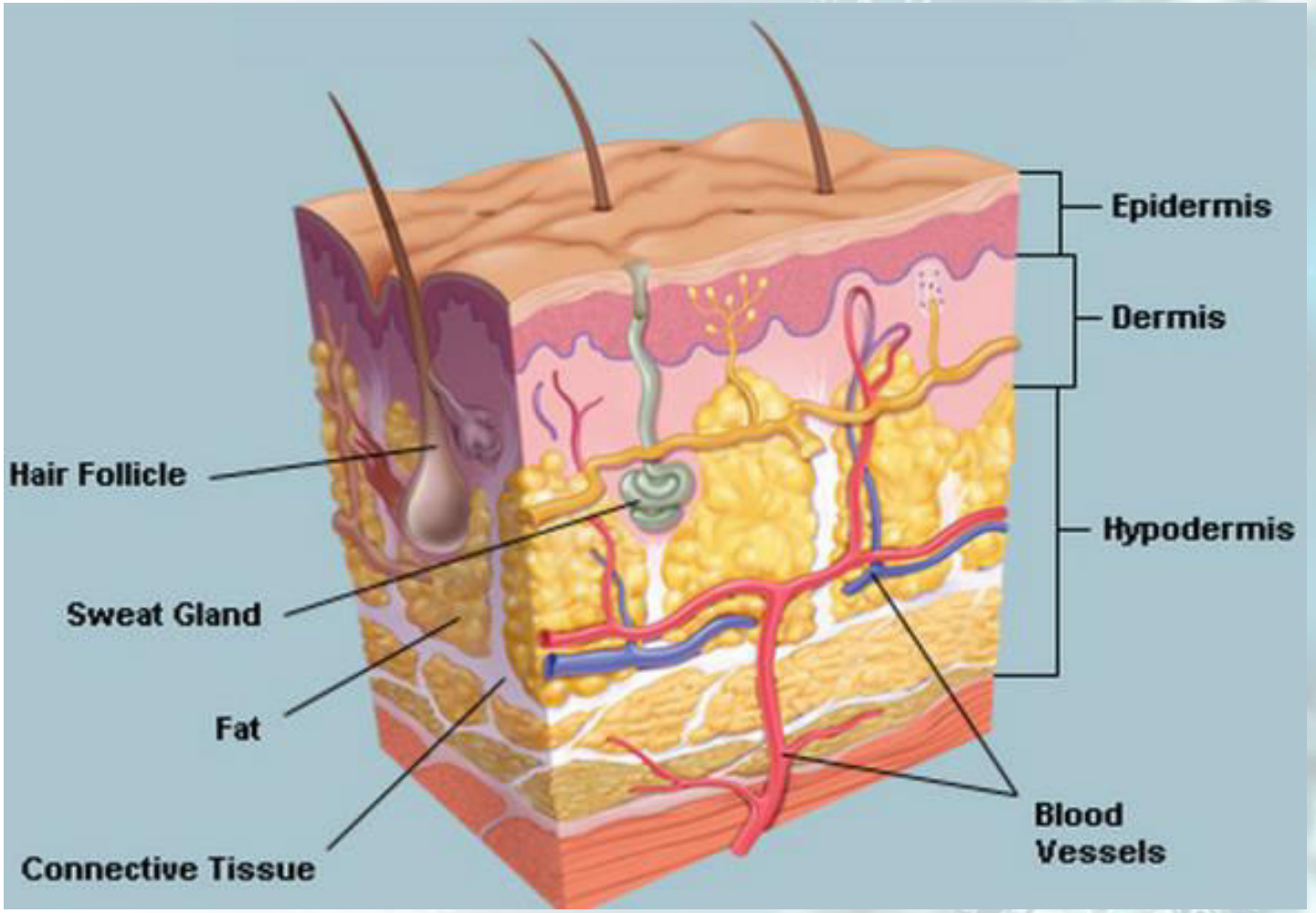
Innate Immunity II

1st Line of Defense

- **Physical Factors:**
  - Urine
  - Earwax
  - Peristalsis
  - Vomiting
  - Diarrhea
  - Defecation

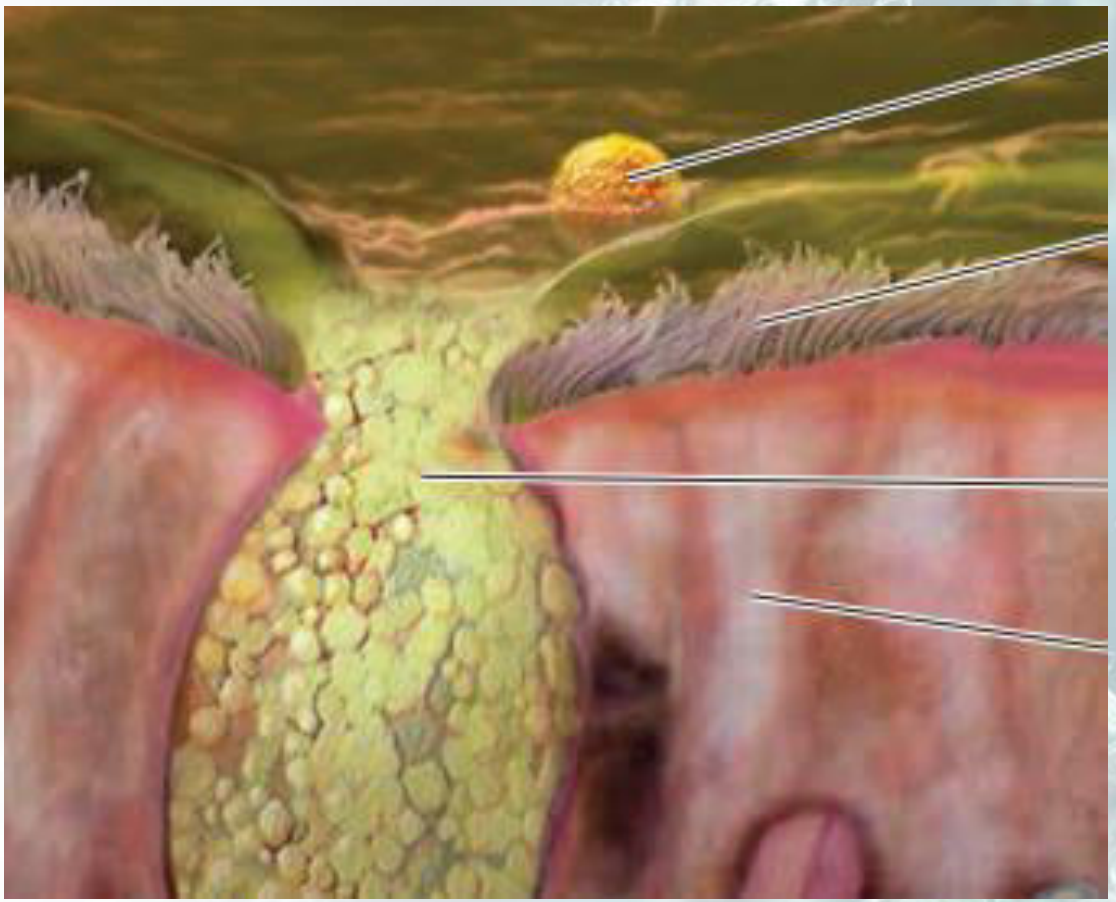
Innate Immunity II

Skin



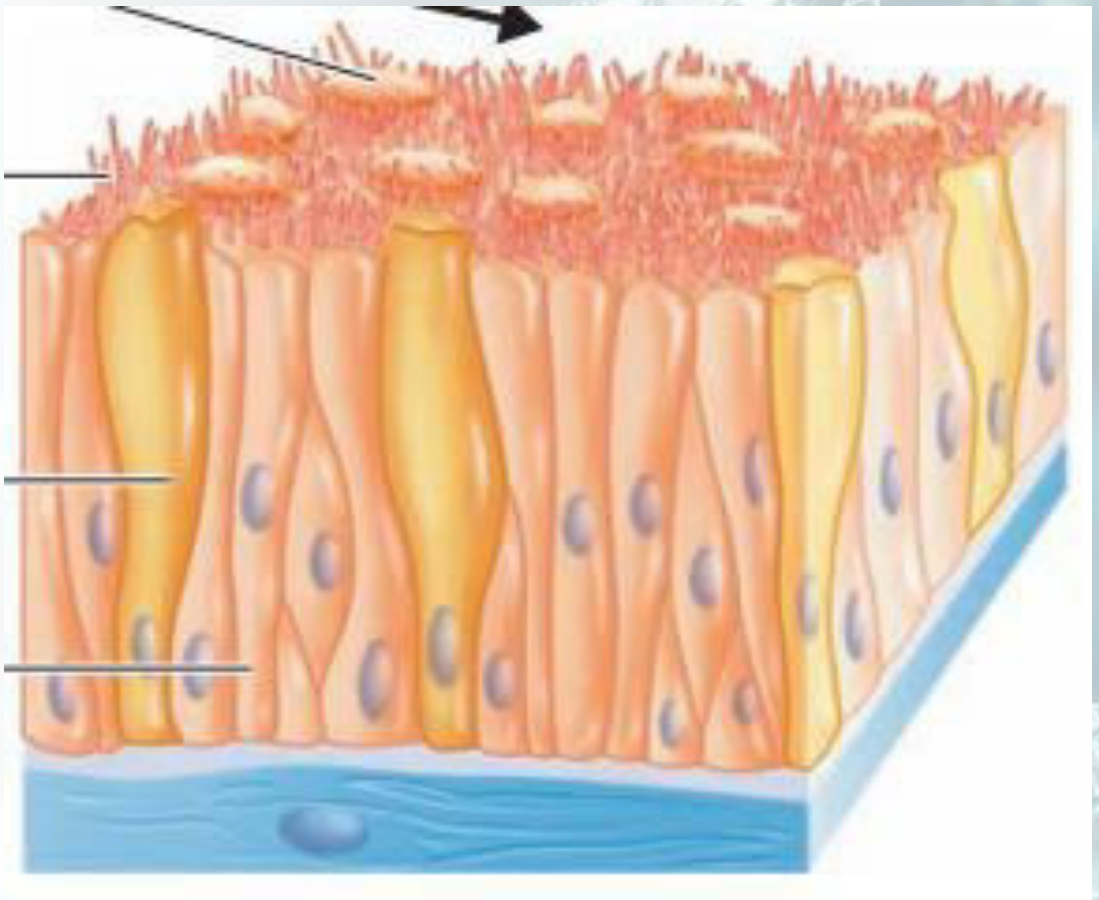
Innate Immunity II

Mucus



Innate Immunity II

Cilia



Innate Immunity II

1st Line of Defense

- **Chemical Factors:**
  - Cebum
  - pH of the skin (3-5)
  - Lysozyme in sweat, tears, saliva, body fluids and urine
  - Acidic pH of stomach

Innate Immunity II

2nd Line of Defense

- Leukocytes
  - Granulocytes
    - Neutrophils
    - Eosinophils
    - Basophils

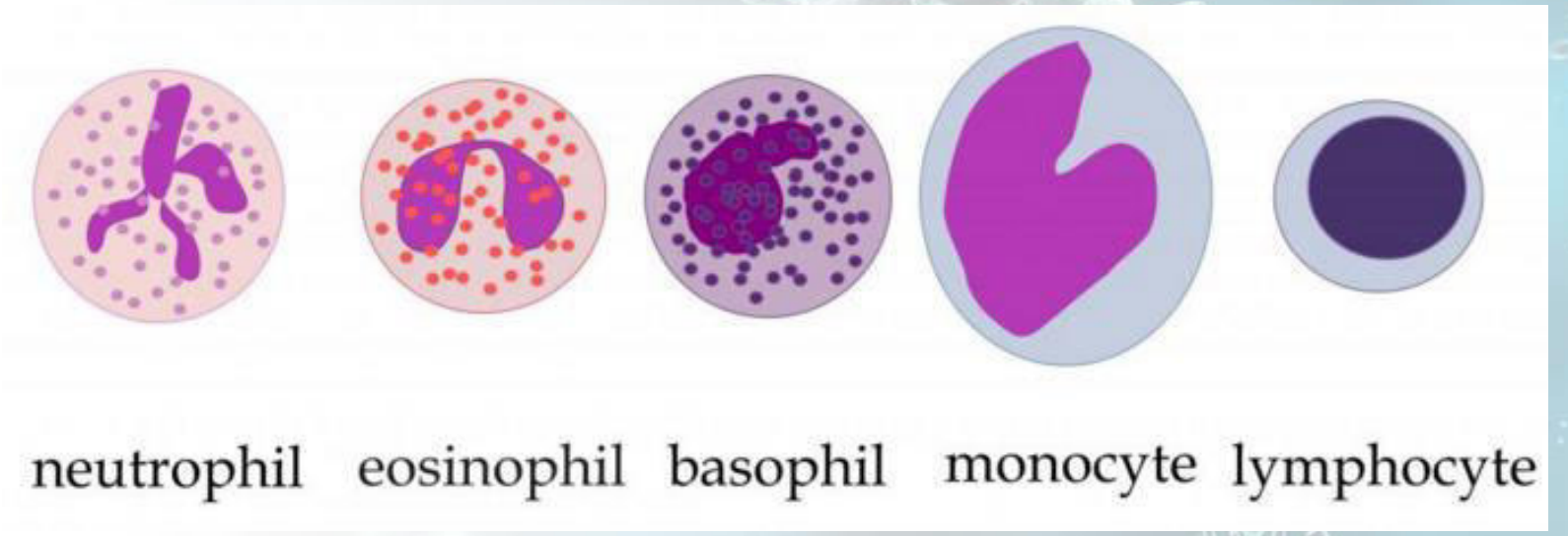
Innate Immunity II

2nd Line of Defense

- Leukocytes
  - Agranulocytes
    - Monocytes
    - Lymphocytes
      - Natural killer
      - B cell
      - T cell

Innate Immunity II

Leukocytes



Innate Immunity II

Summary

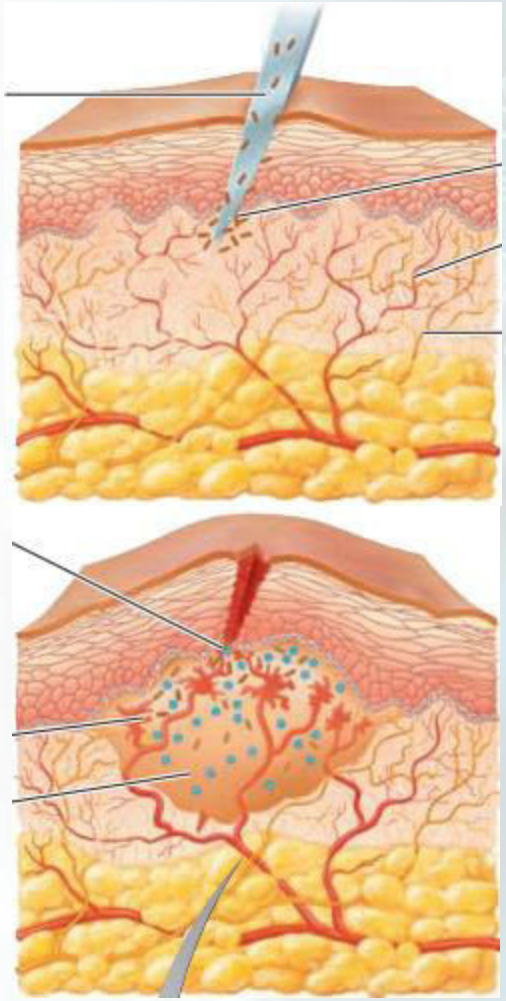
- Various cells of innate immunity
- Some secretions or molecules of innate immunity

Innate Immunity III

Learning Objectives

- Some secretions of innate immunity

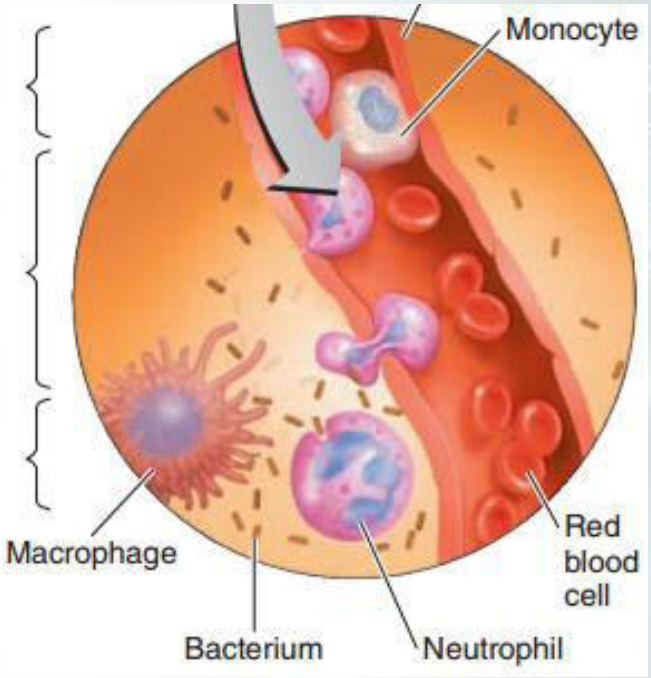
Innate Immunity III



Inflammation

- Damage to body tissues triggers a local defensive response
  - Redness
  - Pain
  - Swelling
  - Heat

Innate Immunity III



Inflammation

- Histamine
- Kinins
- Prostaglandins by damaged cells
- Leukotrienes by mast cells

Innate Immunity III

Inflammation

- Cytokines are released (TNF-alpha)
  - Causes release of acute phase proteins from liver.

Innate Immunity III

Fever

- Systemic response
- Controlled by hypothalmus
- Pyrogens by microbes
- IL-1, TNF alpha
- Growth of microbes arrested

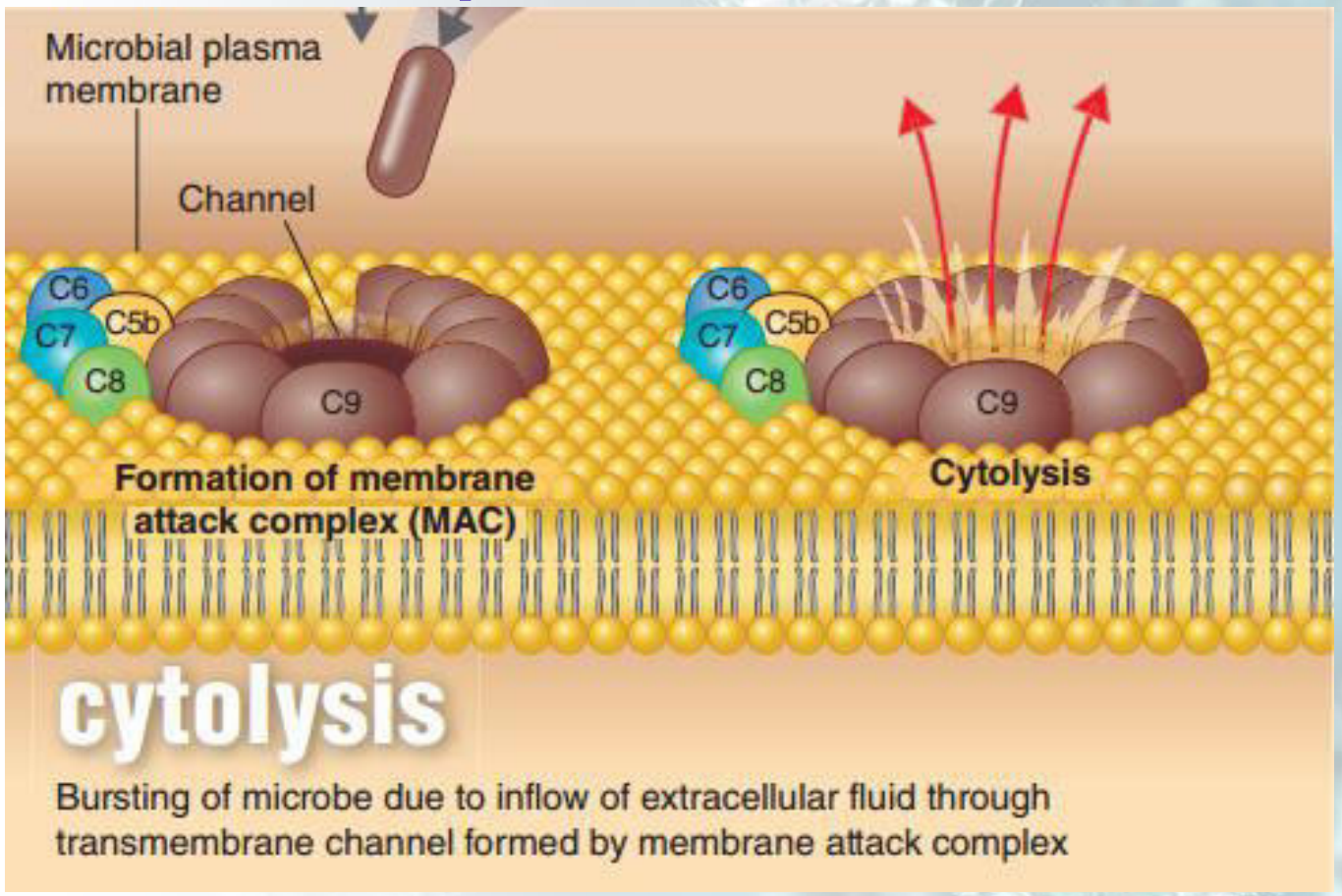
Innate Immunity III

Complement System

- More than 30 difference proteins make complement system
- It complements the cells of the immune system in destroying microbes.

Innate Immunity III

Complement System



Innate Immunity III

Interferons

- Antiviral proteins
- Produced by macrophages and lymphocytes in response to viral infections (fibroblasts)
- Interfere with viral replication

Innate Immunity III

Interferons

- Species specific
- Not viral specific
- Types
  - Alpha (infected cells)
  - Beta (infected)
  - Gamma (lymphocytes)

Innate Immunity III

Antimicrobial Peptides

- 12 - 50 amino acid long peptides
- Defensins by neutrophils
- Dermicidin by sweat glands
- Thrombocidins by platelets

Innate Immunity III

Summary

- Secretions of innate immunity
- Cytokines
- Complement
- Interferones

Adaptive Immunity I

Learning Objectives

- How adaptive immune response was discovered
- What constitute adaptive immune response

Adaptive Immunity I

Adaptive Immunity

- 1st line of defense
  - Skin
- 2nd line of defense
  - Macrophages, neutrophils
- 3rd line of defense
  - B and T cells

Adaptive Immunity I

Adaptive Immunity

- Composed of specialized cells
- B cells: humoral immunity
- T cells: cellular immunity
- This immunity is induced.

Adaptive Immunity I





Adaptive Immunity

- **Observations: Individuals once recovered from smallpox, measles, or chickenpox become immune to it.**
- **Experiments led to antibodies discovery in serum and vaccination.**

Adaptive Immunity I

Immunity Transfer Experiment



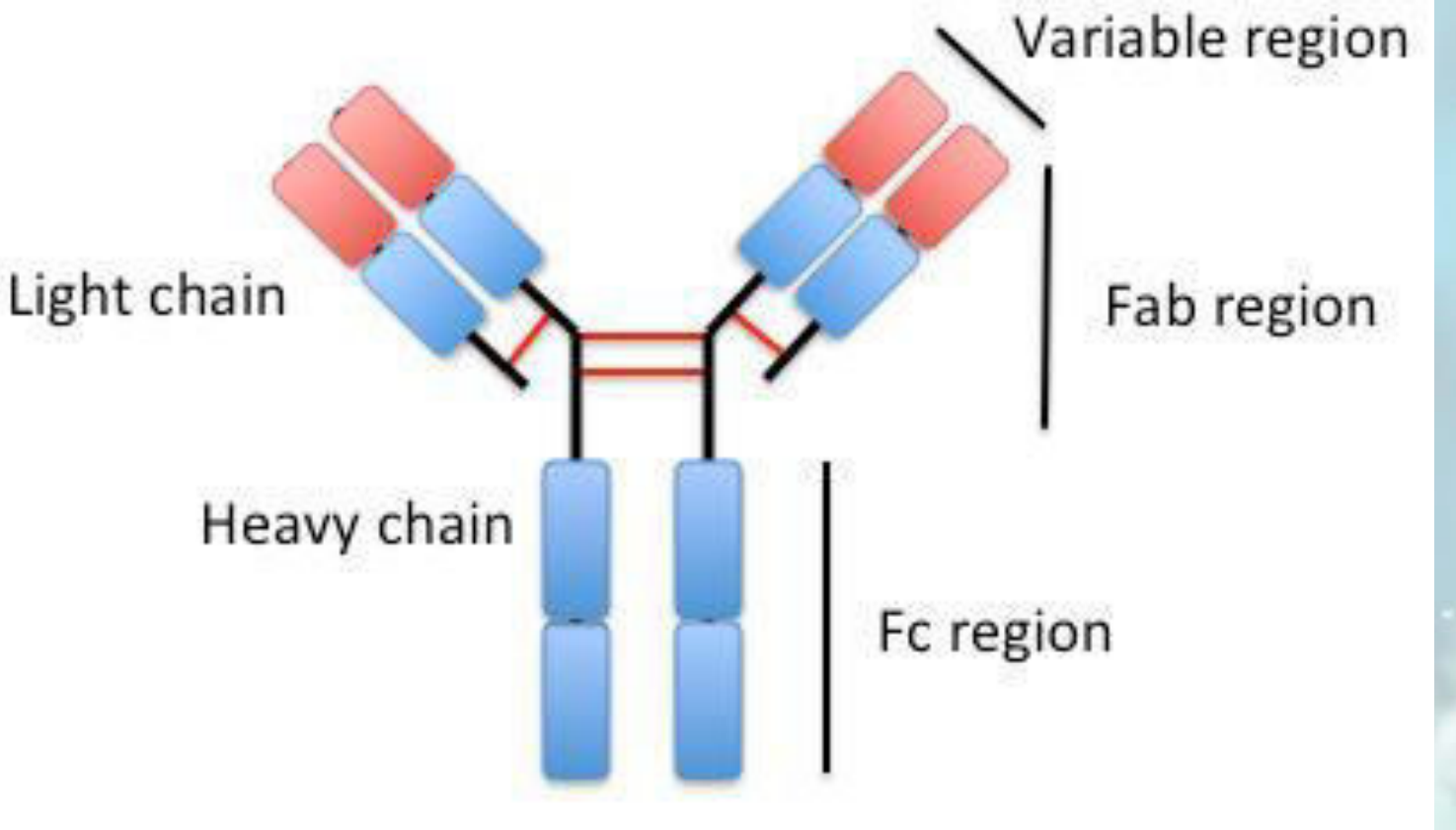






Adaptive Immunity I

Antibodies by B cells



Adaptive Immunity I

Thymus and skin grafting



Adaptive Immunity I

Adaptive Immunity

- T cells were discovered
- Responsible for cellular immunity

Adaptive Immunity I

Summary

- B cells responsible for humoral immunity
- T cells, for cellular immunity

Adaptive Immunity II

Learning Objectives

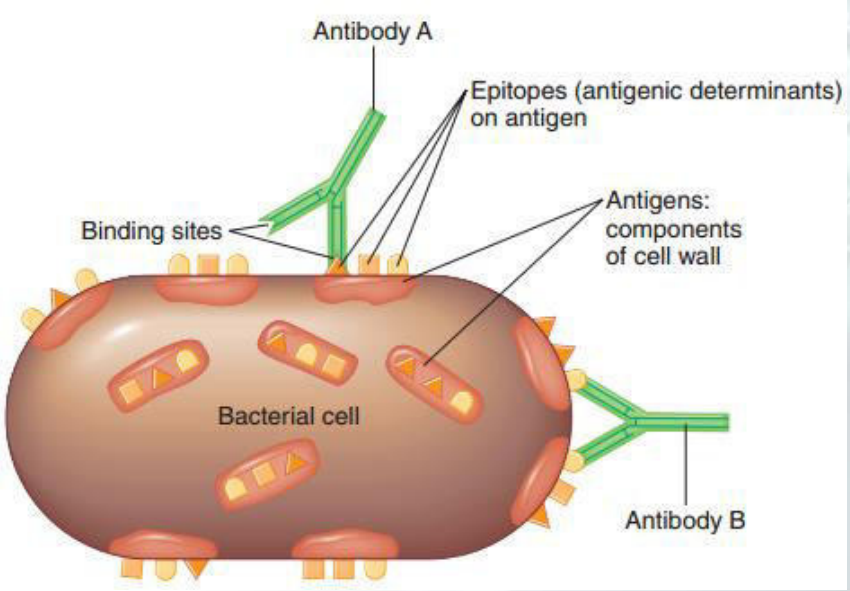
- What are antigens and their epitopes
- What are antibodies and classes of antibodies

Adaptive Immunity II

Adaptive Immunity

- **Antigen:** A foreign substance that provokes immunity
- More specifically called as Immunogen
- Protein in nature
- Could be large polysaccharides

Adaptive Immunity II



Adaptive Immunity

- **Epitopes:** Specific regions on an antigen
  - Also called as antigenic determinants
- Most antigens have MW >10,000 D

Adaptive Immunity II

Adaptive Immunity

- < 10,000 D molecules are poor antigens
  - Haptens
- Need a carrier proteins to make antibodies
- Once Abs are formed, they can bind hapten.
  - Penicillin

Adaptive Immunity II

Antigen Characteristics

- High MW
- Complex in nature
- Protein are better
- Foreign to the body
- Degradable

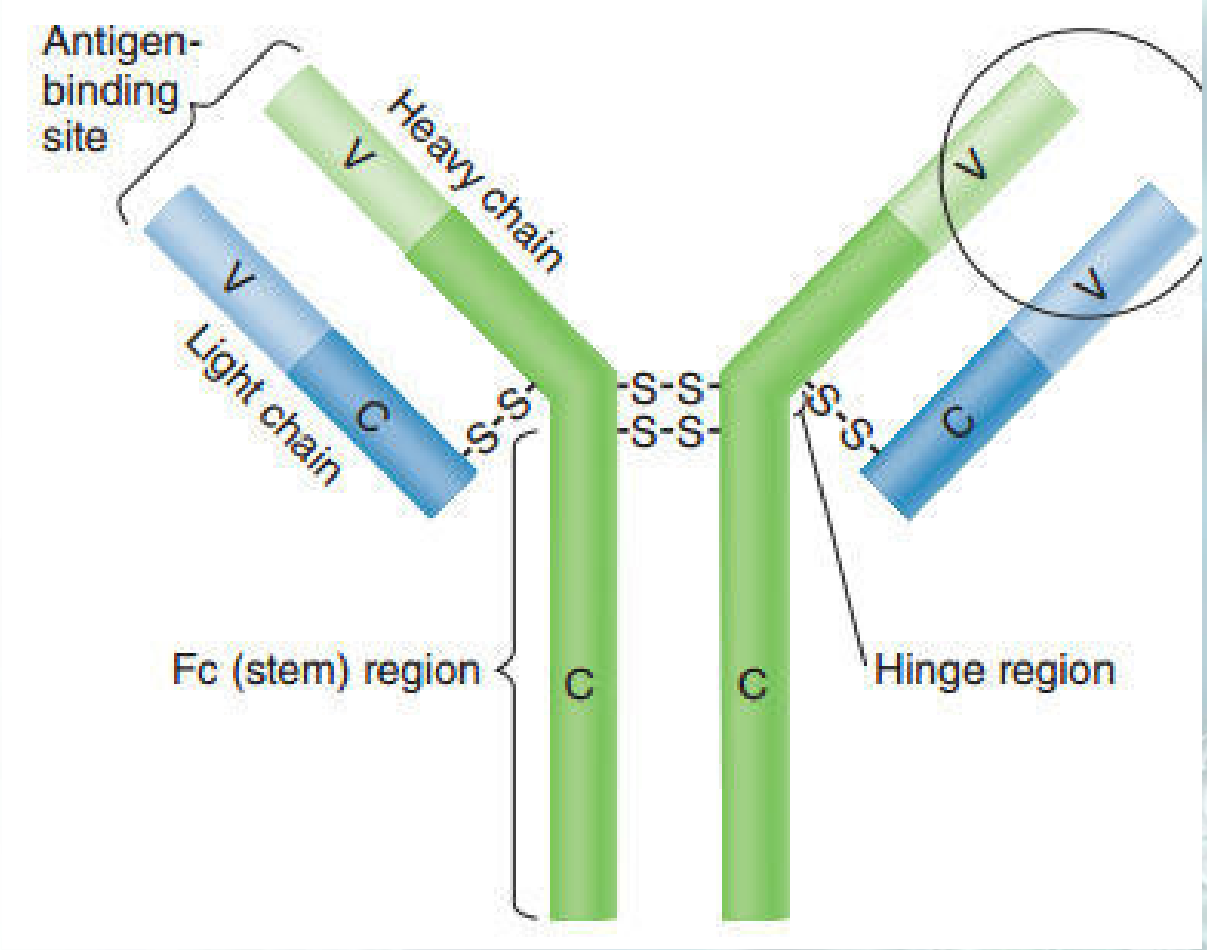
Adaptive Immunity II

Antibodies

- Glycoproteins in nature
- Present in gamma globulins of serum
- IgM: blood
- IgG: serum and body secretions
- IgA (mucus membrane)
- IgE (allergy) and IgD

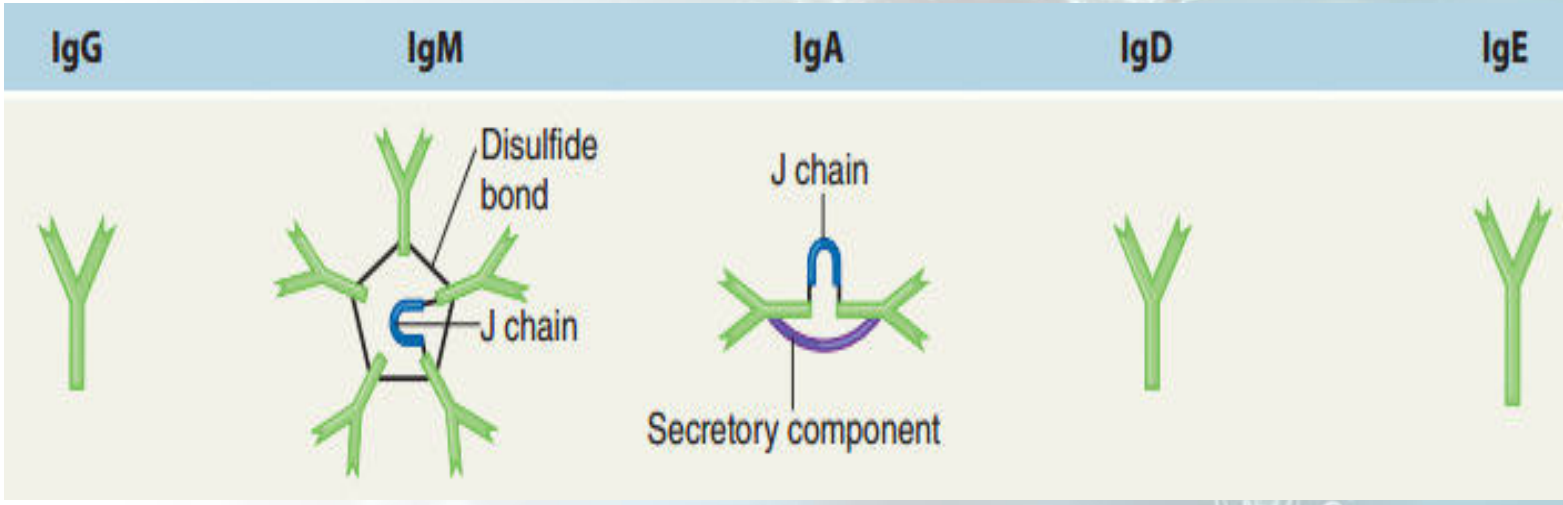
Adaptive Immunity II

Antibodies



Adaptive Immunity II

Antibodies



Adaptive Immunity II

Summary

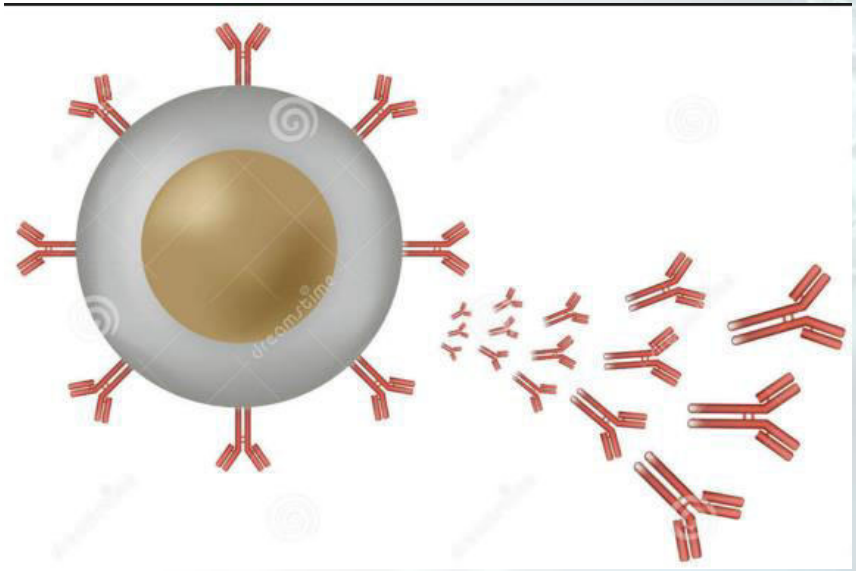
- Proteins are good antigens
- They have antigenic determinants
- Antibodies are glycoproteins
- Five classes of antibodies

Adaptive Immunity III

Learning Objectives

- Clonal selection theory
- How antigen selects a B cell for making antibodies

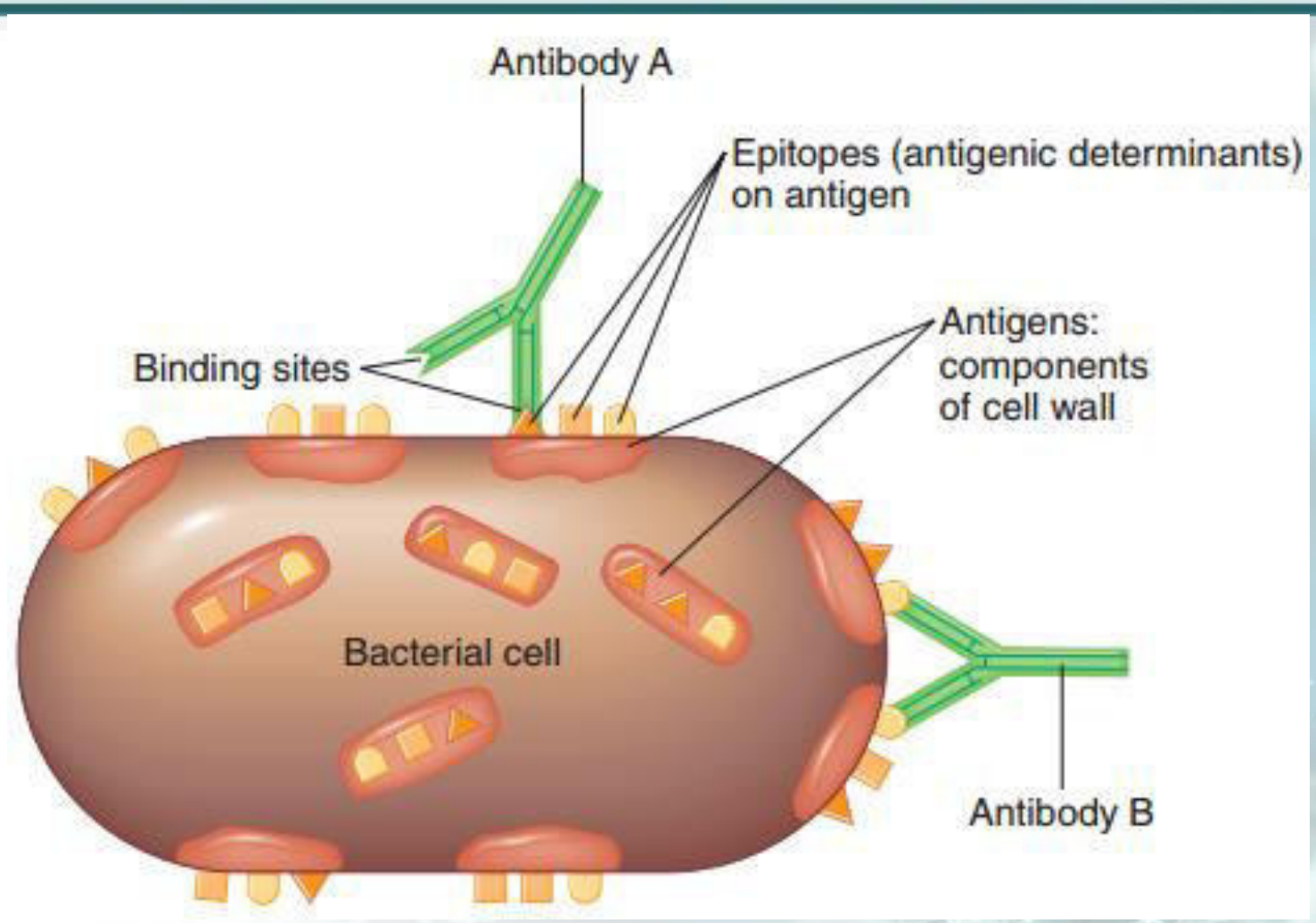
Adaptive Immunity III



Adaptive Immunity

- **B cells:** Each B cell has unique antibody molecules against one epitope only.
- B cell never changes its specificity.
- May change its class

Epitopes on an Antigen



Adaptive Immunity III

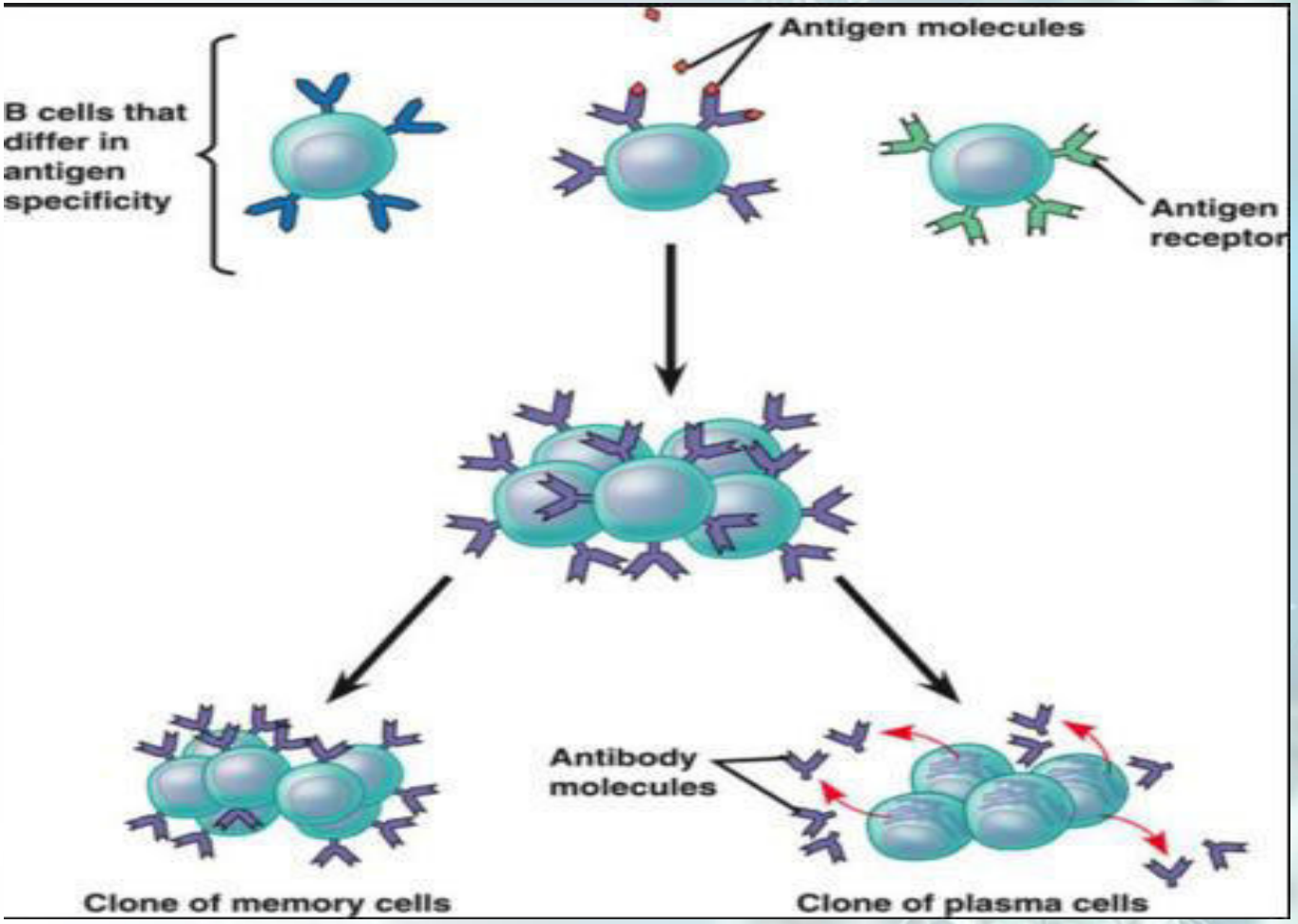


Adaptive Immunity

- **Important:** B cells in bone marrow create antibodies without even seeing the antigens.
- It is like choosing a shoe or a shirt from a shop!

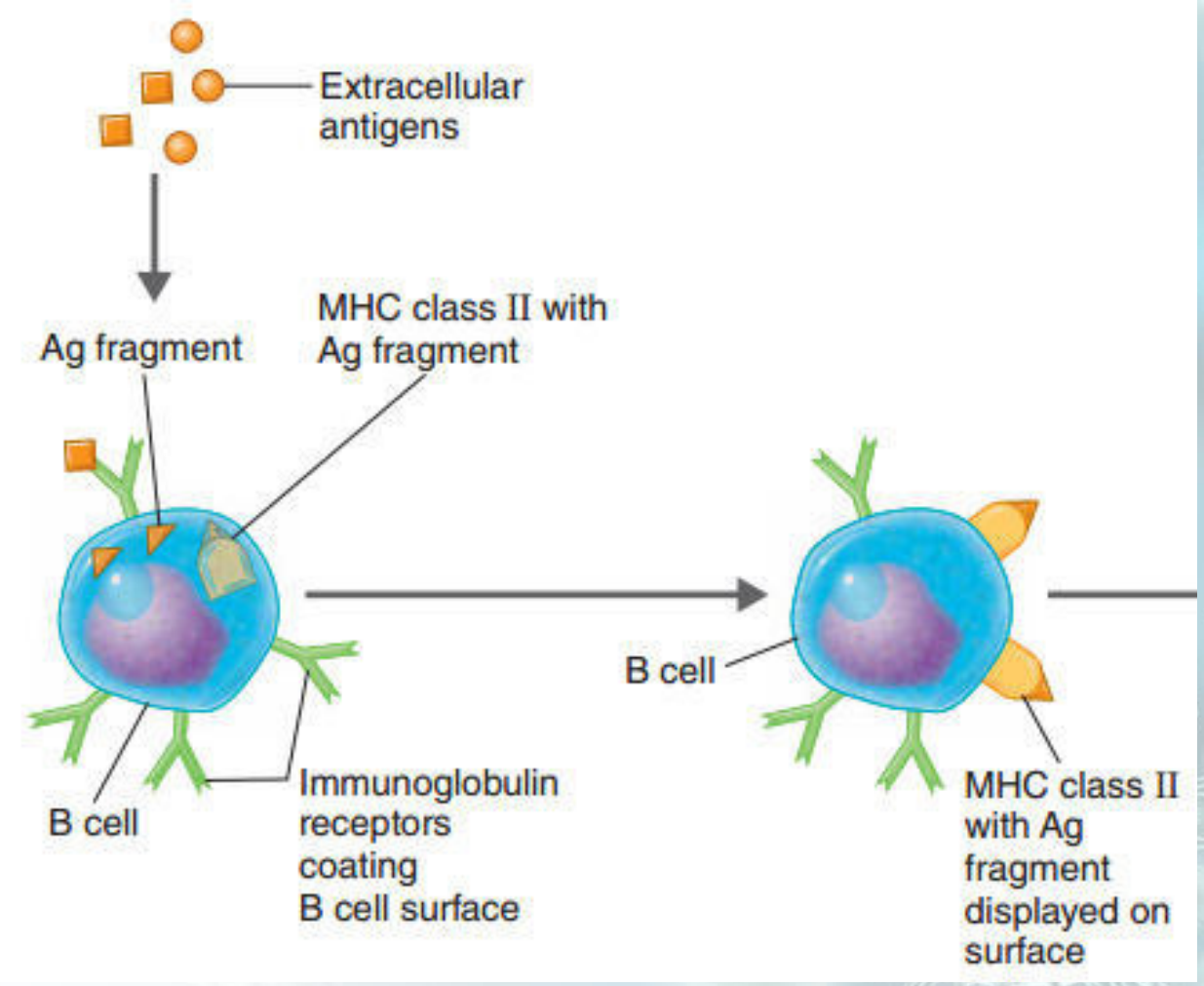
Adaptive Immunity III

Clonal Selection



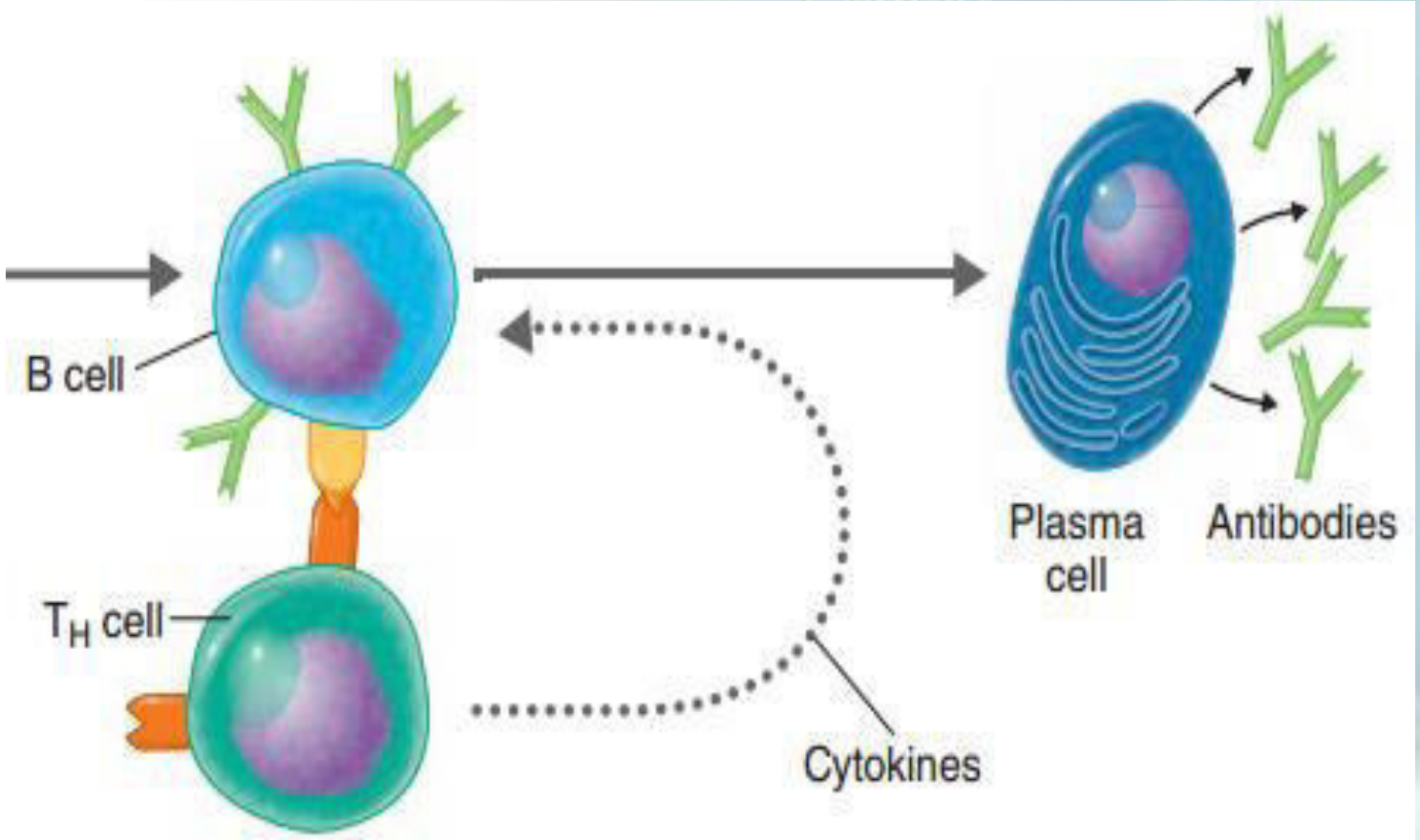
Adaptive Immunity III

Clonal Selection



Adaptive Immunity III

Clonal Selection



Adaptive Immunity III

Summary

- Clonal selection
- B cells are selected by the antigen to secrete antibodies

Adaptive Immunity IV

Learning Objectives

- How antibodies are used as a defense mechanism by the body against microbes (antigens)

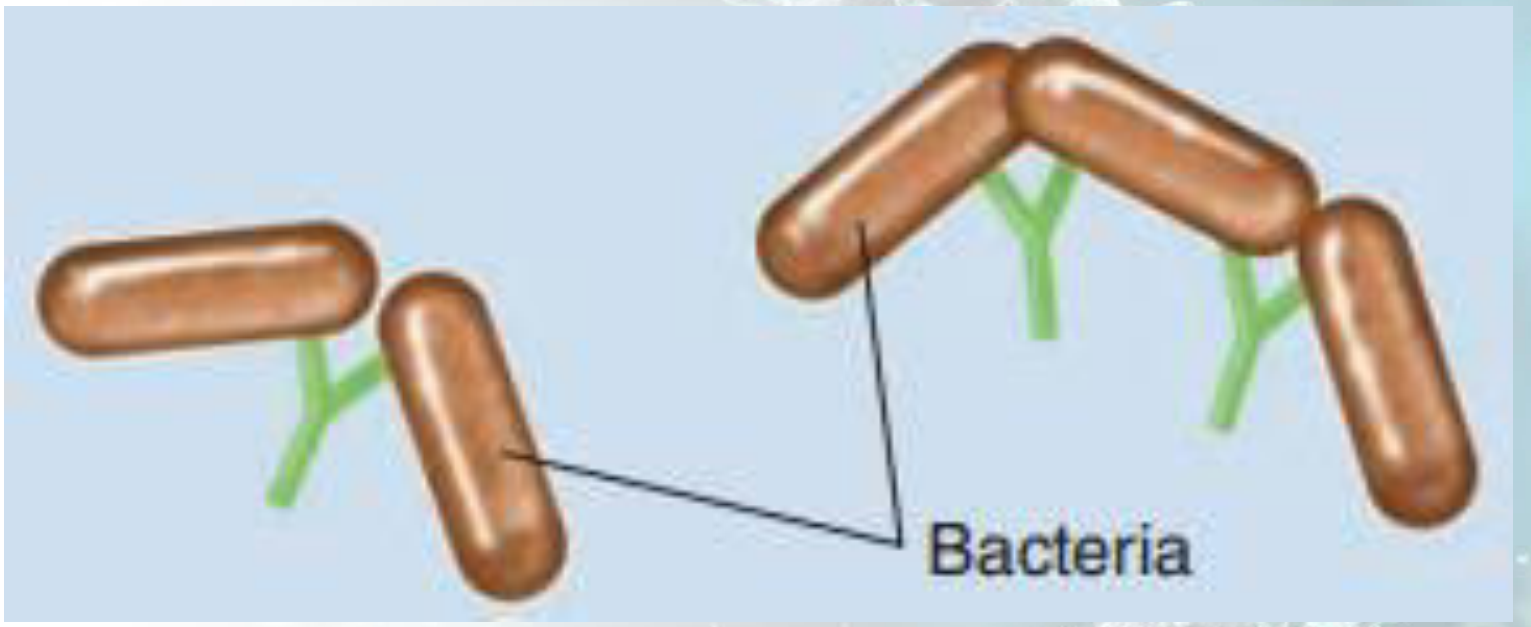
Adaptive Immunity IV

Ag-Ab Binding

- Agglutination
- Opsonization
- Complement Activation
- Ab-dependent cell-mediated cytotoxicity
- Neutralization

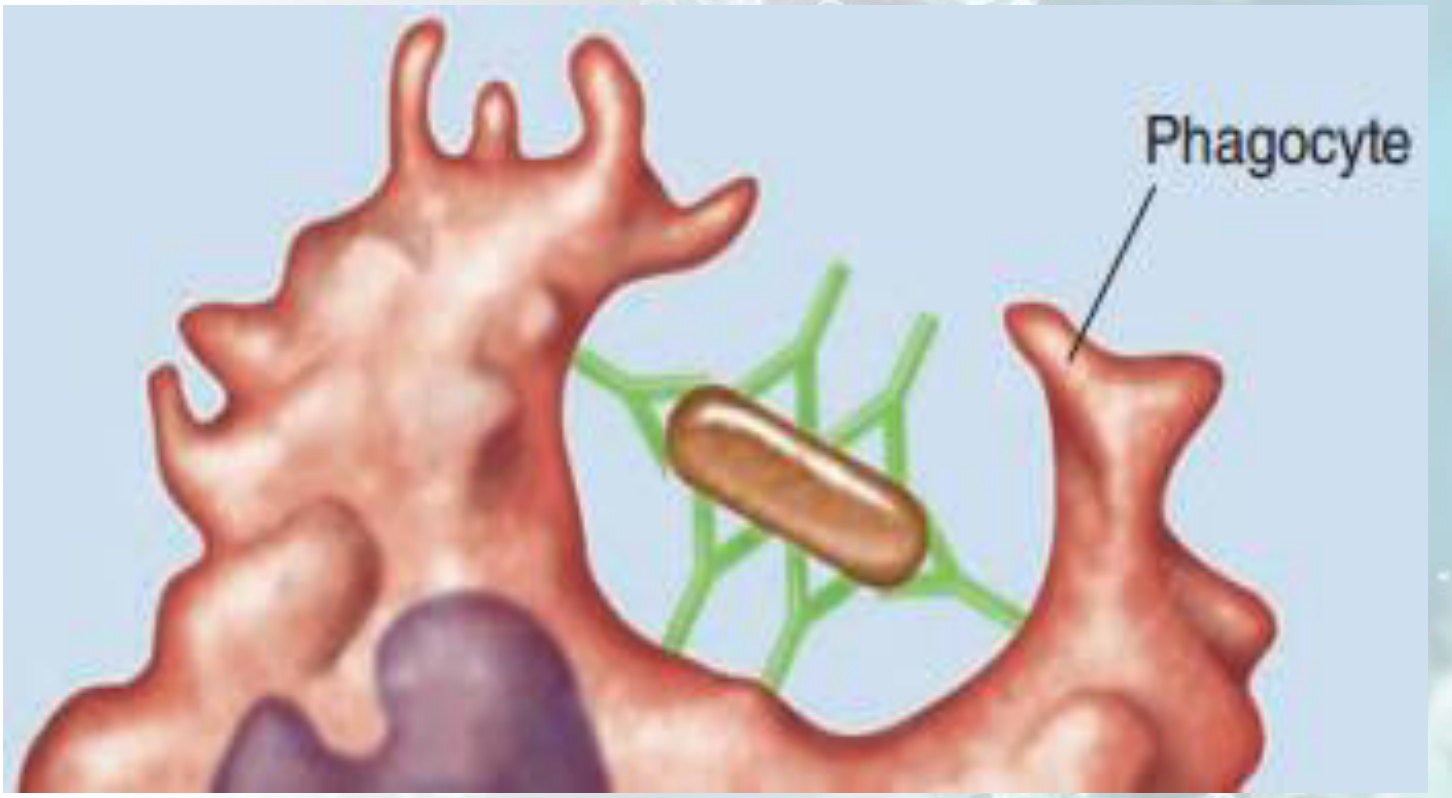
Adaptive Immunity IV

Agglutination



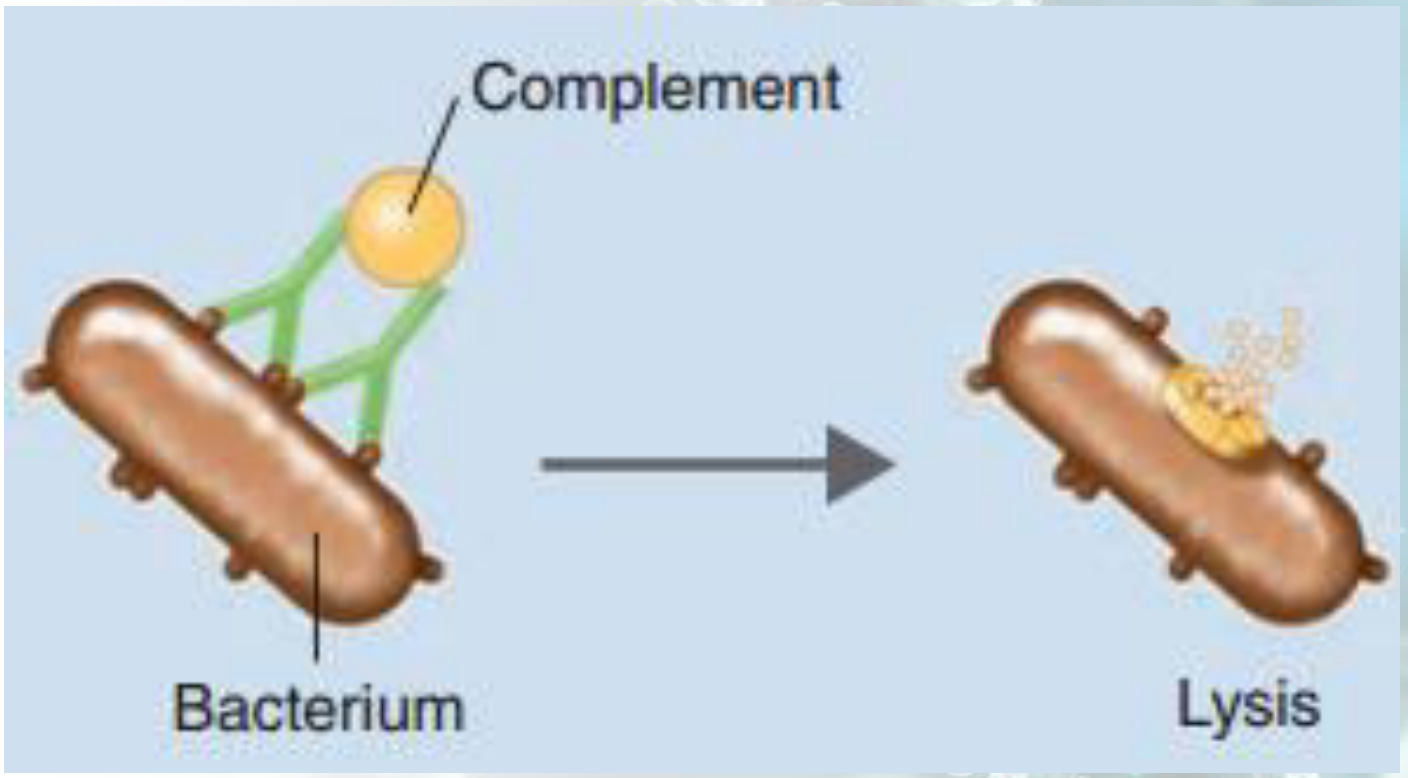
Adaptive Immunity IV

Opsonization



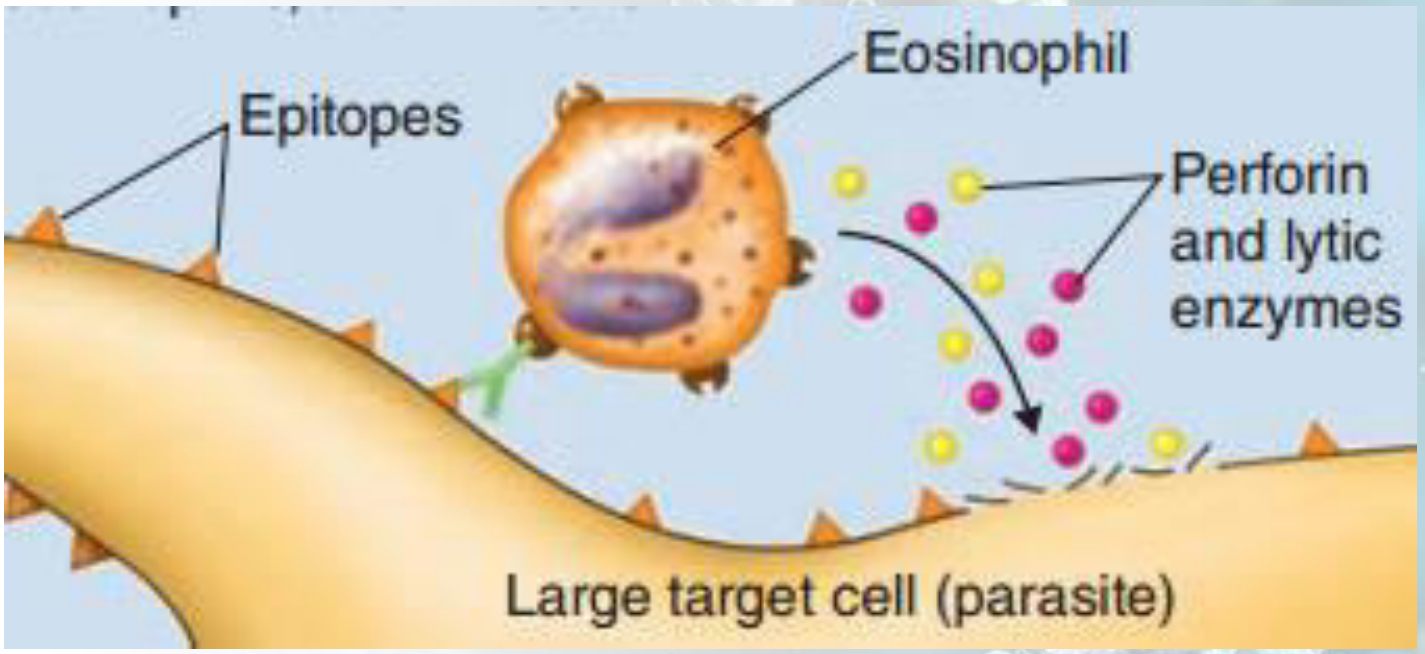
Adaptive Immunity IV

Complement Activation



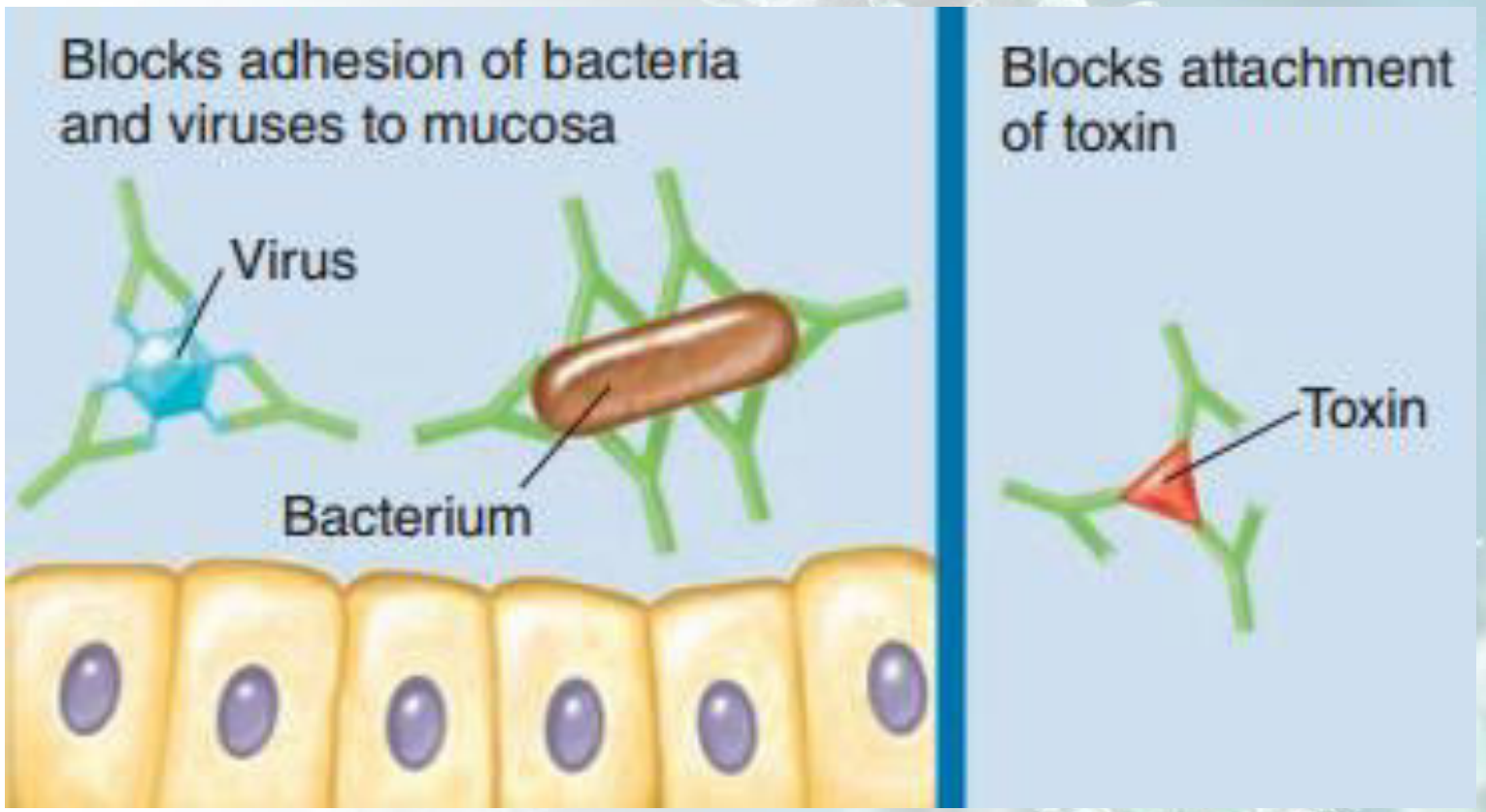
Adaptive Immunity IV

Ab-Dependent Cell-mediated Cytotoxicity



Adaptive Immunity IV

Neutralization



Adaptive Immunity IV

Summary

- Antibodies are used by 5 methods to protect the body from microbes etc.

Adaptive Immunity V

Learning Objectives

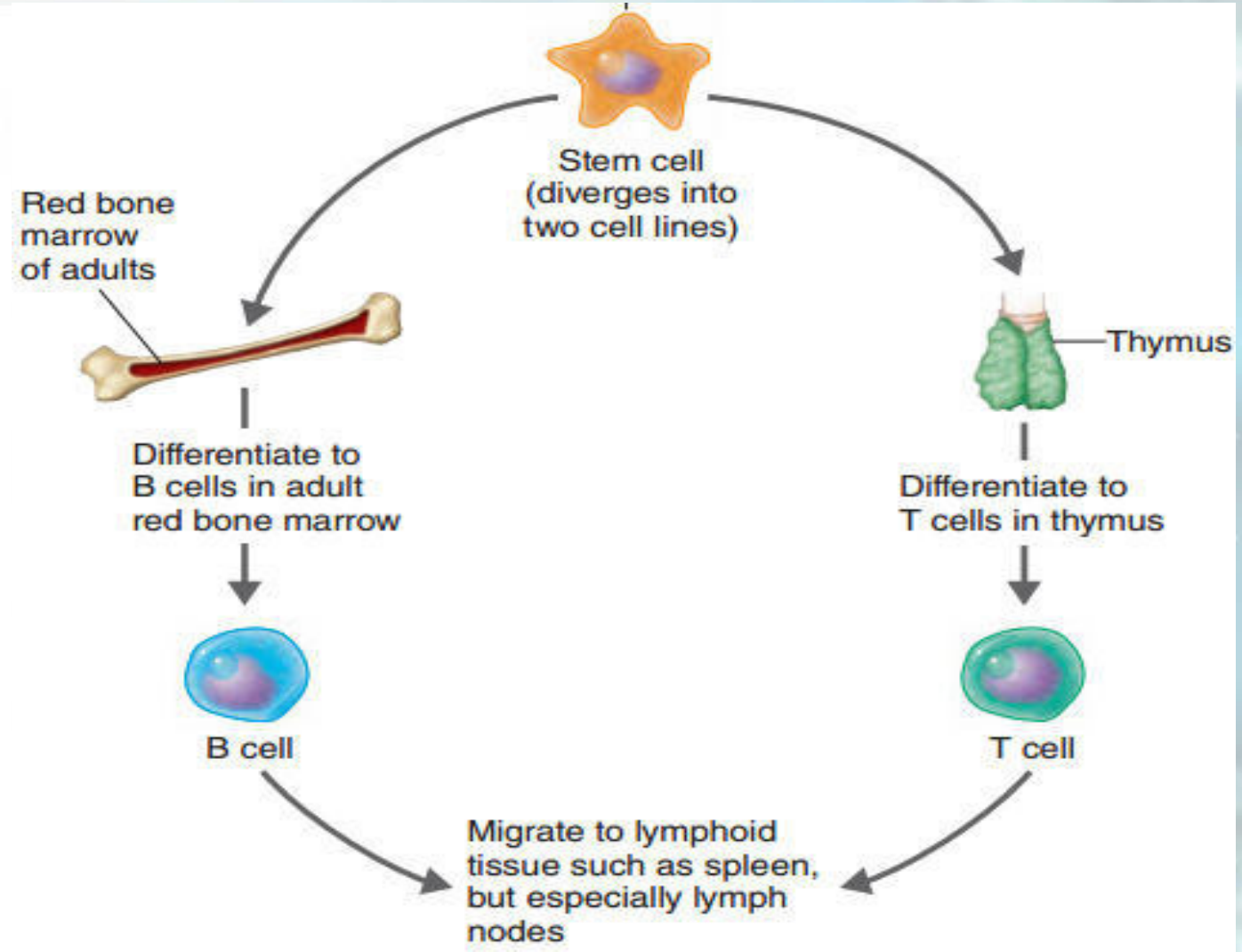
- What constitutes cellular immune response
- How T cells protect the body from antigens that originate endogenously in the body.

Adaptive Immunity V

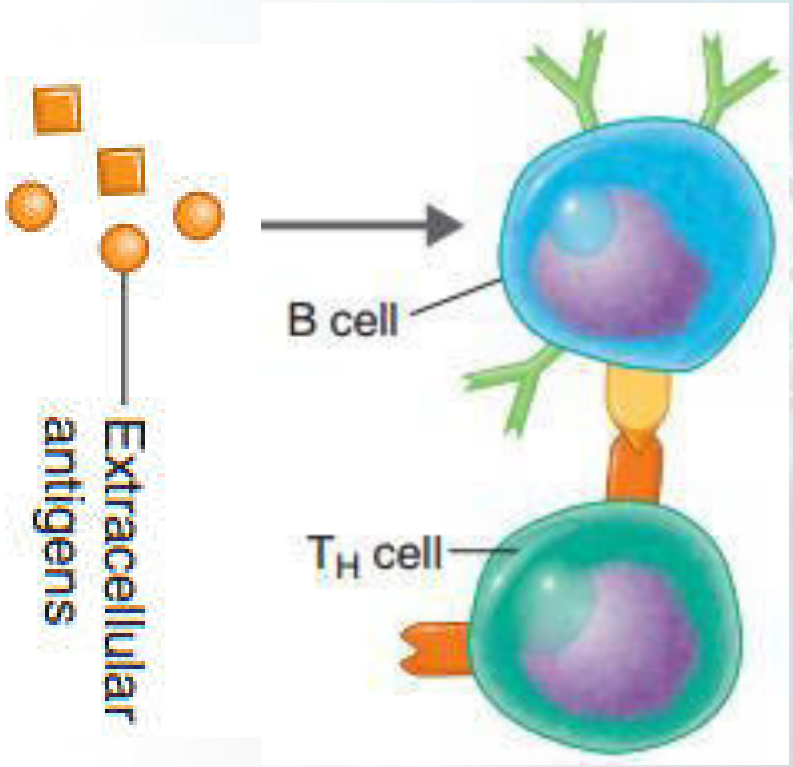
Learning Objectives

- What are APCs
- MHC molecules

Adaptive Immunity V



Adaptive Immunity V



T cells

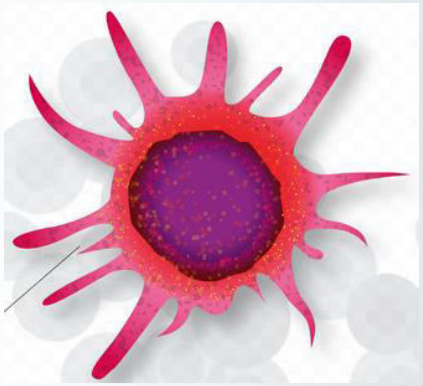
- MHC restricted
- Recognize Ag only when the Ag is associated with MHC molecule.
- Do not attach to free antigens.
- Have TCRs

Adaptive Immunity V

MHC Molecules

- MHC I: Present on all nucleated cells
- MHC II: Present only on APCs
- Antigen Presenting Cells

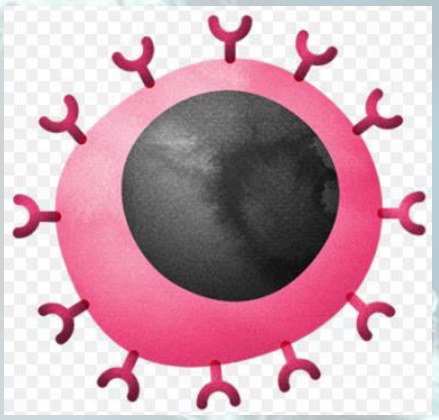
Adaptive Immunity V



APCs

- Dendritic cells
- Macrophages





- B cells

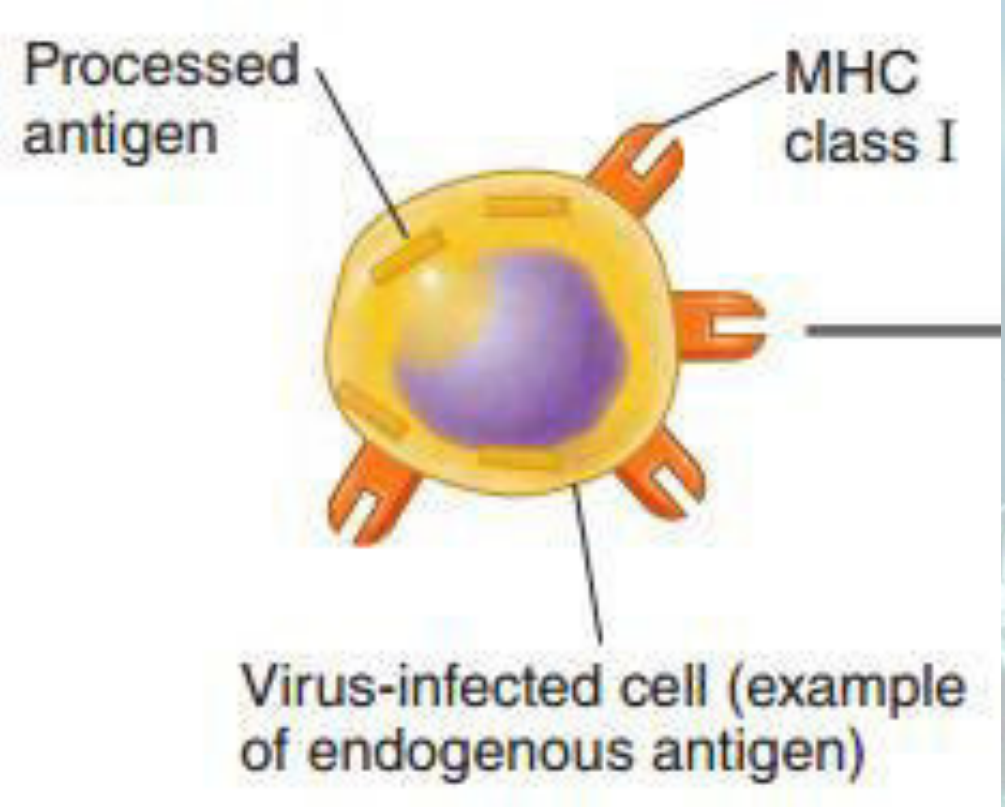
Adaptive Immunity V

T cells

- T helper cells
  - TH1 ($CD4^+$)
  - TH2($CD4^+$)
- T cytotoxic cells ($CD8^+$)

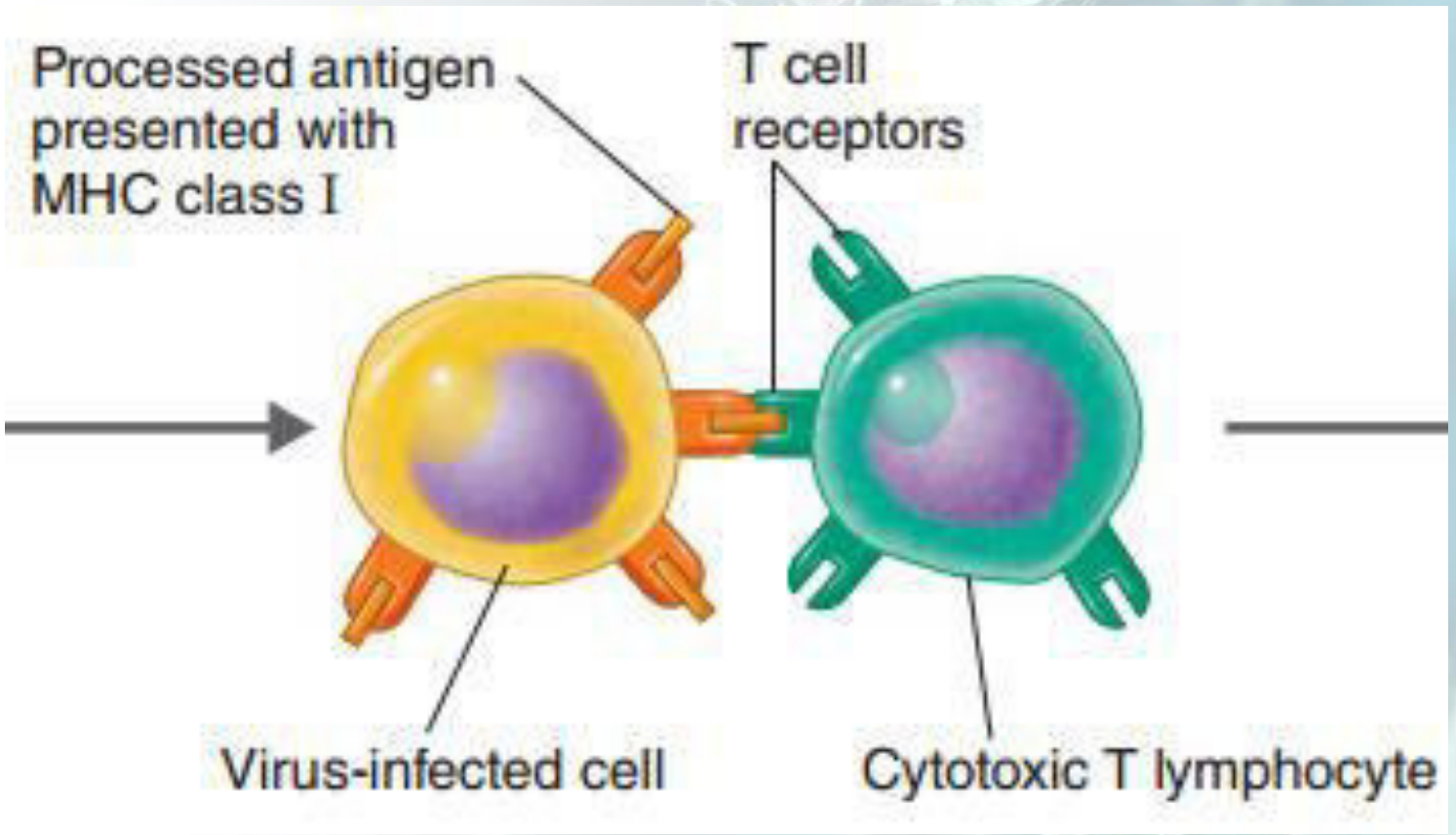
Adaptive Immunity V

Virus Infected Cells



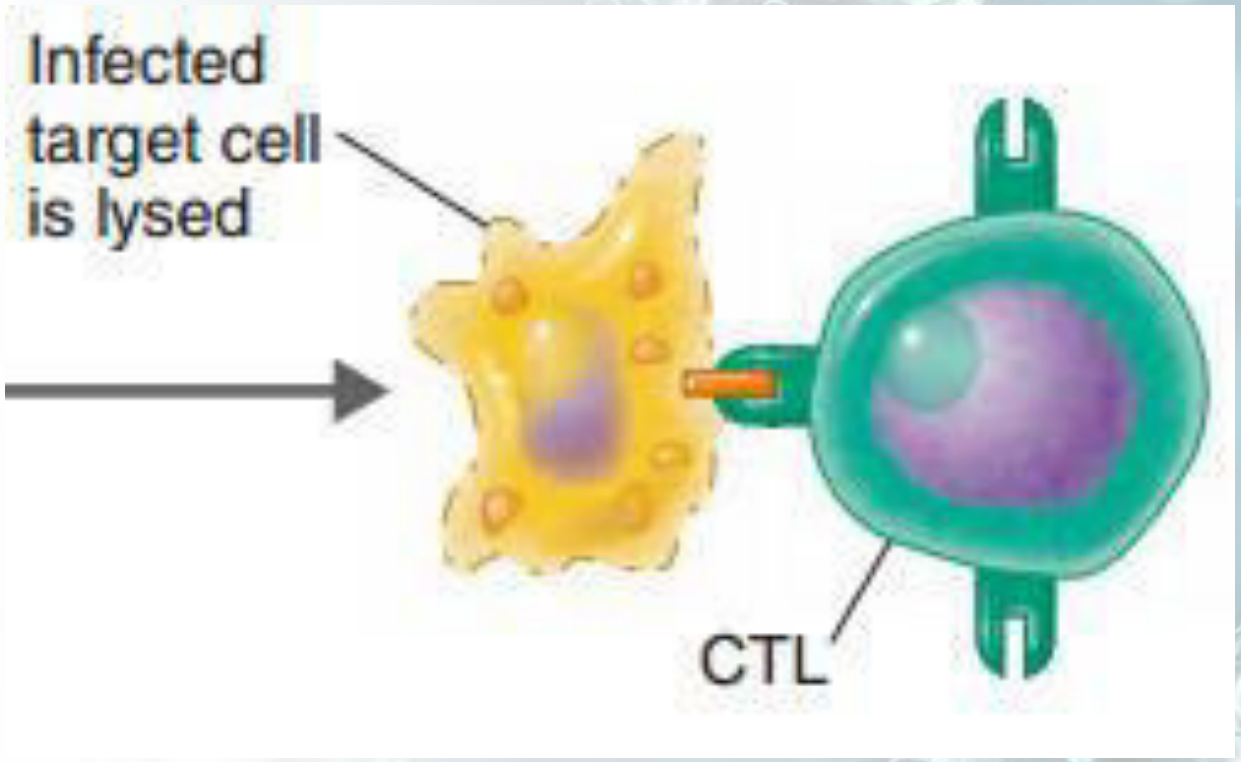
Adaptive Immunity V

Cytotoxic T cells



Adaptive Immunity V

Cytotoxic T cells



Adaptive Immunity V

Summary

- T cells are responsible for cellular immunity.
- They eliminate endogenous antigens.
- APC are MHC II cells

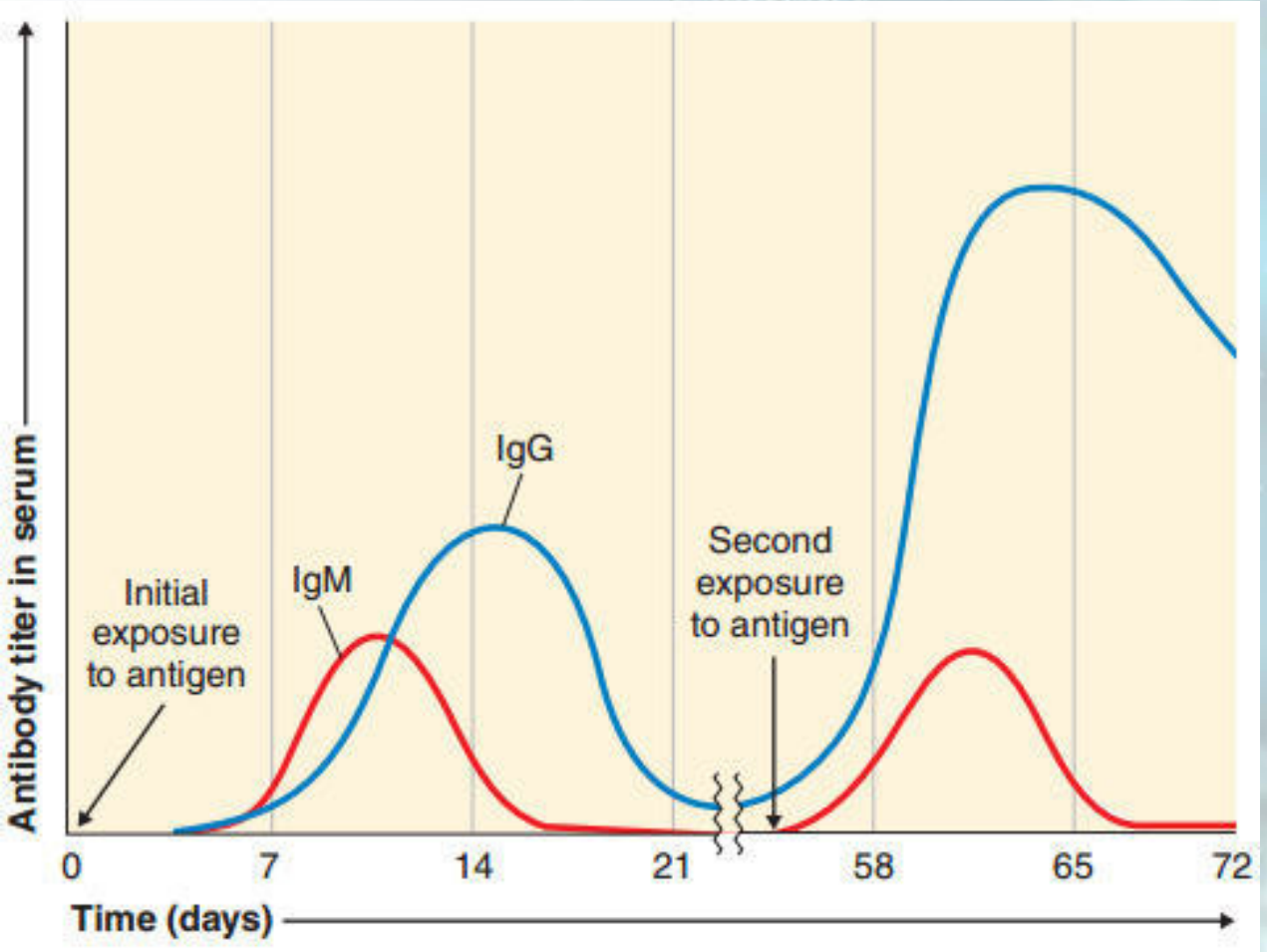
Adaptive Immunity VI

Learning Objectives

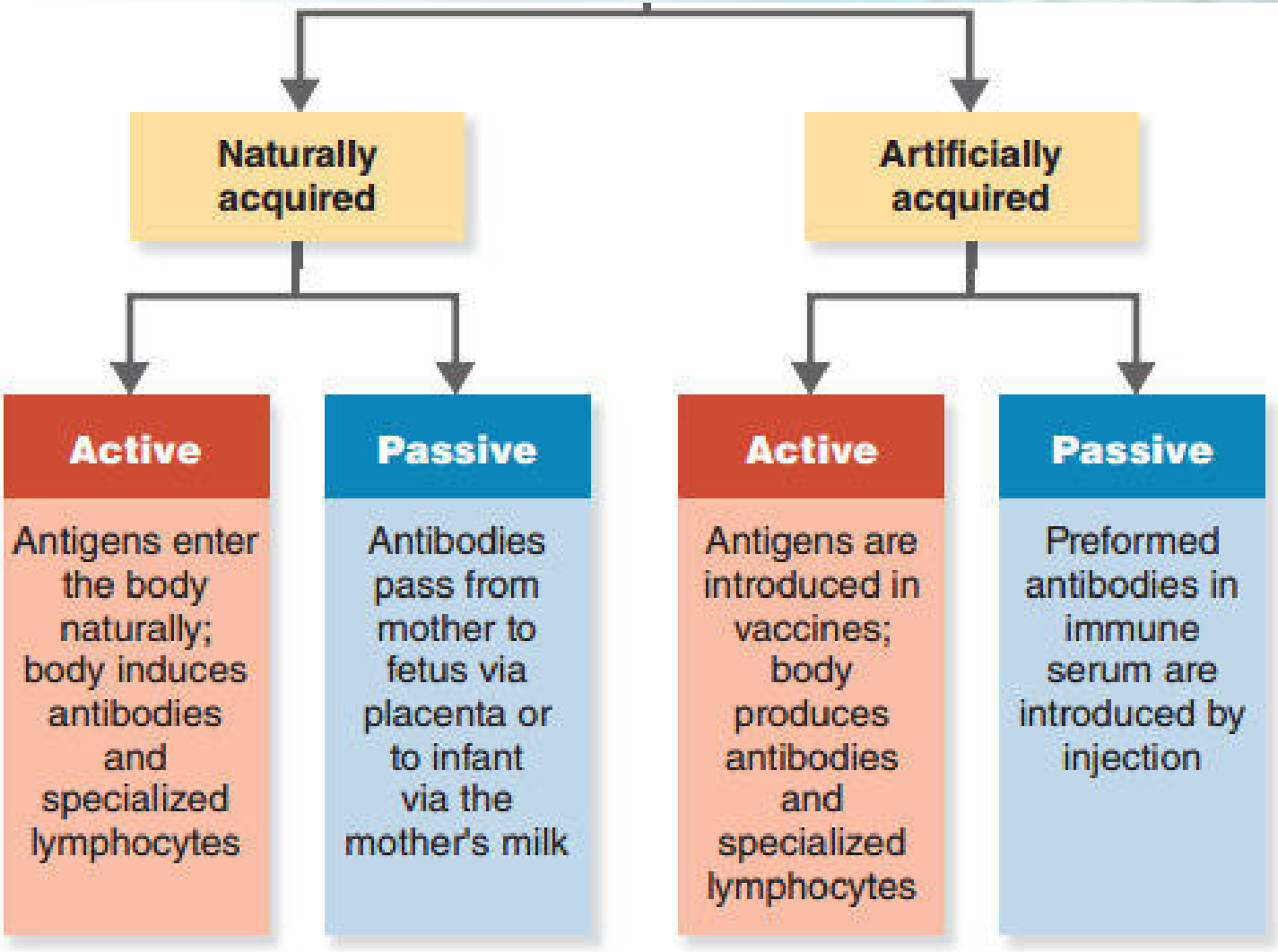
- What is a primary and a secondary immune response
- Active and passive immunity
- Summary of immune responses

Adaptive Immunity VI

Primary and Secondary Immune Response

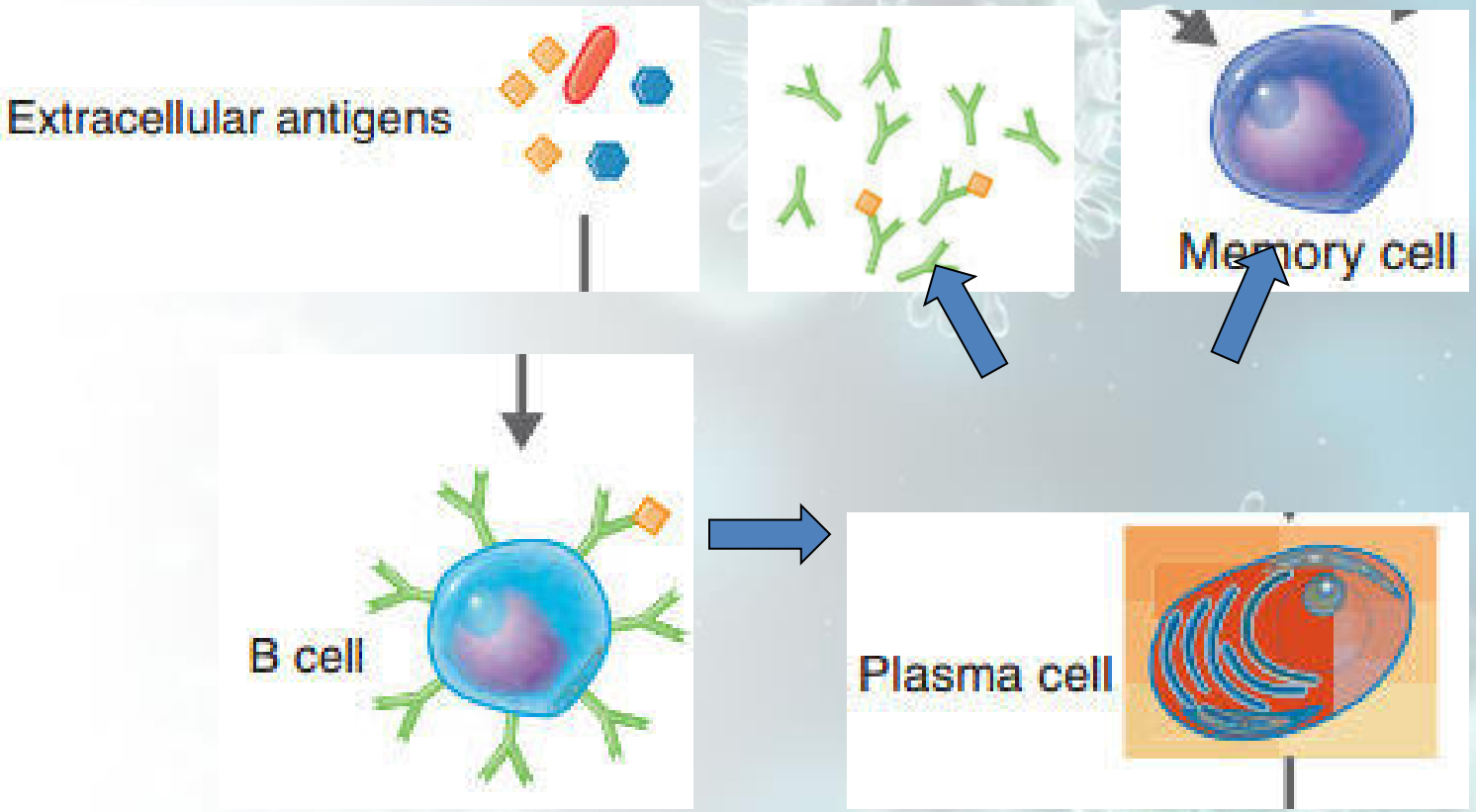


Adaptive Immunity VI



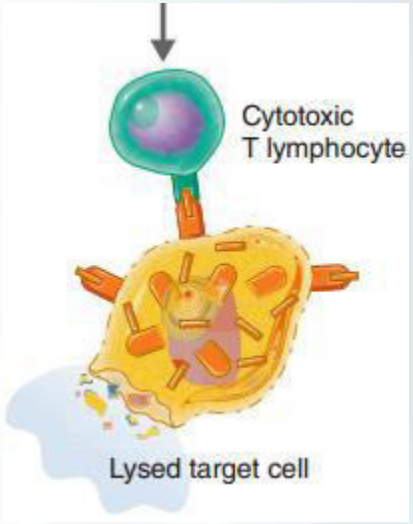
Adaptive Immunity VI

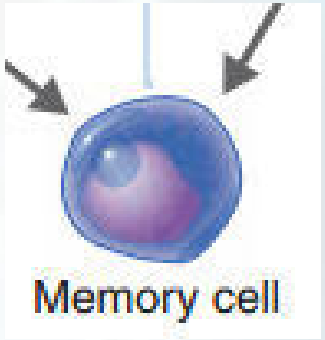
Humoral Immune Response Summary

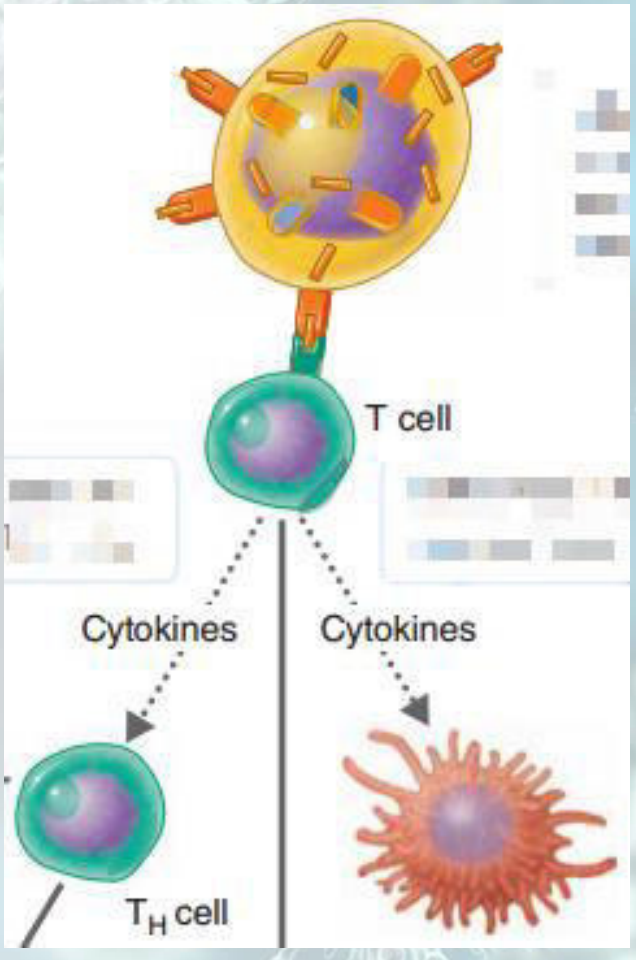


Adaptive Immunity VI

Cellular Immune Response Summary







Adaptive Immunity VI

MHC Molecules

- MHC I: Present on all nucleated cells
- MHC II: Present only on APCs
- Antigen Presenting Cells

Adaptive Immunity VI

Summary

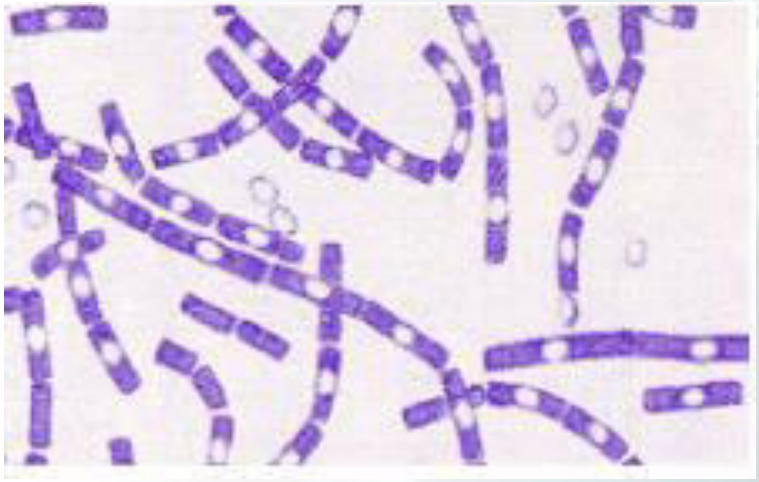
- Primary and secondary immune responses
- Their importance and applications in vaccinations

Special Staining I

Learning Objectives

- Why we stain organisms
- Spore staining

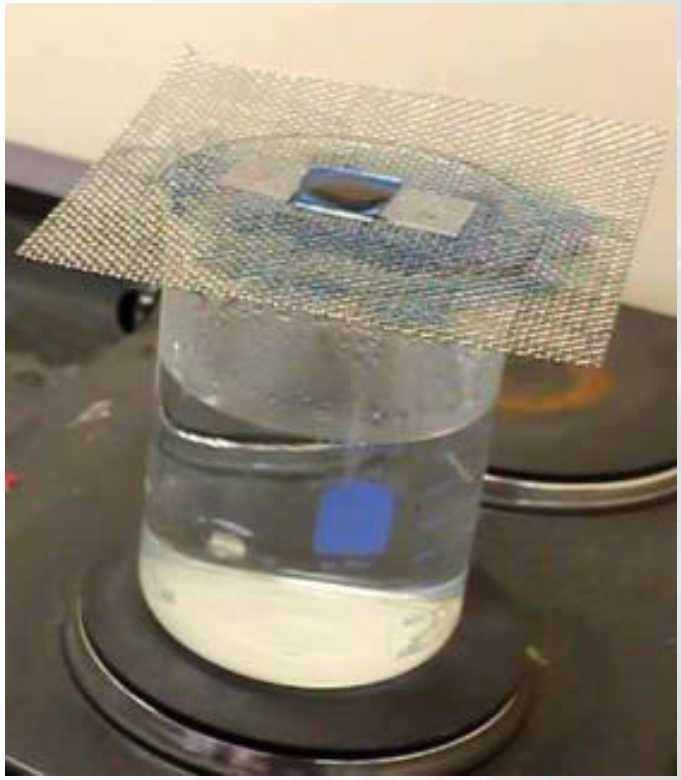
Special Staining I



Spore Staining

- Spores contain dipicolinic acid + Ca++
- Spores becomes heat resistant
- So, strong stains are used with heat

Special Staining I



Spore Staining

- Make a smear and air dry it.
- Fix the smear by heat.
- Heat water in a beaker.
- Place slide with smear side up.

Special Staining I

Spore Staining

- Pour malachite green on the slide.
- Keep it there on the boiling water beaker for 5 min.
- Wash the smear with dH2O.

Special Staining I



Spore Staining

- Counterstain with safranine for 30 sec.
- Observe under oil-immersion lens

Special Staining I

Summary

- Spore can be seen with special stains.
- Heating is need for the stain to penetrate the cell.

Special Staining II

Learning Objectives

- Why we stain organisms
- Flagellar staining

Special Staining II

Flagella Stain

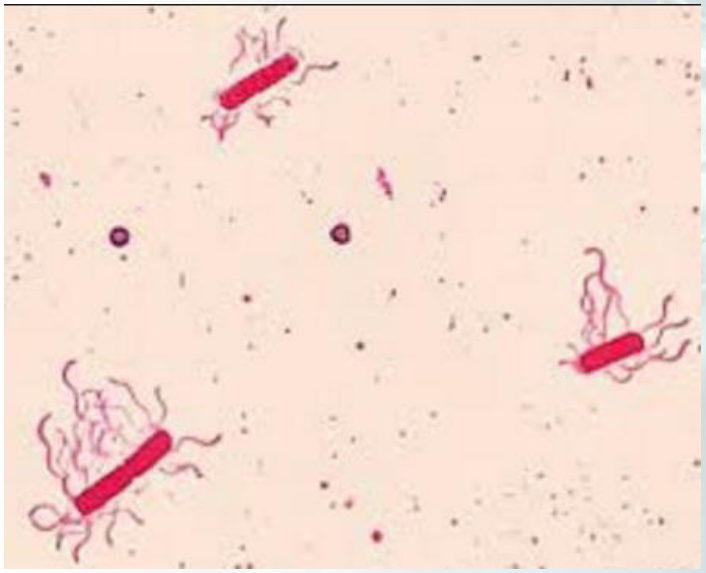
- Transfer a drop of fresh culture on one end of a slide.
- Allow the drop to trickle down slowly
- Air dry the smear. Do not heat fix it.

Special Staining II

Flagella Stain

- Cover the smear with mordant (tannic acid 10% in 5% NaCl) for 10 to 30 min.
- Rinse the smear gently with H_2O.
- Add carbol fuchsin and let it sit for 5-15 min.

Special Staining II



Flagella Stain

- Rinse off the stain with water
- Air dry the smear
- See under the microscope

Special Staining II

Summary

- Flagella can be seen with or without the stain (by observing motility).

Special Staining III

Learning Objectives

- Why we stain organisms
- Acid fast staining

Special Staining III

Acid Fast Staining

- Differential stain
- 1882, Paul Ehrlich observation on Mycobacterium
- Not easily stained
- Once stained do not loose the stain even with acid alcohol

Special Staining III

Acid Fast Staining

- Mycobacterium have mycolic acid (waxy)
- Makes cell wall impermeable to most stains
- Ziehl Neelsen method is commonly used

Special Staining III

Acid Fast Staining

- Make a smear
- Carbolfuchsin on slides
- Heat the slide for 5-10 min
- Wash with water and then with acid alcohol for 1 min.

Special Staining III



Acid Fast Staining

- Counterstain with methylene blue for 1 min.
- See under the microscope.

Special Staining III

Summary

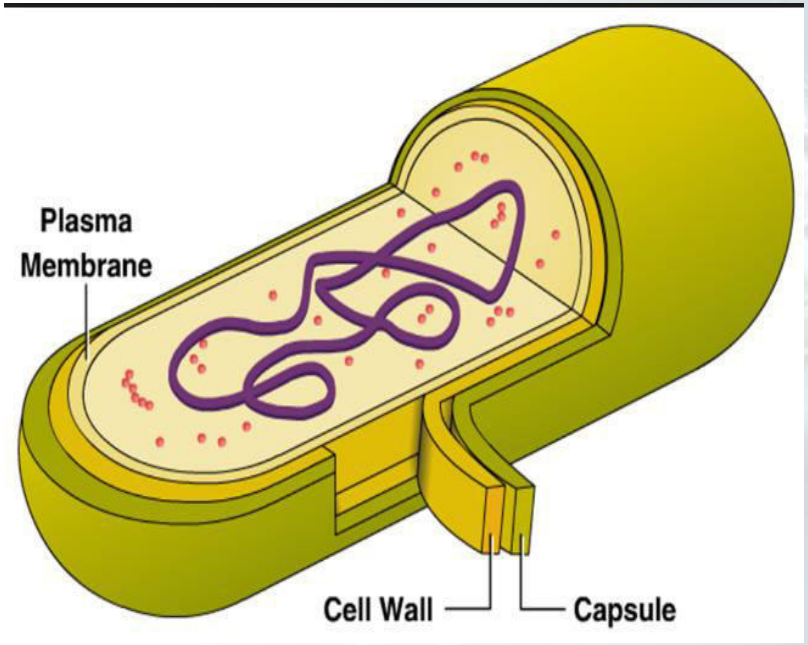
- Acid fast bacteria have mycolic acid that can be stained by heating the organisms with the stain.

Special Staining IV

Learning Objectives

- Why we stain organisms
- Capsular staining

Special Staining IV



Capsular Staining

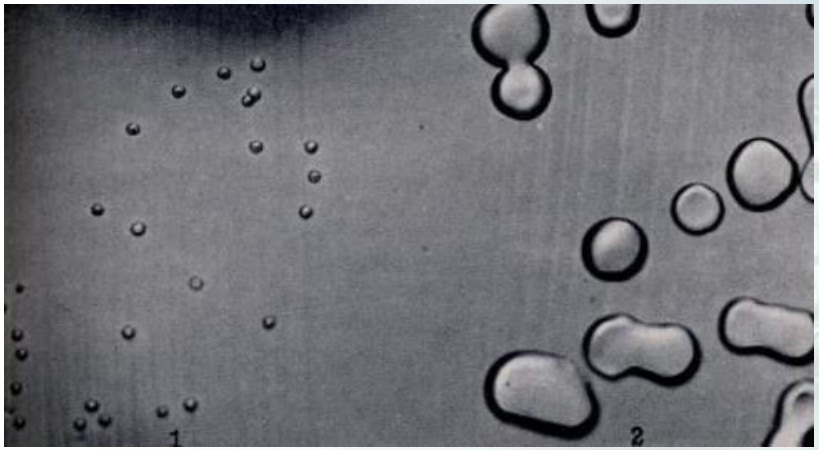
- It is extracellular material (glycocalyx).
- When regular and thick, called capsule
- When loose and not organized, called slime

Special Staining IV

Capsular Staining

- Thickness of the capsule depends upon the culture conditions.
- Mostly are water soluble.
- Capsular organisms usu make the broth viscous and stringy

Special Staining IV



Rough Vs Smooth Colonies

Capsular Staining

- Colonies look moist, glistening, and sticky.
- Involves in virulency
- Those lacking capsule are rough strains.

Special Staining IV

Capsular Staining

- Prepare a thick smear in a loopful of congo red (1%) stain
- Fixation in acid alcohol for 15 sec
- Wash with dH2O
- Cover it with acid fuchsin for 1 min.

Special Staining IV



Capsular Staining

- Wash with water
- Bacteria stains red
- Capsule colorless
- Dark blue background

Special Staining IV

Summary

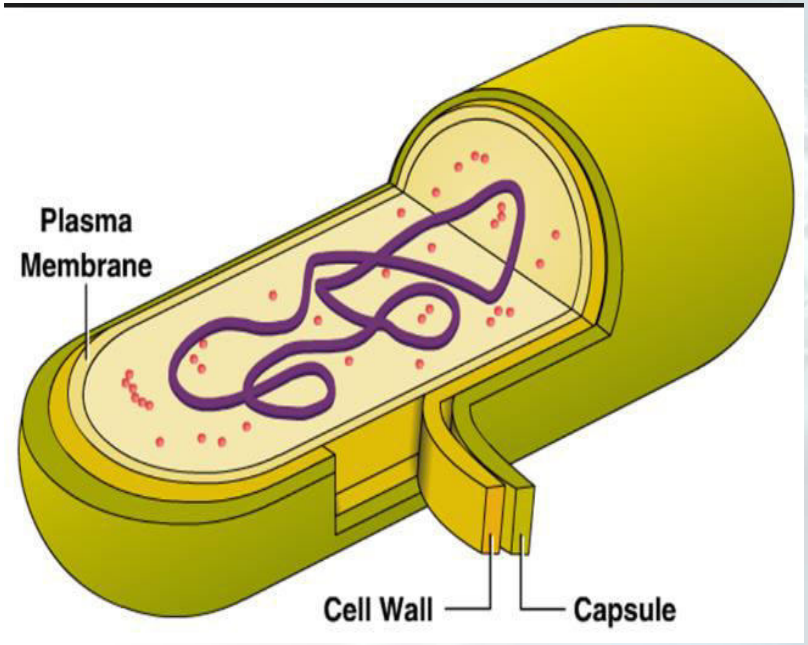
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- Helps in identifying organisms.

Special Staining IV

Learning Objectives

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- Capsular staining

Special Staining IV



Capsular Staining

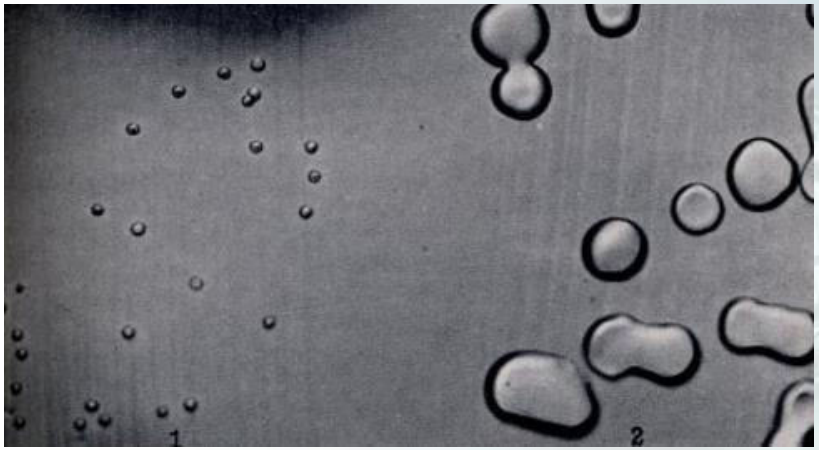
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Special Staining IV

Summary

- Capsule can be seen with staining.
- Helps in identifying organisms.

Hypersensitivity I

Learning Objectives

- What is anaphylaxis
- How it is initiated
- How to treat

Hypersensitivity I

Hypersensitivity?

- Abnormal antigenic response
- Allergy: another name
  - Those antigens are called allergens.
  - Immune system acts in a damaging manner.

Hypersensitivity I

Types of Hypers

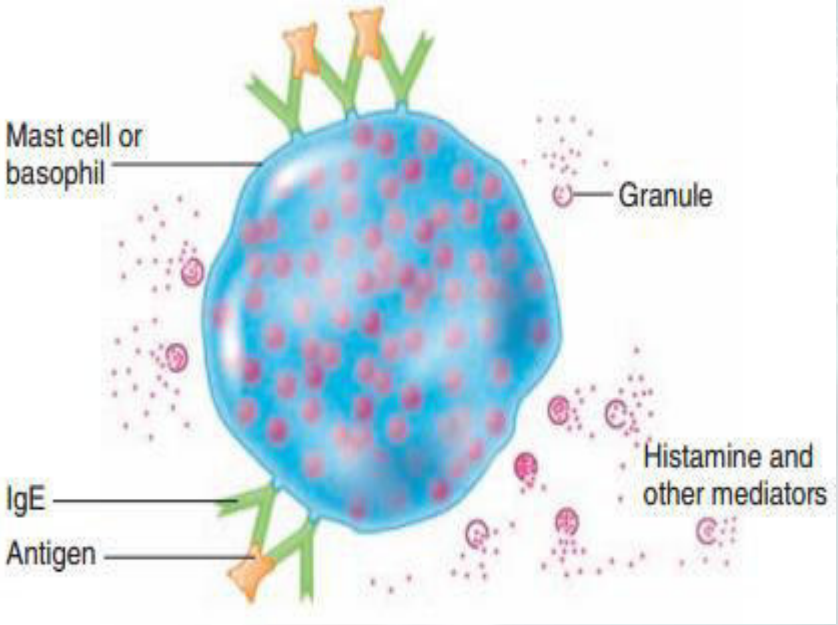
- Type I (Anaphylactic):
- Type II (Cytotoxic):
- Type III (Immune complex):
- Type IV (delayed cell-mediated):

Hypersensitivity I

Anaphylaxis?

- How to recognize it?
  - Difficulty in respiration
  - Shock
  - Hypotension
  - Increase in heart rate

Hypersensitivity I



Anaphylactic

- Happens in <30 min.
- IgE is involved which binds to mast cells
- Mast cells degranulate and secretes mediators of inflammation.
  - Histamine, leukotrienes, PG

Hypersensitivity I

Anaphylaxis

- Shock from drug reactions
- Venoms
- Common allergens causing asthma

Hypersensitivity I

Anaphylaxis

- Diagnosis by demonstration of IgE by i/d injections of allergens one by one.
- Quantitation of IgE by ELISA
- Rx: Epinephrine & antihistaminic drugs

Hypersensitivity I

Summary

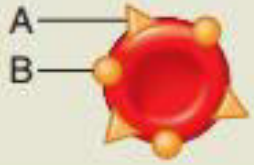
- Anaphylaxis involves IgE antibodies
- Histamine and other mediators mediate the reaction.
- Could be treated by antihistaminic drugs.

Hypersensitivity II

Learning Objectives

- How cytotoxicity (Type II) occurs
- How to diagnose it?
- Rx.

Hypersensitivity II

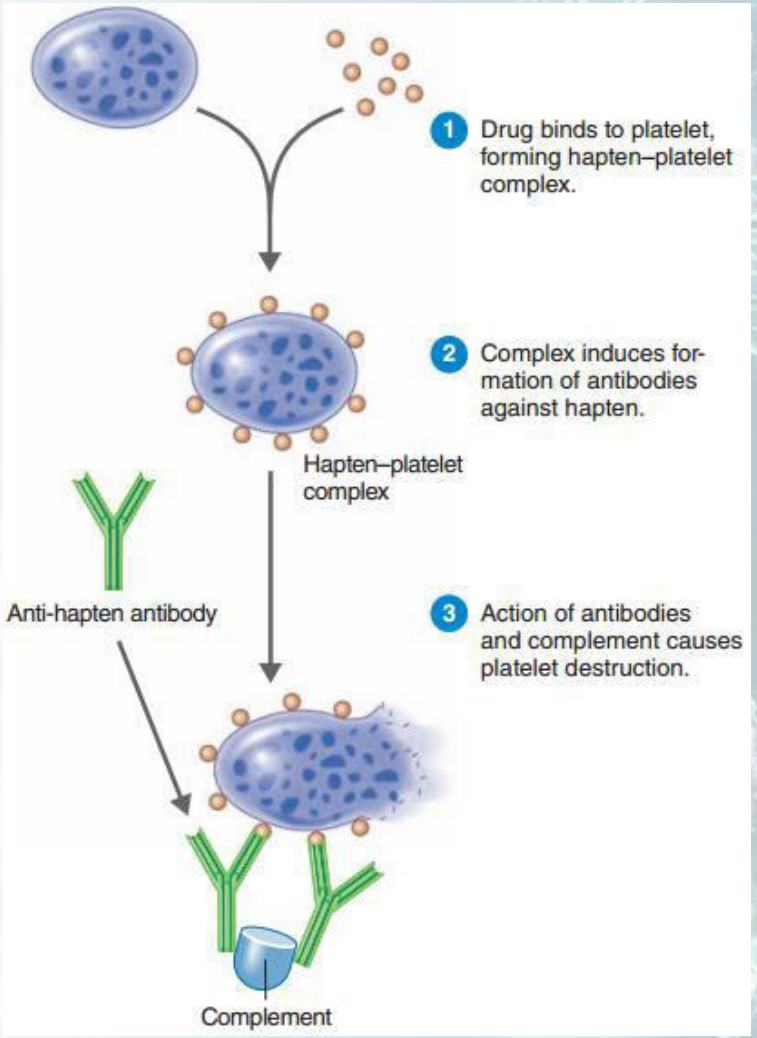
Illustration	Plasma Antibodies
	Neither anti-A nor anti-B antibodies
	Anti-A
	Anti-B
	Anti-A and Anti-B

Type II: Cytotoxic

- **Involves**
  - **Antigenic cell**
  - **IgG or IgM**
  - **Complement**
- **Blood transfusion reaction is an example.**

Hypersensitivity II

Drug induced cytotoxicity



Hypersensitivity II

Diagnosis

- Cell deficiencies
- Low number of RBCs due to hemolysis
- Thrombocytopenia if platelets are involved.

Hypersensitivity II

Rx

- Avoid those drugs that cause it.
- Blood transfusion with correct blood type
- Symptomatic treatment

Hypersensitivity II

Summary

- IgG or IgM involved
- Causes damage to the cells.
- Treated according to the cause.

Hypersensitivity III

Learning Objectives

- How immune complexes are formed
- How to diagnose it?
- Rx.

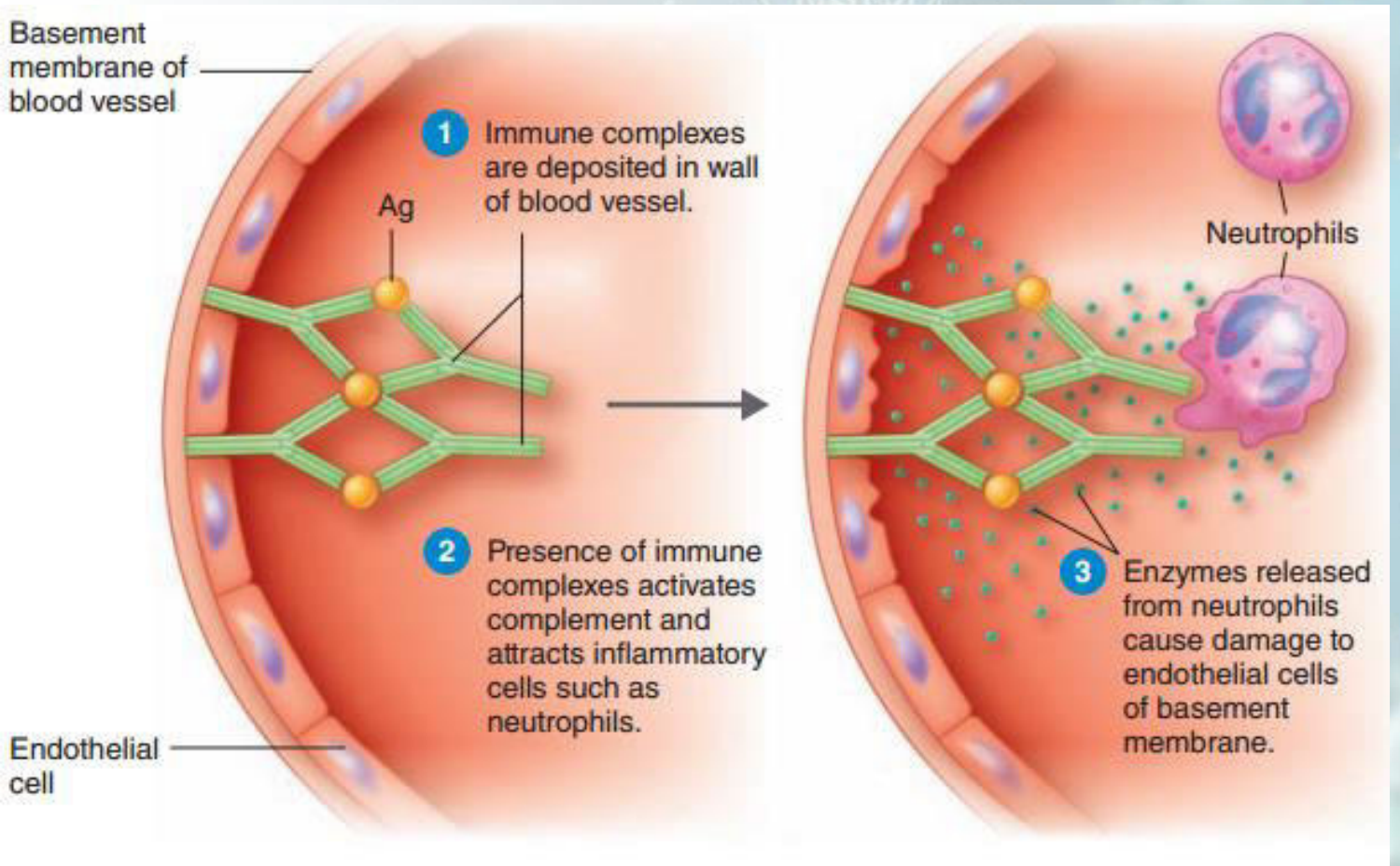
Hypersensitivity III

Immune complex

- Involves
  - Soluble Antigen circulating in the serum
  - IgG or sometimes IgM
  - Complement

Hypersensitivity III

Immune complex



Hypersensitivity III

Immune complex

- Diagnosis by history and symptoms
  - Glomerulonephritis
- Confirmed by biopsy and testing of presence of IgG in kidney cells
- Rx. Corticosteroids

Hypersensitivity III

Summary

- Immune complexes are formed which when get stuck in capillaries cause damage by compliment activation.

Hypersensitivity IV

Learning Objectives

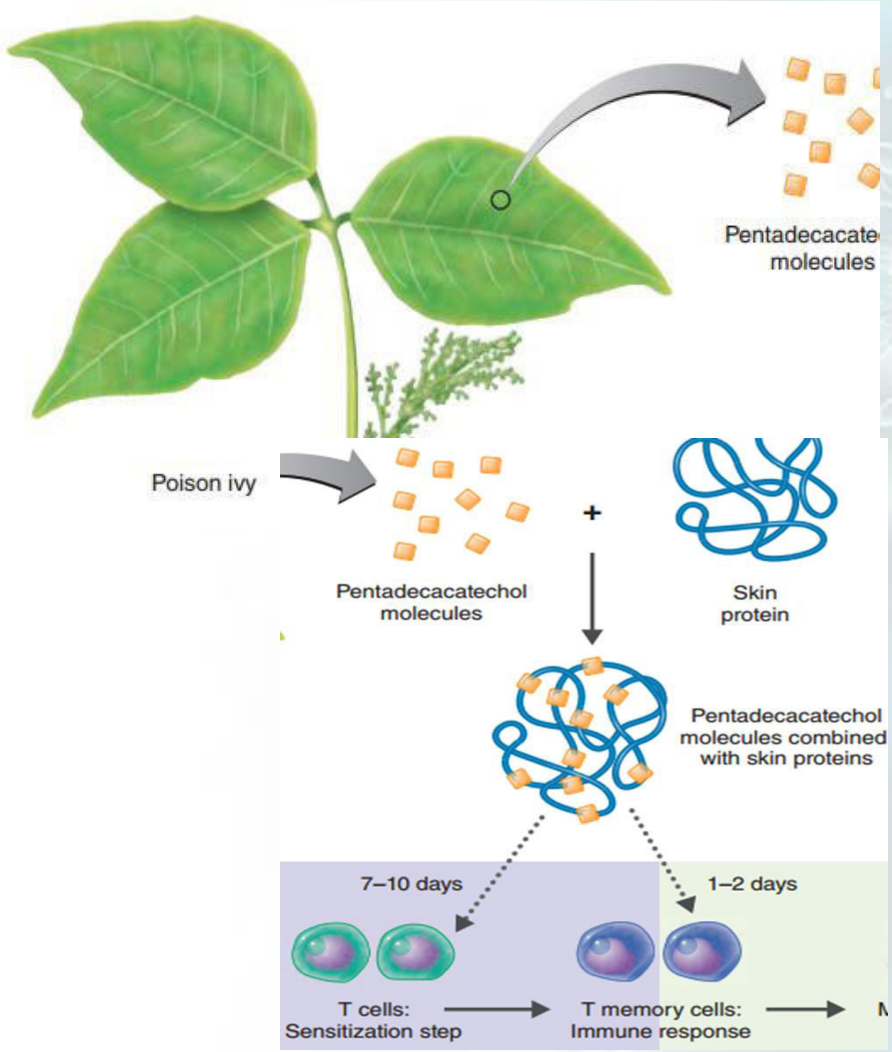
- What is type IV hypersensitivity?
- Impact on body
- Rx

Hypersensitivity IV

Type IV

- Originally called tuberculin reaction
- Robert Koch performed the test.
- Contact dermatitis is the most common example.

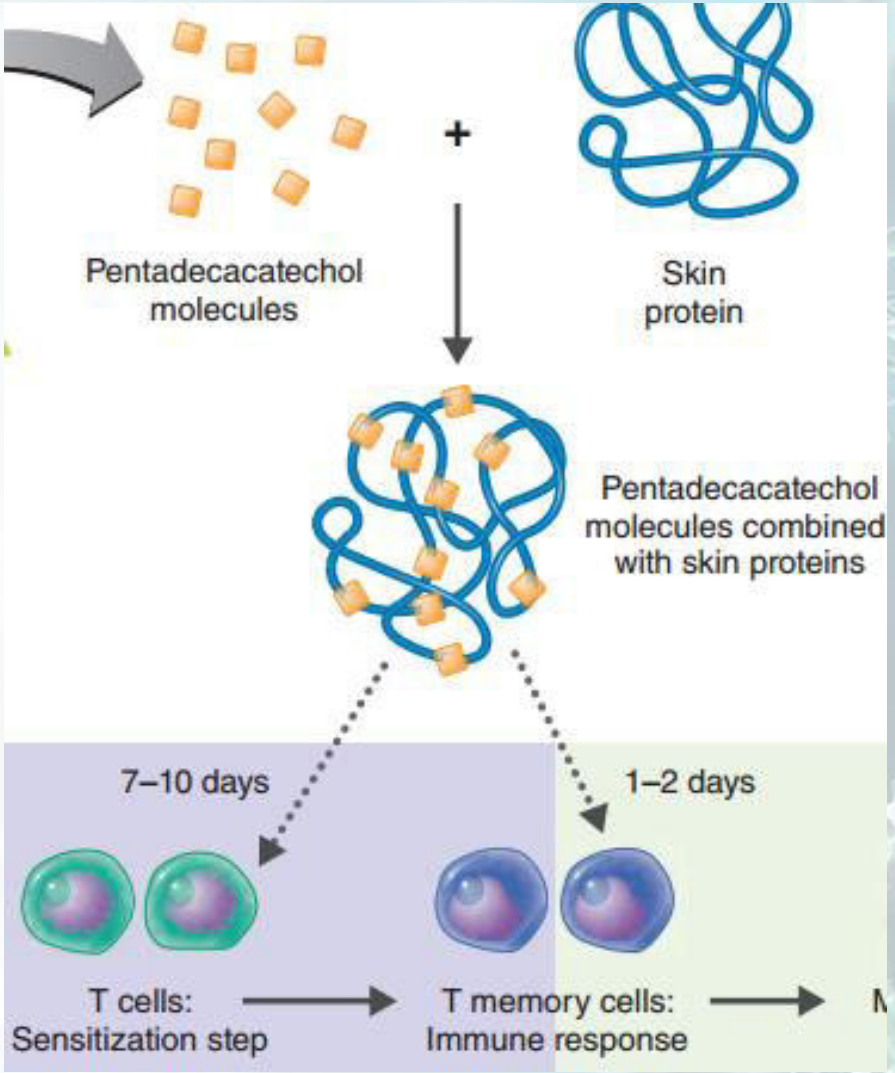
Hypersensitivity IV



Type IV

- **Involves:**
  - Antigen
  - T cells
  - Macrophages
- Transplant rejection is good example.
- Reaction to hair dye

Hypersensitivity IV



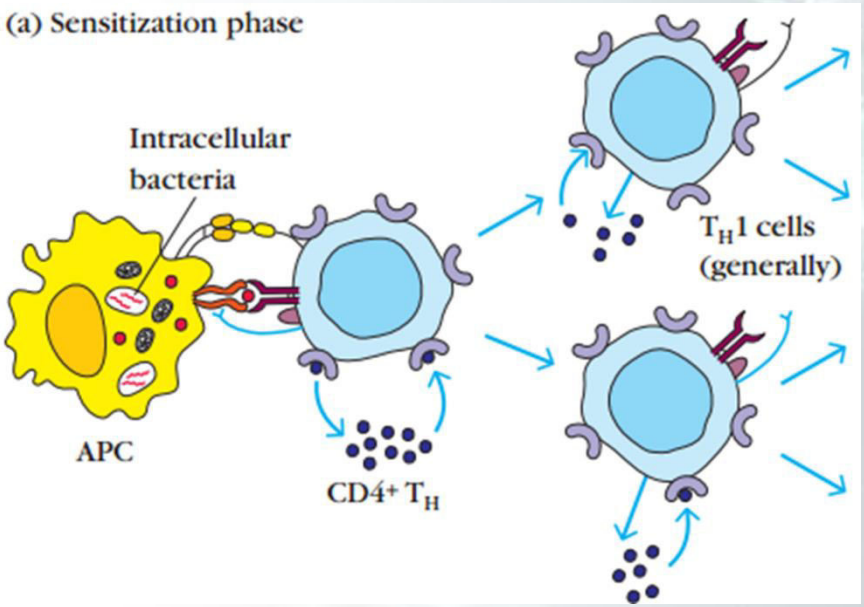
Hypersensitivity IV

Type IV: Ivy & Latex Glove Reaction





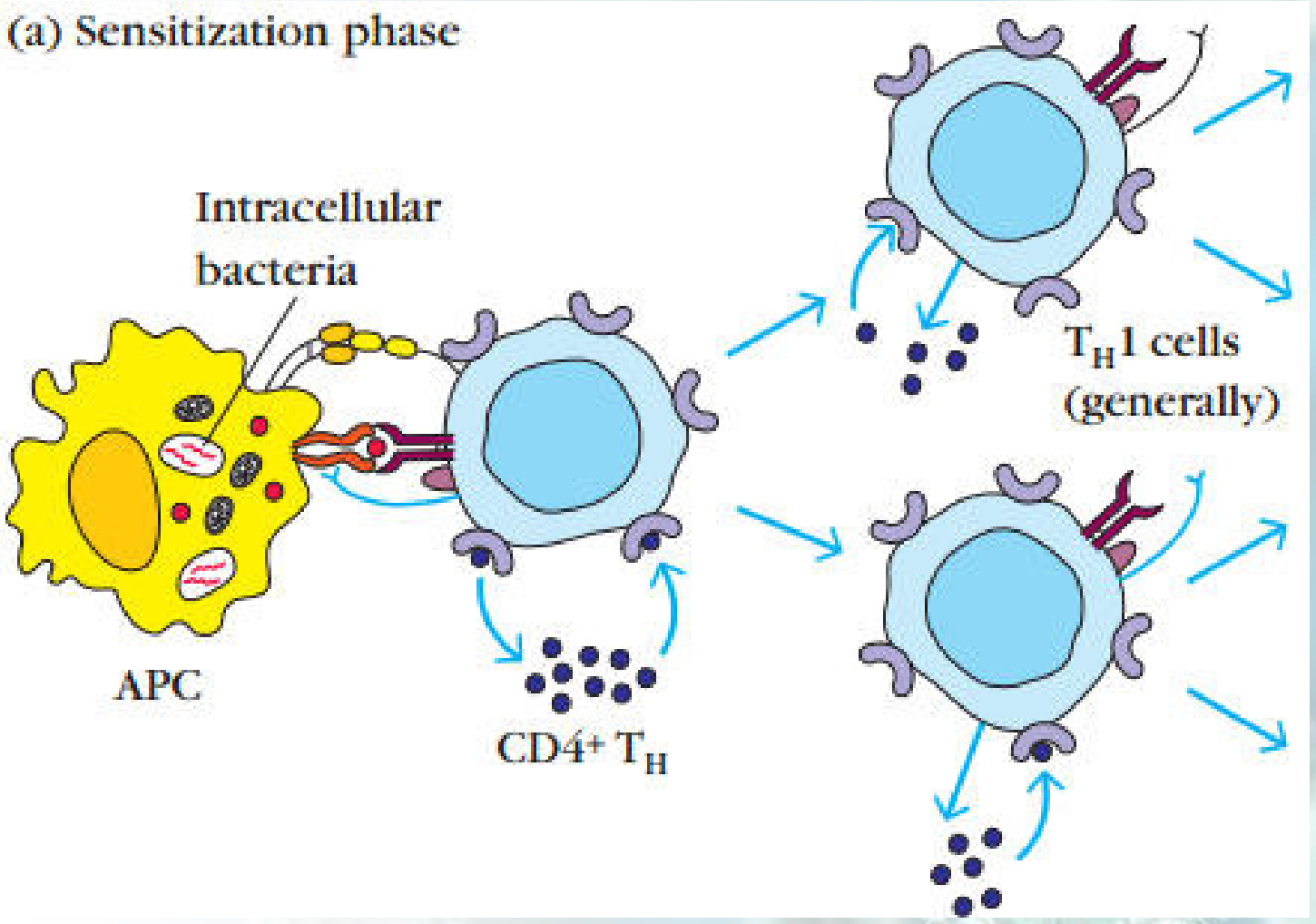
Hypersensitivity IV



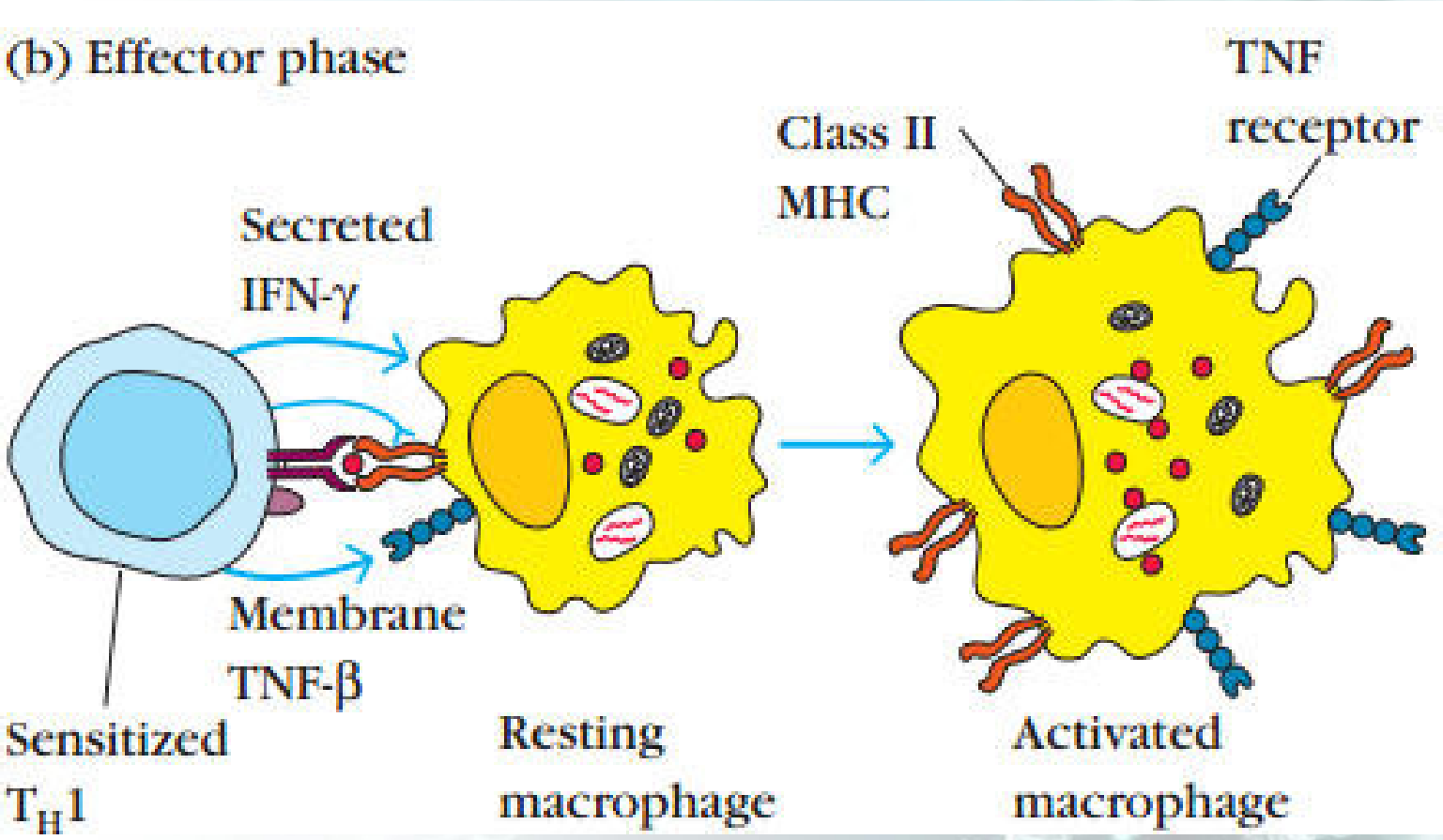
Type IV

- **How it happens:**
  - Macrophages take up the antigen
  - Present it to CD^{4+} Th1 cells
  - Th1 cells secrete cytokines that activate macrophages

Hypersensitivity IV



Hypersensitivity IV



Hypersensitivity IV



Type IV

- Diagnosis:
  - Skin testing
- Rx: symptomatic
- Avoid contact

Hypersensitivity IV

Summary

- T helper cells are involved mostly.
- Rx. Avoid the contact or symptomatic Rx

Vaccine I

Learning Objectives

- What is a vaccine
- Types of vaccines

Vaccine I

Concept of Vaccines

- People recovered from infections are immune to them.
- **Vaccine:** Suspension of organism or fraction of organism that induces immunity

Vaccine I



Principle of Vaccines

- It mimics the infection by the organism but without causing the disease

Vaccine I

Types of Vaccines

- **Live attenuated:**
  - Weakened organisms
  - Louis Pasteur was the first to do that.
    - *Pasteurella multocida*

Vaccine I

Benefits of Live Vac

- Better immunity: mimics actual infection
  - Humoral
  - Cellular
- Virus replicates increasing the Ag dose
- Long term immunity

Vaccine I

Types of Vaccines

- **Inactivated Killed:**
  - Organism killed with formalin or phenol
  - But why??
- Cannot be weakened
- Need frequent boosters

Vaccine I

Inactivated Killed

- Examples
  - Rabies
  - Polio
  - Influenza
- Mostly induce humoral immunity

Vaccine I

Other Vaccines

- **Toxoids:** Toxins are inactivated and used as a vaccine.
  - Tetanus toxoid
- **Conjugated** Vaccines:
  - Poor antigens attached to proteins
  - *Haemophilus influenzae* type b

Vaccine I

Other Vaccines

- DNA vaccines:
  - DNA injected into cells
  - Proteins are expressed which stimulate immunity
  - West Nile Virus
- Polysaccharides have poor success

Vaccine I

Modern Vaccines

- **Subunit vaccines:**
  - Portion of viral or bacterial proteins that can induce immunity
- Also called **recombinant vaccines**

Vaccine I

Modern Trends

- Organisms in bulk are needed to make vaccines.
- Bacteria can be grown in flasks by providing medium of growth
- Cell culture for viral vaccines

Vaccine I

Modern Trends

- Chicken embryos can be used to grow viruses
  - Influenza vaccine
- Animals can be used to grow viruses
- Plants can be used in recombinant technology

Vaccine I

Summary

- Vaccines mimic infection but without harmful effects.
- Various types are available.
- Booster is required for better protection.

Antimicrobial I

Learning Objectives

- What are antimicrobials and antibiotics
- Who makes them
- Problems in finding them

Antimicrobial I



A Little History

- Paul Ehrlich coined the word "chemotherapy"
- His observation: Organisms stained without staining the background.

Antimicrobial I



Word, Antibiotics coined.

A Little History

- 1928: Alexander Fleming working with *Staph aureus* observed colony of a mold with inhibitory effect.
  - *Penicillium notatum*
  - Later, *P chrysongenum*

Antimicrobial I

A Little History

- **Disinfectant:**
  - Toxic to all
- **Antimicrobials used for chemotherapy:**
  - Selectively toxic to microbes sparing body cells

Antimicrobial I

Microbe	Antibiotics
Bacillus subtilis	Bacitracin
Paenibacillus polymyxa	Polymyxin
Streptomyces nodosus	Amphotericin B
Streptomycems venezuelae	Chloramphenicol
Streptomyces aureofaciens	Chlortetracycline and tetracycline
Saccharopolyspora erythraea	Erythromycin
Streptomyces fradiae	Neomycin
Streptomyces griseus	Streptomycin
Micromonospora purpurea	Gentamicin

Antimicrobial I

Antibiotic Discovery

- Must not be toxic to the body
  - Many antibiotics are produced by microbes, but only a few are selectively toxic to microbes.

Antimicrobial I

Summary

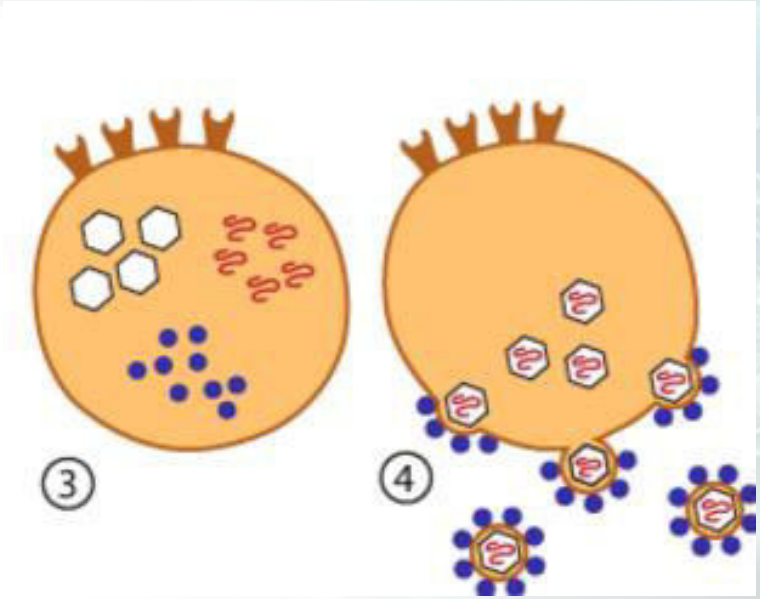
- Antibiotics are products of microbes against microbes.
- Must not be toxic to the body.

Antimicrobial II

Learning Objectives

- Difficulties in developing effective antimicrobials
- Mode of action

Antimicrobial II



A Little History

- Developing drugs against prokaryotes is relatively easy
- Against eukaryotes is difficult
  - Viruses and fungi
  - Problem is at the cellular level

Antimicrobial II

Spectrum of Drugs

- **Narrow:** targets only a group of organisms
  - Penicillin for G+ve
- **Wide:** Targets many groups of microbes
  - Tetracycline

Antimicrobial II

Spectrum of Drugs

- **Narrow:** well targeted
  - Spare microflora
- **Wide:** Targets every organism
  - Destroy normal microflora as well

Antimicrobial II

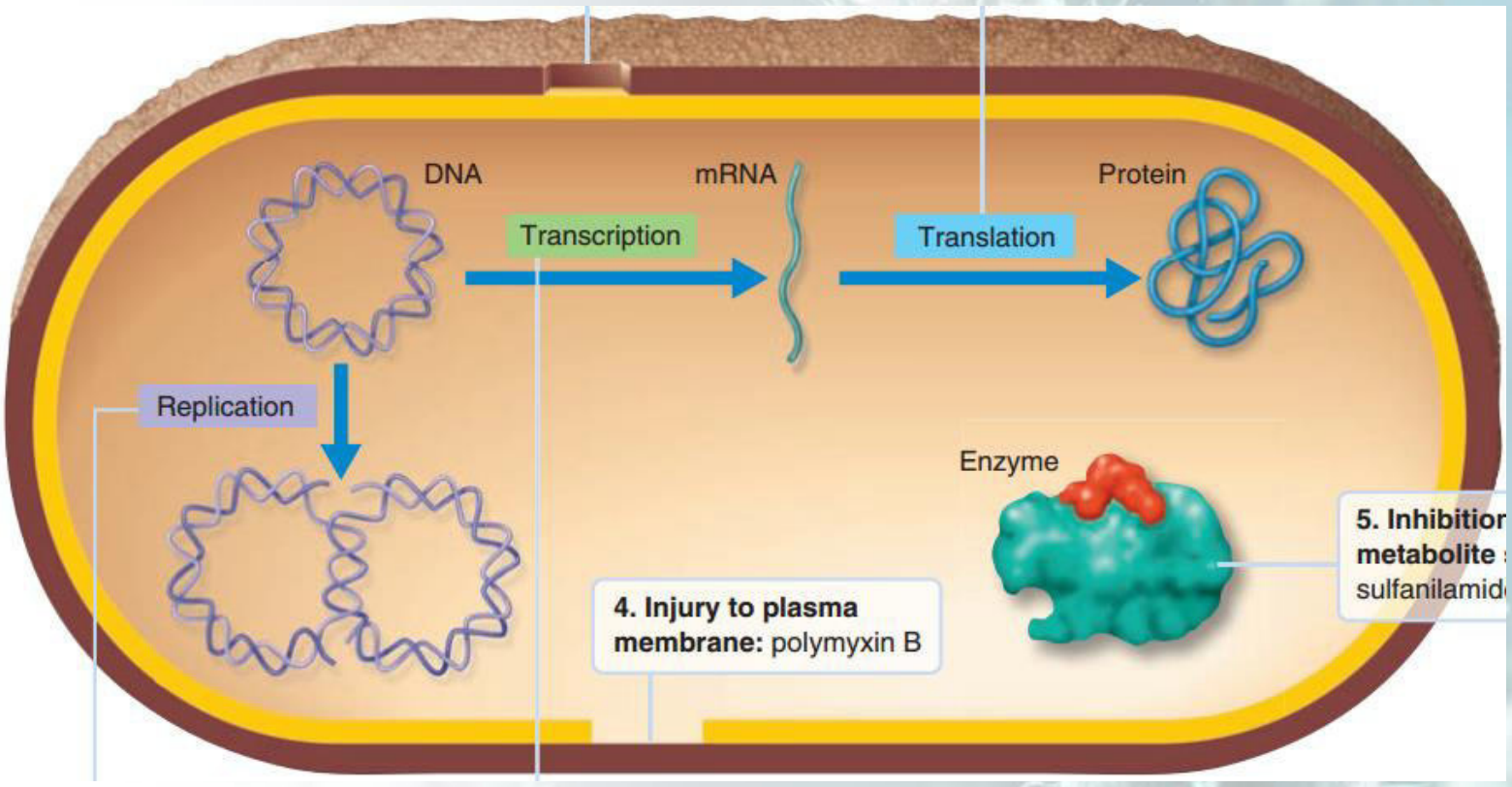


Spectrum of Drugs

- Candida albicans
  - Grows in the absence of normal microflora esp. in children

Antimicrobial II

Drug Targets

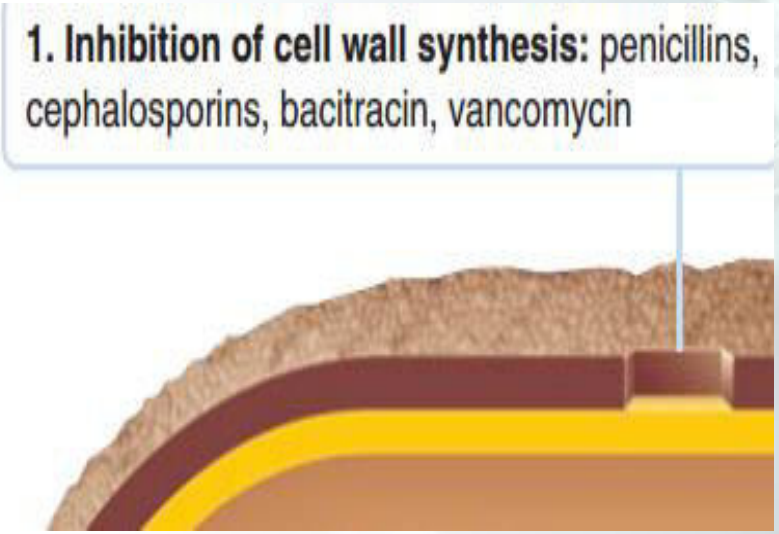


Antimicrobial II

Target: Cell Wall

- Inhibition
  - Penicillins
  - Cephalosporins
  - Bacitracin
  - Vancomycin

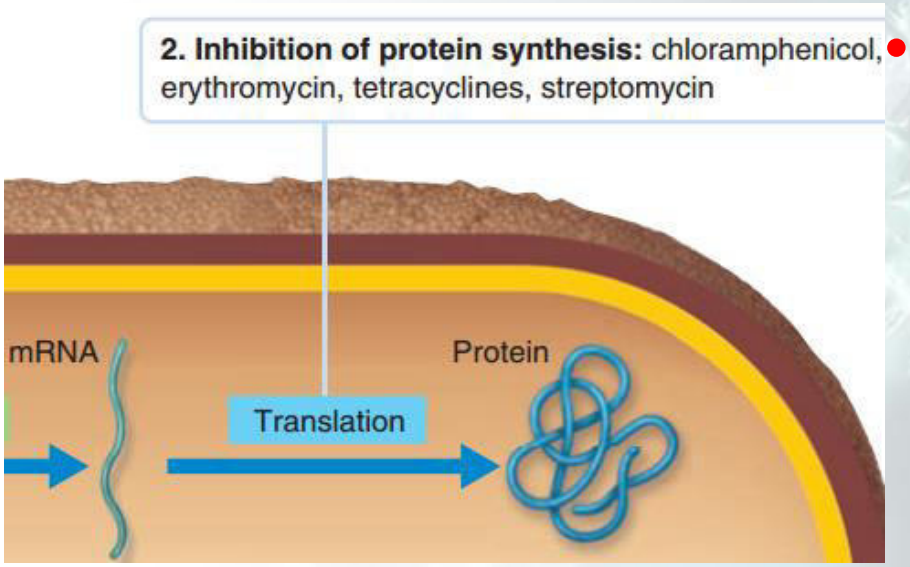
Antimicrobial II



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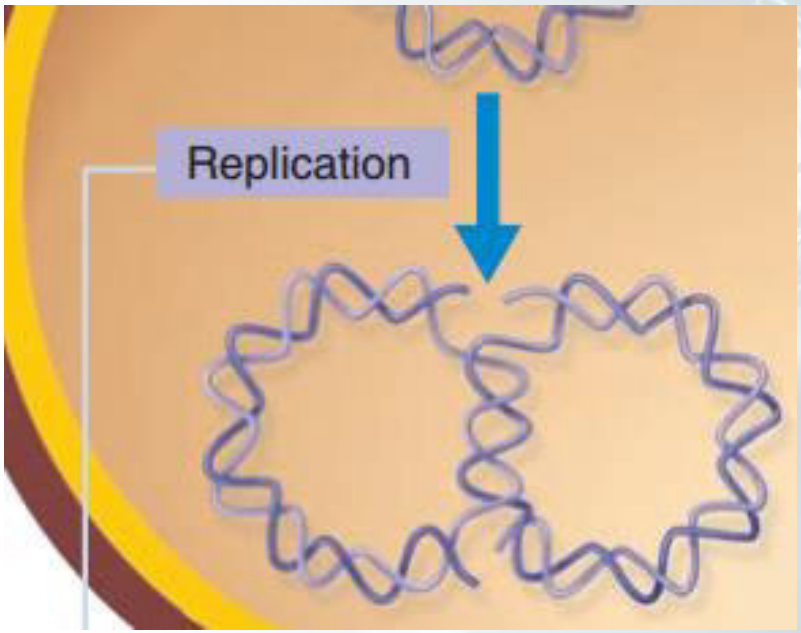
Antimicrobial II



Target: Protein Synthesis

- **Inhibition**
  - Chloramphenicol
  - Erythromycin
  - Tetracyclines
  - Streptomycin

Antimicrobial II



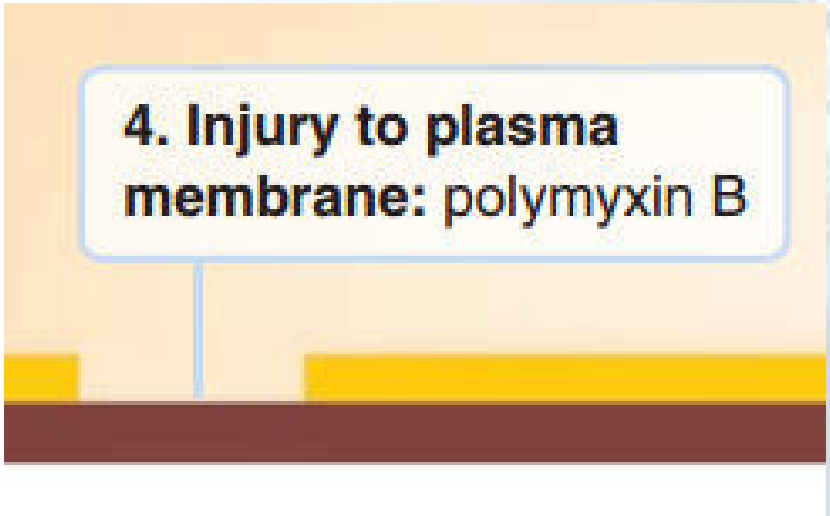
Target: DNA Replication

- **Inhibition**
  - **Quinolones**
  - **Rifampin**

Antimicrobial II

Target: Plasma Membrane

- **Injury**

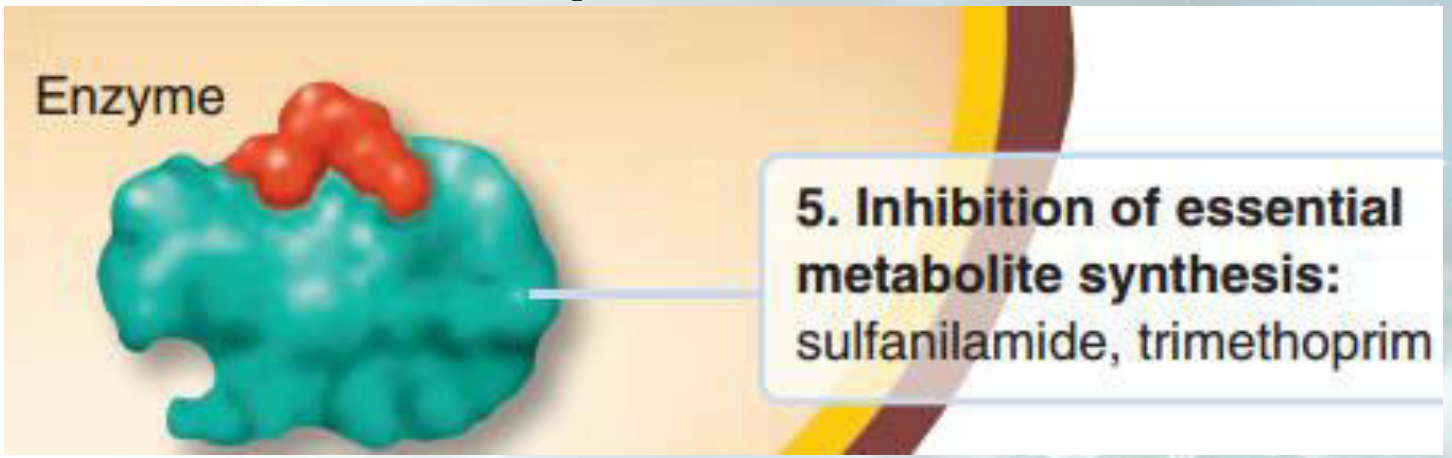


– **Polymixin B**

Antimicrobial II

Target: Ess. Metabolite

- **Inhibition:**
  - **Sulfanilamide**
  - **Trimethoprim**



Antimicrobial II

Summary

- Antibiotics selectively inhibit various components of microbes.

Antimicrobial III

Learning Objectives

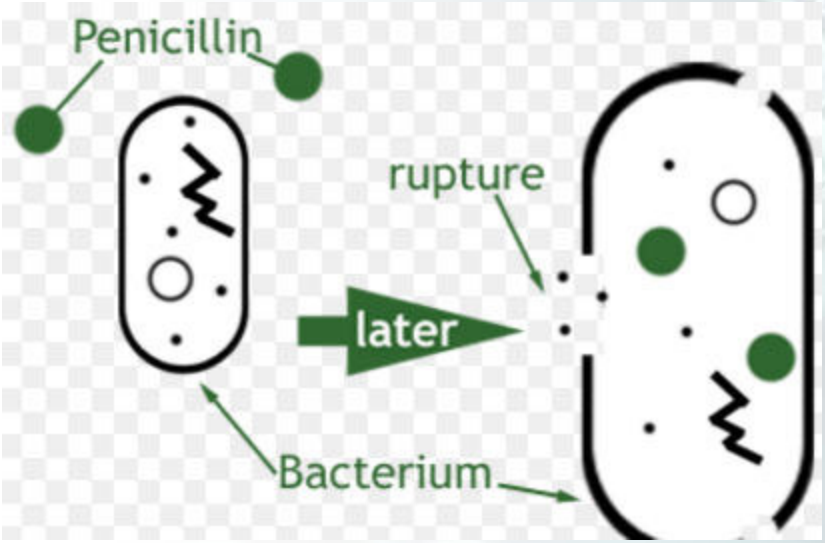
- Bactericidal
- Bacteriostatic

Antimicrobial III

Terminology

- **Bactericidal:**
  - Kill microbes directly.
- **Bacteriostatic:**
  - Prevent growth of microbes

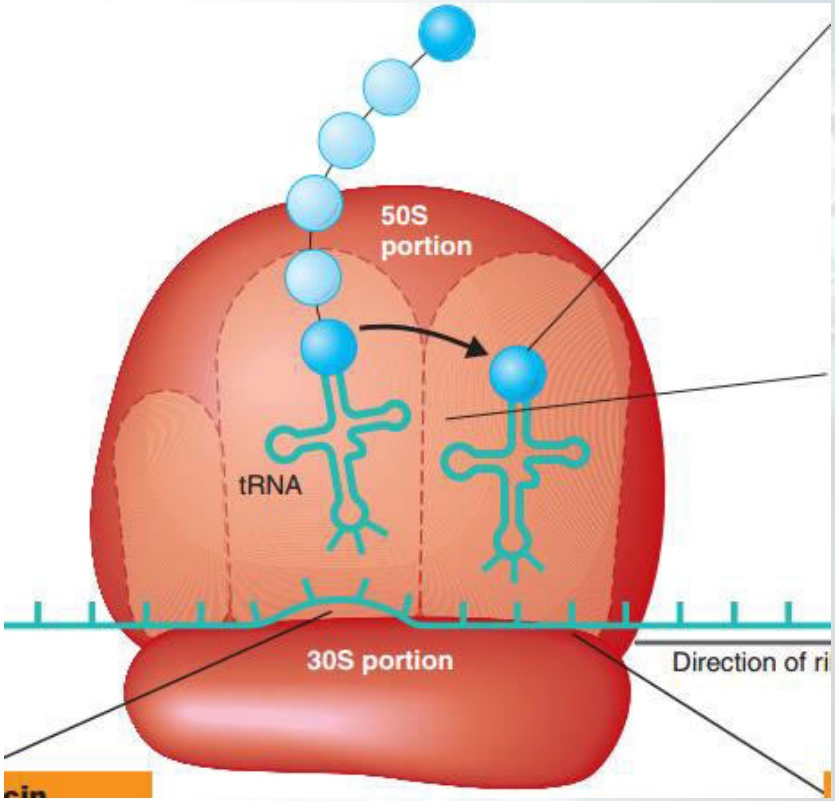
Antimicrobial III



Cell Wall Inhibition

- Target is peptidoglycan
  - Cell wall does not get synthesized
  - Cell lysed as a result (cidal).
  - Targets growing cells
    - Penicillin

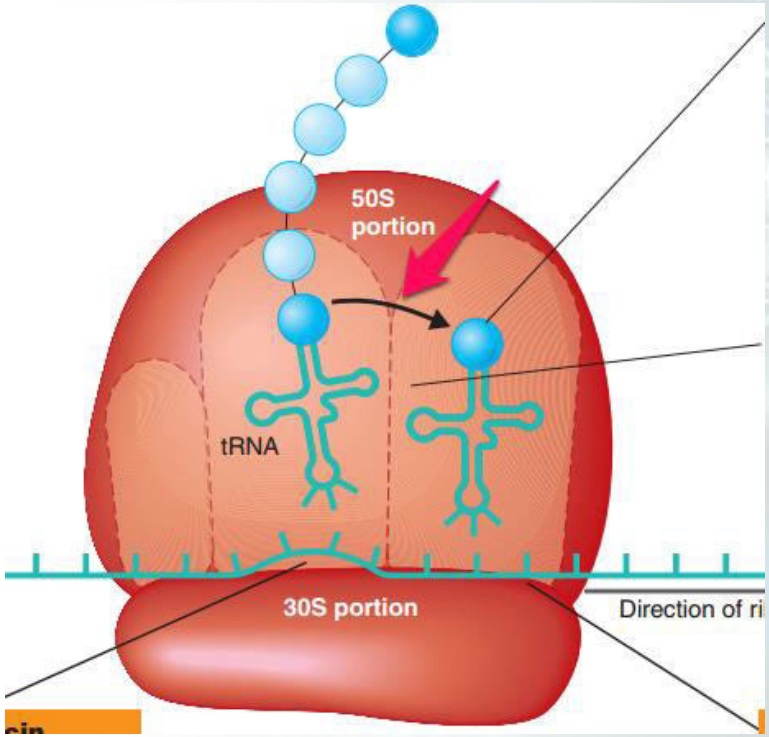
Antimicrobial III



No Protein Synthesis

- Based on ribosome structural differences in pro and eukaryotes
  - Eukaryotes: 80S
  - Prokaryotes: 70S

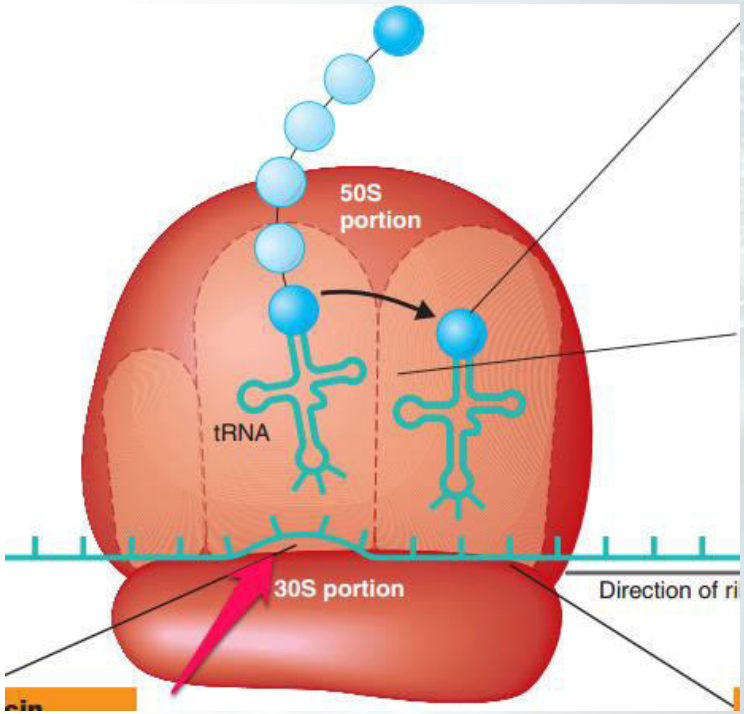
Antimicrobial III



No Protein Synthesis

- Chloramphenicol:
  - Binds to 50S
  - Inhibits peptide bonding
  - Bacteriostatic

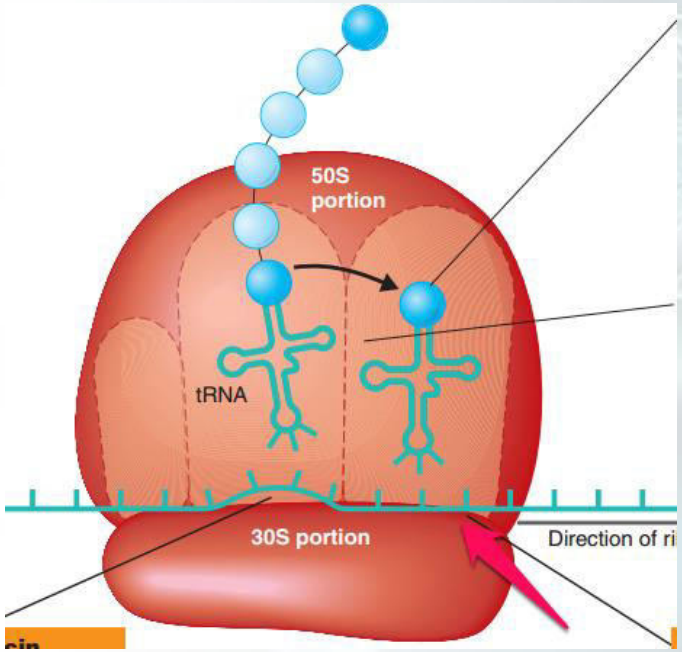
Antimicrobial III



No Protein Synthesis

- **Streptomycin:**
  - **Changes shape of 30S**
  - **Misreading of RNA codes**
  - **Bacteriostatic**

Antimicrobial III



No Protein Synthesis

- **Tetracyclines:**
  - **Interfere with attachment of tRNA to mRNA ribosome complex**
  - **Bacteriostatic**

Antimicrobial III

Plasma Membrane

- Polypeptide antibiotics brings changes in the permeability of plasma membrane
- Result in the loss of imp metabolite from the cell

Antimicrobial III

Plasma Membrane

- Antifungal drugs are examples
  - Amphotericin B
  - Ketoconazole
- Combine with sterols in the plasma membrane
- Bacteria lack sterol

Antimicrobial III

Nucleic Acid Synth

- Interfere with DNA replication and transcription
  - Very limited b/c can interfere with body cells

Antimicrobial III

Essential Metabolites

- Sulfanilamide and para-aminobenzoic acid (PABA)
- PABA is a substrate for an enzyme that is involved in folic acid synthesis
- FA is needed for purine and pyrimidine synthesis

Antimicrobial III

Summary

- Antibiotics target various cell components
  - Structural integrity
  - DNA or RNA
  - Protein

Antimicrobial IV

Learning Objectives

- Antibiotic sensitivity testing
- Minimum inhibitory concentration
- Is the drug bactericidal

Antimicrobial IV

Drug Sensitivity

- Physicians usu prescribe drugs that they know will work.
- If the organism has been identified, selection of drug is easy.

Antimicrobial IV

Drug Sensitivity

- **Disk Diffusion Method:**

- **Standardized inoculum spread on agar plate**

- **Filter paper disks soaked with antibacterials**

Antimicrobial IV



Drug Sensitivity

- **Disk Diffusion Method:**
- **Zone of inhibition is recorded**
- **Zone compared to a standard Table**

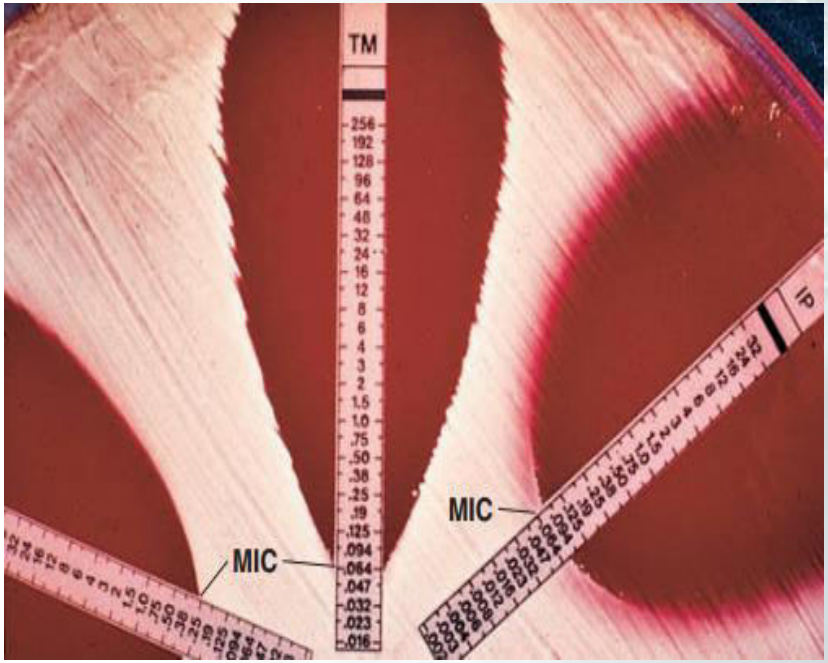
Antimicrobial IV



Drug Sensitivity

- **Disk Diffusion Method:**
- **Report:**
  - **Sensitive**
  - **Intermediate**
  - **Resistant**

Antimicrobial IV



Drug Sensitivity

- **Minimum Inhibitory Concentration (MIC):**
- **Also called E test:**
  - **The lowest conc. that inhibits visible growth**

Antimicrobial IV

Drug Sensitivity

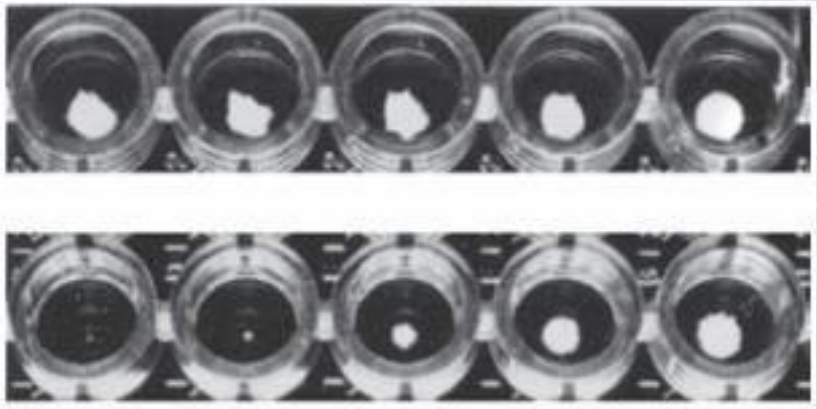
- **Broth Dilution Method:**
- **Diffusion method has limitation**
  - **Does not tell you if the drug is bacteriostatic or bactericidal**

Antimicrobial IV

Drug Sensitivity

- **Broth Dilution Method:**
- A broth dilution test is the answer.
- Various dilutions of the drug made and tested with test organism

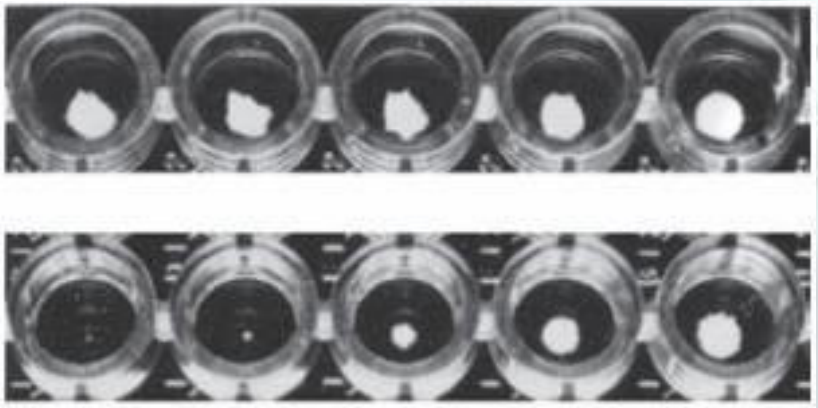
Antimicrobial IV



Drug Sensitivity

- **Broth Dilution Method:**
- Tells both **MIC** and minimal bactericidal concentration (**MBC**)
- The wells that do not show growth are important.

Antimicrobial IV



Drug Sensitivity

- **Broth Dilution Method:**
- **Culture the well that is one well higher in concentration than MIC.**
  - **If growth occurs, the drug is bacteriostatic, otherwise cidal.**

Antimicrobial IV

Drug Sensitivity

- Antibiograms are important.
- Susceptibility of organisms is recorded over a period to see if there is a change in response.

Antimicrobial IV

Summary

- Disk diffusion
- Broth dilution
- Antibiogram

Antimicrobials V

Learning Objectives

- Examples of antibiotics and their mode of action

Antimicrobials V

Cell Wall

- Penicillin
- Cephalosporins
  - Cefixime (4th gen)
- Polypeptide antibiotics
  - Bacitracin (topical)
  - Vancomycin (Pen-R)

Antimicrobials V

Cell Wall

- Antimycobacterials (inhibit mycolic acid synthesis)
  - Isoniazids
  - Ethambutol

Antimicrobials V

Protein Synthesis

- Chloramphenicol (potentially cytotoxic)
- Aminoglycosides
  - Streptomycin
  - Neomycin (topical)
  - Gentamycin (Pseudomonas)
- Tetracyclines

Antimicrobials V

Protein Synthesis

- Tetracyclines
  - Against chlamydias and rickettsias
  - Animal feed additives
- Macrolides
  - Erythromycin
  - Azithromycin (ssyn)

Antimicrobials V

Plasma Membrane

- Polymixin B
  - Topical for G-ve

Antimicrobials V

Nucleic Acid Syn

- Rifampin
  - Inhibits synthesis of mRNA
  - Rx for tuberculosis

Antimicrobials V

Nucleic Acid Syn

- Quinolones and Flouroquinolones
  - Nalidixic acid
  - Norfloxacin
  - Ciprofloxacin
  - Used preferably for urinary tract infections

Antimicrobials V

Essential Metabolites

- Sulfonamides
  - Sulfa drugs
  - Trimethoprim
  - Sulfamethoxazole
  - Common combination used

Antimicrobials V

Summary

- Various antibiotics target various components of the cell.

Antimicrobials VI

Learning Objectives

- Antifungal drugs
- Antiviral drugs

Antimicrobials VI

Antifungal Drugs

- Fungal Sterols (PM)
  - Polyenes
    - Amphotericin B
    - Used for systemic infections

Antimicrobials VI

Antifungal Drugs

- Fungal Sterols (PM)
  - Azoles (inhibits synthesis of PM)
    - Clotrimazole (topical)
    - Ketoconazole
    - Voriconazole
      - Crosses BBB

Antimicrobials VI

Antifungal Drugs

- Nucleic Acids
  - Flucytocine
    - Inhibits synthesis of RNA and DNA in fungal cells

Antimicrobials VI

Antifungal Drugs

- Other Antifungals
  - Griseofulvin
    - Inhibition of mitotic microtubules
    - Rx of skin infections

Antimicrobials VI

Antiviral Drugs

- Nucleoside and nucleotide analogs
  - Acyclovir
  - Ganciclovir
  - Ribavirin
  - Used mostly for herpesviruses

Antimicrobials VI

Antiviral Drugs

- Attachment and uncoating (flu virus)
  - Zanamivir
    - Inhibits neuraminidase
  - Amantadine
  - Zimantadine
    - Inhibits uncoating

Antimicrobials VI

Summary

- Drugs can selectively kill fungi and viruses
- Not all viruses could be killed by drugs

Antimicrobials VII

Learning Objectives

- Antiprotozoan drugs
- Antihelminthic drugs

Antimicrobials VII

Antiprotozoan Drugs

- Chloroquine
  - Inhibits DNA synthesis
  - Used against malaria

Antimicrobials VII

Antiprotozoan Drugs

- Metronidazole
  - Interferes with anaerobic metabolism
  - Used for giardiasis, amebiasis, trichomoniasis

Antimicrobials VII

Antihelminthic Drugs

- Praziquantel
  - Alters permeability of PM
  - Used for tapeworm and fluke infections

Antimicrobials VII

Antihelminthic Drugs

- Albendazole, Mebendazole
  - Inhibit absorption of nutrients
  - Used against intestinal roundworms

Antimicrobials VII

Antihelminthic Drugs

- Ivermectin
  - Paralyzes worms
  - Used against intestinal roundworms
  - Mites and lice

Antimicrobials VII

Summary

- Protozoa and helminths can be killed too with drugs

Antimicrobials VIII

Learning Objectives

- What is drug resistance
- How it develops

Antimicrobials VIII

Drug Resistance

- Resistance begins with mutations
- Resistance transmitted by conjugation and transduction

Antimicrobials VIII

Drug Resistance

- Blocking Entry:
- Inactivating by enzymes:
- Alteration of target molecule
- Efflux of antibiotics

Antimicrobials VIII

Superbugs

- Bacteria resistant to many antibiotics
  - MRSA (most commonly)
  - *Staph aureus*
  - *Enterococcus faecium*
  - *Klebsiella pneumoniae*

Antimicrobials VIII

Summary

- Microbes can develop various ways to resist antimicrobials
- Superbugs are resistant to many drugs

Final Term Syllabus
Lecture 53 to 104

Efforts By

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Contact us to join our esteemed group

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SAY NO CURRUPTION

HELP TO OTHERS

Regards,

AARISH ALI & TEAM

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